ESET 350 Entirely

Hi! Here’s how you’re allowed to study <3

This is ground rules because time needs to be spent well and not unwell

- Stay hydrated, like go get water if you don’t have water
- Get in a good head/study space
    - When you’re studying now, THAT’s what you’re doing and THAT’s it
- Set clear goals for what you’re wanting to study and understand
    - Broad studying isn’t a good idea tbh
- Do things specific to that and note how well it went
- Take like 5-10 minute breaks every 50 mins or so
- Recognize that studying at all is progress and just do your best <3

All Concepts

Diodes
Diode physics

- Valence electrons
- Electron and hole current
- p-type and n-type semiconductors
- doping, pentavalent and trivalent impurities
- majority and minority carriers
- the pn junction
- depletion region
- barrier potential

Forward and reverse bias
Reverse breakdown
V-I characteristic curves (knee)
Diode models

- Ideal Model
- Practical Model
- Complete Model

Rectifiers

- Transformers
- Half wave
- Full wave
  - Center tapped
  - Bridge
- Peak inverse voltage
- Capacitor-Input filter
- Ripple Voltage, ripple factor
- Line, Load regulation (percentage)

Diode limiters
Biased diode limiters
Diode clampers
Voltage doubler
Zener diode

- Zener regulation
- Zener limiter

Light emitting diode (LED)

Operational Amplifiers
Ideal vs practical characteristics

- Open loop gain
- Input impedance
- Output impedance
- Bandwidth

Terminals and connections
Differential mode
Common mode
Common mode rejection ratio
Maximum output voltage (saturation)
Input offset voltage
Input bias current
Slew rate
Op amp bandwidth limitations / Gain bandwidth product
Negative feedback
Closed loop gain
Non-inverting amplifier
Inverting amplifier
Voltage follower / Unity gain buffer
Comparator
Node-voltage analysis method

- Finding all voltages and currents in an op amp circuit

Summing amplifier
Difference amplifier
Integrating amplifier
Differentiator
Complex impedances
Filters

- Ideal vs real filter responses
- Passive RC filter circuits
- Active filter circuits
- Why use active filters?
- Analysis
- Transfer functions
- Poles
- Magnitude, phase response
- 3-dB cutoff frequency
- Lowpass filters
- Highpass filter
- Cascaded filters
- Bandpass filters
- Band-reject / notch filters
- Sallen-key topology
- Butterworth filters
- Chebyshev filters
- Passband ripple
- Bode plot

Transistors
Bipolar Junction Transistors (BJTs)

- npn and pnp transistors
- Base
- Collector
- Emitter
- Currents
- Current controlled current source

Biasing

- Forward-active
- Saturation
- Cutoff
- Reverse-active

DC Beta βDC
Collector Characteristic Curves
DC Load Line
Voltage amplification (e.g., common emitter amplifier)
BJT as a switch
DC operating point or quiescent point (Q-point)
Voltage divider bias

- Thevenin's theorem

Linear operation

- Waveform distortion (saturation, cutoff)
- Coupling capacitors

Field Effect Transistor (see slides)

- Voltage controlled current source
- High input resistance
- Non-linear
- JFET – p-channel and n-channel
  - Vp, VGS(off) and IDSS
  - ID as a function of VGS
  - Universal Transfer Characteristic
- MOSFET
  - E-MOSFET (no structural conduction channel)
  - D-MOSFET (has a channel)
  - Basic MOSFET switching circuit (refer to last lab)
  - Basic idea behind CMOS (logic)
- FET transistor switch
- Amplifier Classes (A, B, AB, C, D)

Formulas

$$i = I_S\left(e^{v/nV_T} - 1\right)$$

$$i \cong I_S e^{v/nV_T}$$

$$V_T = \frac{kT}{q} \cong 25\text{ mV}$$

$$V_{r(pp)} \cong \left(\frac{1}{fR_LC}\right)V_{p(rect)}$$

$$V_{DC} \cong \left(1 - \frac{1}{2fR_LC}\right)V_{p(rect)}$$

$$r = \frac{V_{r(pp)}}{V_{DC}}$$

$$\frac{V_p}{N_p} = \frac{V_s}{N_s}$$

$$\text{Line regulation} = \left(\frac{\Delta V_{OUT}}{\Delta V_{IN}}\right) \times 100\%$$

$$\text{Load regulation} = \left(\frac{V_{NL} - V_{FL}}{V_{FL}}\right) \times 100\%$$

$$\text{CMRR} = 20\log\left|\frac{A_{dm}}{A_{cm}}\right|$$

$$V_o(t) = -\frac{1}{C_fR_s}\int_0^t V_i(\tau)\,d\tau$$

$$V_o(t) = -RC\frac{dV_i(t)}{dt}$$

$$Q = \frac{f_0}{\Delta f_{3\text{dB}}}$$

$$R_{IN(BASE)} = \frac{\beta_{DC}V_B}{I_E}$$

$$I_E = \frac{V_{TH} - V_{BE}}{R_E + R_{TH}/\beta_{DC}}$$

$$R_{TH} = \frac{R_1R_2}{R_1 + R_2}$$

$$V_{TH} = \left(\frac{R_2}{R_1 + R_2}\right)V_{CC}$$

$$r_e' = \frac{25\text{ mV}}{I_E}$$

$$R_{in(base)} = \beta_{ac}r_e'$$

$$A_v \cong \frac{R_c}{r_e'}$$

$$R_{in(base)} = \beta_{ac}(r_e' + R_{E1})$$

$$A_v \cong \frac{R_c}{R_{E1}}$$

$$R_c = R_C||R_L$$

$$Z_Z = \frac{\Delta V_Z}{\Delta I_Z}$$

Cascaded First Order Filters

For Lowpass: $\frac{\omega_{c1}}{\omega_{cn}} = \frac{1}{\sqrt{\sqrt[n]{2}-1}}$

For Highpass: $\frac{\omega_{c1}}{\omega_{cn}} = \sqrt{\sqrt[n]{2} - 1}$

Two Pole Sallen-Key:

$$H_{LP}(s) = \frac{\frac{1}{R^2C_1C_2}}{s^2 + \frac{2s}{RC_1} + \frac{1}{R^2C_1C_2}}$$

$$H_{HP}(s) = \frac{s^2}{s^2 + \frac{2s}{R_2C} + \frac{1}{R_1R_2C^2}}$$

Three Pole Sallen-Key:

$$H_{LP}(s) = \frac{\frac{1}{R^3C_1C_2C_3}}{s^3 + 2s^2\frac{C_1 + C_2}{RC_1C_2} + s\frac{C_2 + 3C_3}{R^2C_1C_2C_3} + \frac{1}{R^3C_1C_2C_3}}$$

$$H_{HP}(s) = \frac{s^3}{s^3 + s^2\frac{3R_2 + R_3}{R_2R_3C} + 2s\frac{R_1 + R_2}{R_1R_2R_3C^2} + \frac{1}{R_1R_2R_3C^3}}$$

Diodes

Diodes
Diode physics

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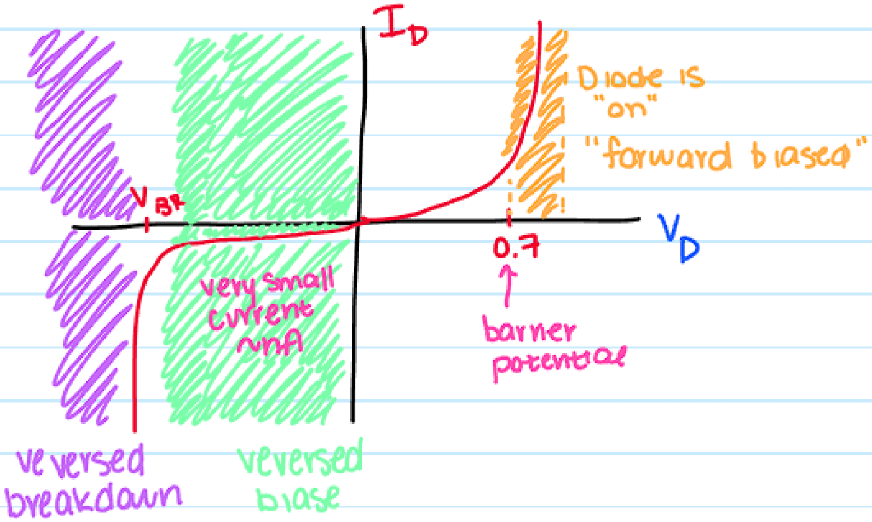
Light emitting diode (LED)

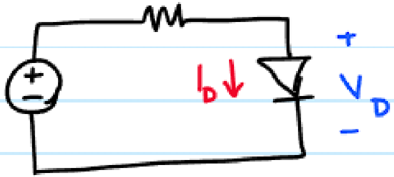
Diode Background Info

- The basis of diodes is silicon sharing its electrons covalently to create a crystalline structure within molecules
- @Room temperature, electrons jump between the valence ring and the conduction band leaving holes in the valence shell which are used later
- Recombination is when conduction band electrons fall back into their corresponding shell
- N Type Semiconductor:
  - Add pe**n**tavalent impurity (5 valence electrons)
  - Enables conduction
  - Gives a donor atom and does not leave a hole in the valence band
  - Major carriers are electrons, holes are minority carriers
  - **N** type -> electro**n** is major carrier
- P Type:
  - Adds a trivalent impurity, increases holes in the valence
  - All 3 valence are bonded so the holes are interior
  - Impurity atom is an acceptor atom
  - Holes are major carriers
  - Think that it's trivalent strictly because its backward
- N region becomes positive and P region becomes - which is super cool
- Depletion Region: Middle of the PN junction where the charge carriers become depleted
- There's more because of course there is but this is the basics. Can read up more if genuinely have time

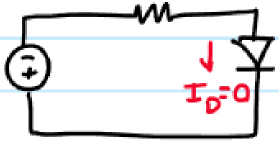
Basic Diode Characteristics Intro



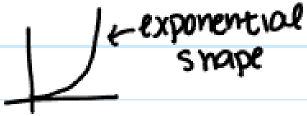
forward biased

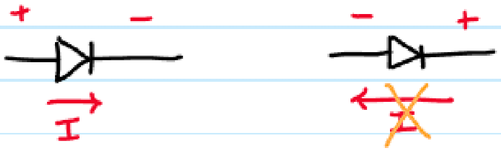


reverse biase case



Diode Circuit Analysis





* diode allows one direction of current

Shockley Diode Model Question

Shockley diode model

↳ based on physics

$$i = I_s\left(e^{\frac{V}{nV_T}} - 1\right)$$

i = diode current
V = voltage across diode
I_s = reverse saturation current
n = diode identity factor 1 to 2
V_T = thermal voltage = for S_i = 25 mV $\frac{KT}{q}$

example:

Consider a silicon diode with $n = 1.5$ i_1

Find ΔV if current changes from 0.1 mA to 10 mA $\Delta V = V_2 - V_1$

$$i \approx I_s e^{\frac{V}{nV_T}}$$

Shockley Diode Model Question

Shockley diode model

↳ based on physics

$$i = I_s\left(e^{\frac{v}{nV_T}} - 1\right)$$

i = diode current
v = voltage across diode
I_s = reverse saturation current
n = diode identity factor 1 to 2
V_T : thermal voltage = for Si = 25 mV $\frac{KT}{q}$

example:

Consider a silicon diode with $n = 1.5$

Find ΔV if current changes from 0.1 mA (i_1) to 10 mA (i_2) $\quad \Delta V = V_2 - V_1$

$$i \approx I_s e^{\frac{v}{nV_T}} \qquad i_1 = I_s e^{\frac{V_1}{nV_T}} \qquad i_2 = I_s e^{\frac{V_2}{nV_T}}$$

$$\frac{i_2}{i_1} = \frac{10\,mA}{0.1\,mA} = 100 = \frac{\cancel{I_s} e^{\frac{V_1}{nV_T}}}{\cancel{I_s} e^{\frac{V_2}{nV_T}}} \Rightarrow 100 = e^{\frac{(V_2 - V_1)}{nV_T}}$$

$$\ln(100) = \frac{V_2 - V_1}{nV_T} \Rightarrow \Delta V = nV_T \cdot \ln(100) \Rightarrow \boxed{\Delta V = 173\ mV}$$

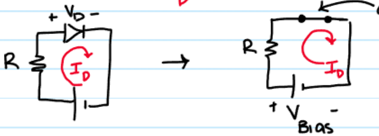
3 Diode Models

Ideal Model:

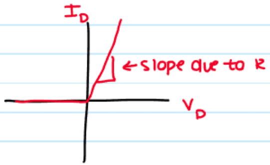
treats diode as simple switch
↳ either open or closed
neglects voltage drop across diode (barrier potential)

forward bias - diode is "on"

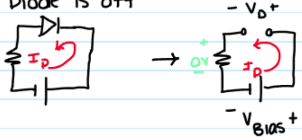
→ $V_D \geq 0$ diode is on
$V_D < 0$ diode is off



$I_D = \frac{V_{Bias}}{R}$



Reverse Bias
Diode is off

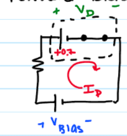


$I_D = 0A$
$V_D = V_{Bias}$

Practical Model

- constant voltage drop model
includes barrier potential $V_D = 0.7$
needs $\geq 0.7V$ to turn it on

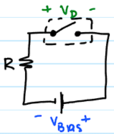
Forward Bias - diode on



$I_D = \frac{V_{Bias} - 0.7}{R}$

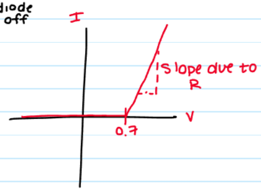
$V_{Bias} > 0.7$

Reverse Bias - diode off



$V_{Bias} < 0.7$

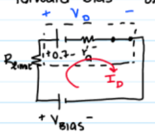
$I_D = 0$



"Complete" Diode Model

more accurate approximation
includes barrier potential
accounts for small forward dynamic resistance r_d'
accounts for large internal reverse resistance r_R'

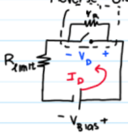
forward Bias - diode is ON



$I_D = \frac{V_{Bias} - 0.7}{R_{limit} + r_d'}$

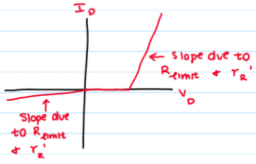
$V_D = 0.7 + r_d' \cdot I_D$

Reverse Bias - diode is OFF



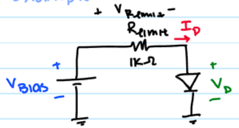
$I_D = \frac{V_{Bias}}{r_R' + R_{limit}}$

$V_D = I_D \cdot r_R'$



Basic Example (but a question that get's asked a lot)

Example:



IDEAL MODEL

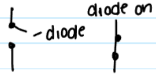
V_{Bias}	Diode status on, off, turning on	I_D	V_D	V_{Rlimit}
−5				
−1				
0				
0.5				
0.7				
5				
10				

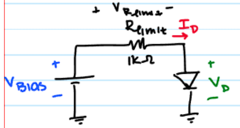
* no voltage across diode when on b/c short

* b/c KVL

recall:

diode off



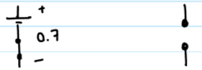


PRACTICAL MODEL

V_{Bias}	Diode status on, off, turning on	I_D	V_D	V_{Rlimit}
−5				
−1				
0				
0.5				
0.7				
5				
10				

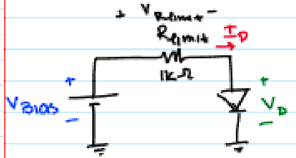
Diode ON

Diode OFF



COMPLETE MODEL

$I_D \cdot 1M$



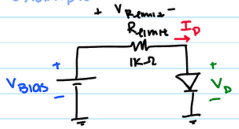
$r_d' = 10\,\Omega$

$r_R' = 1\,M\Omega$

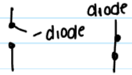
V_{Bias}	Diode status on, off, turning on	I_D	V_D	V_{Rlimit}
−5				
−1				
0				
0.5				
0.7				
5				
10				

Basic Example (but a question that get's asked a lot)

Example:



recall:
diode off
diode on



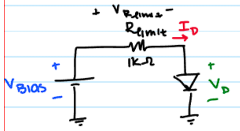
IDEAL MODEL

V_{Bias}	Diode status on, off, turning on	I_D	V_D	V_{Rlimit}
−5	off	0A	−5V	0V
−1	off	0A	−1V	0V
0	just turn on	0A	0V	0V
0.5	on	0.5mA	0V	0.5V
0.7	on	0.7mA	0V	0.7V
5	on	5mA	0V	5V
10	on	10mA	0V	10V

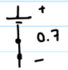
V=IR

* no voltage across diode when on b/c short
* b/c KVL

PRACTICAL MODEL



Diode ON



Diode OFF

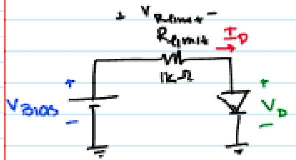
V_{Bias}	Diode status on, off, turning on	I_D	V_D	V_{Rlimit}
−5	off	0A	−5V	0V
−1	off	0A	−1V	0V
0	off	0A	0V	0V
0.5	off	0A	0.5V	0V
0.7	just turn on	0A	0.7V	0V
5	on	4.3mA	0.7V	4.3V
10	on	9.3mA	0.7V	9.3V

[$\frac{V_{Bias} - 0.7}{r_{limit}}$

$V_{Bias} - 0.7$

$I_D \cdot R_{limit}$

COMPLETE MODEL



$r_d' = 10\Omega$

$r_R' = 1M\Omega$

$I_D \cdot 1M$

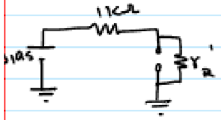
V_{Bias}	Diode status on, off, turning on	I_D	V_D	V_{Rlimit}
−5	off	−4.995μ	−4.995V	−5 mV
−1	off	−0.999μ	−0.99V	−1 mV
0	off	0	0	0
0.5	off	0	0.5	0
0.7	just turn on	0	0.7	0
5	on	4.25 m	0.7426V	4.257 V
10	on	9.21 m	0.7921V	9.207V

$V_{Bias} - V_D$ or $I_D \cdot R_{limit}$

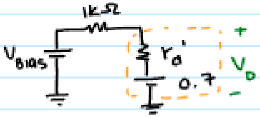
$0.7 + I_D \cdot 10\Omega$

Diode Off

reverse bias



Diode On

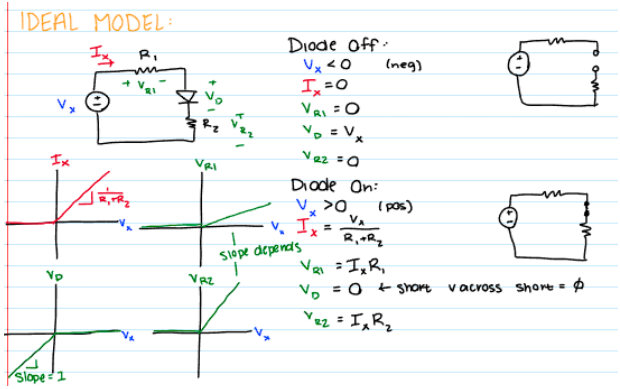


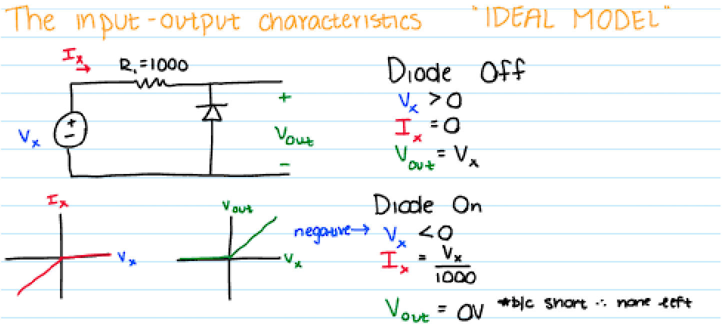
divide reverse bias

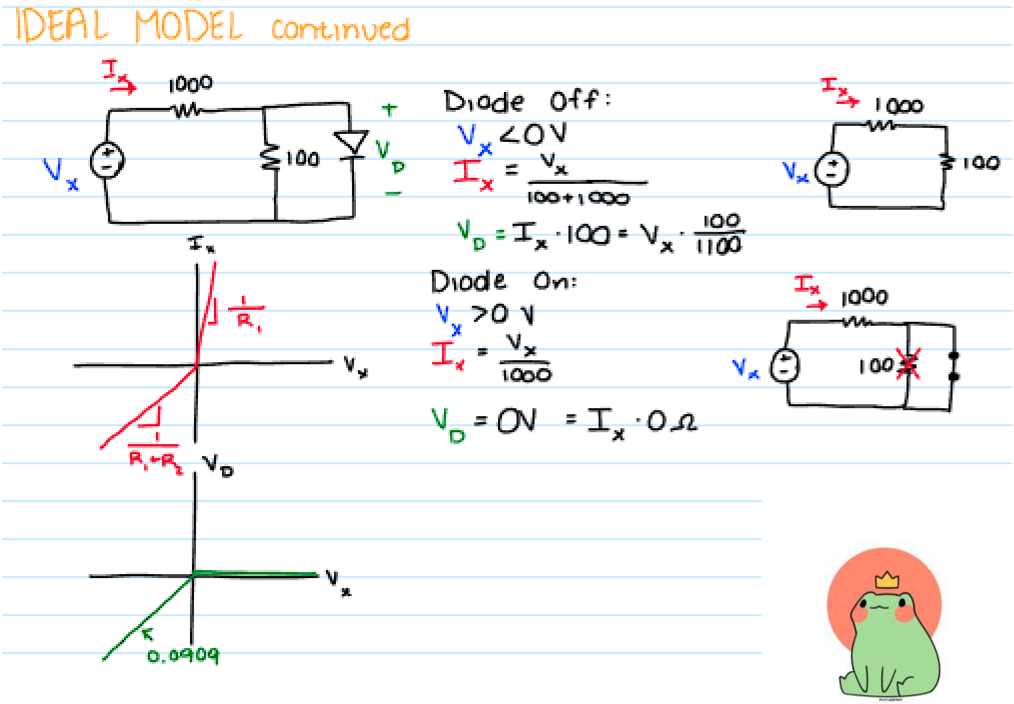
$I_D = \frac{V_{Bias}}{1k + 1M}$ (negative) (off)

$I_D = \frac{V_{Bias} - 0.7}{1k + 10}$ (positive) (on)

Ideal Model Basic Examples (Skim don't work too hard on)







Basic Diode Analysis

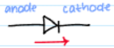
PRACTICAL | CONSTANT VOLTAGE DROP MODEL

principles of diode analysis

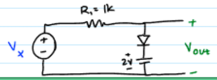
① begin by assuming certain states for all diodes
↳ check final result against assumptions

② If a diode is about to turn on or off, its voltage $V_D = 0.7$, but its current $I_D = 0\,A$

③ If a diode is on, and carries a current it must flow from anode → cathode



PRACTICAL MODEL



what if the order of the components is changed?

what if we flipped the diode?

For each:

- Plot I and V_o
- For on and off
  - Find V_x
  - Find I_x
  - Find V_O

Basic Diode Analysis

PRACTICAL | CONSTANT VOLTAGE DROP MODEL

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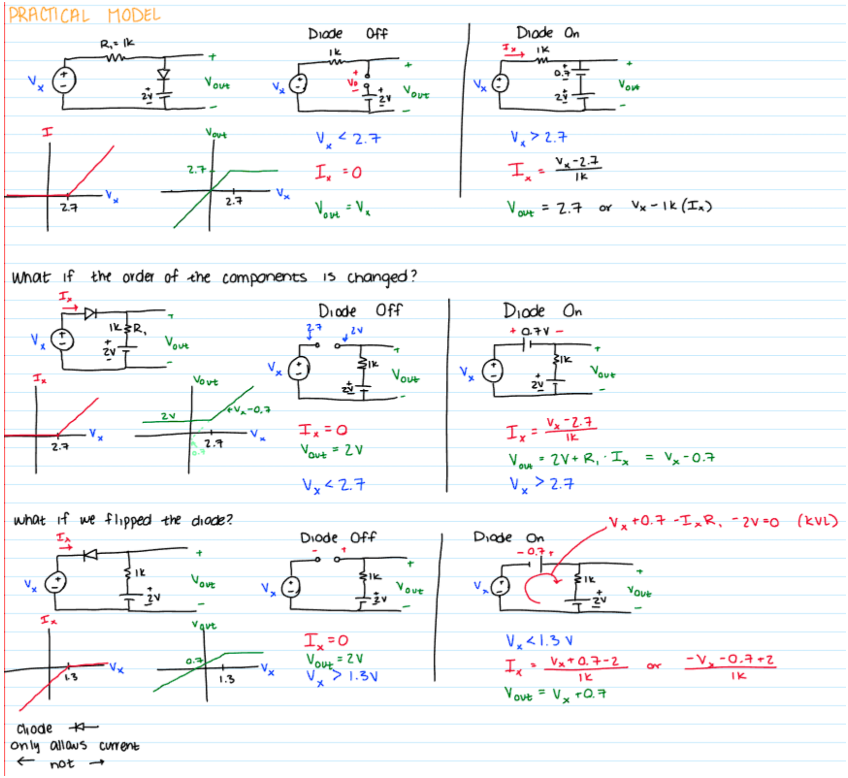
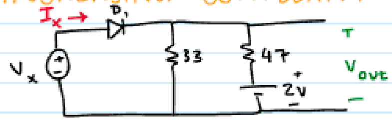
③ If a diode is on, and carries a current it must flow from anode → cathode

anode cathode



Diode Analysis More Complex

INCREASING COMPLEXITY

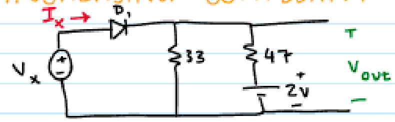


Find for on and off:

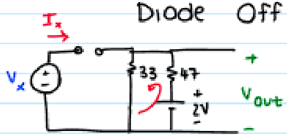
- V_X
- V_o
- I_x

Diode Analysis More Complex

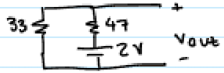
INCREASING COMPLEXITY

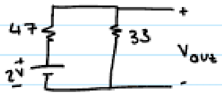


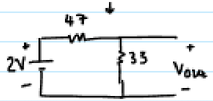
Diode Off



$I_x = 0$







$$V_{out} = \frac{33\,(2V)}{47+33}$$

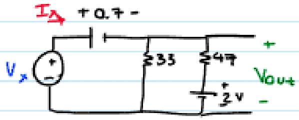
$$= 0.825\ V$$

$V_{out} = 0.825\ V$

$V_x < 1.525\ V$

$(0.7)+0.825$

Diode On



$V_x > 1.525\ V$

$V_{out} = V_x - 0.7$

$I_x = I_1 + I_2$

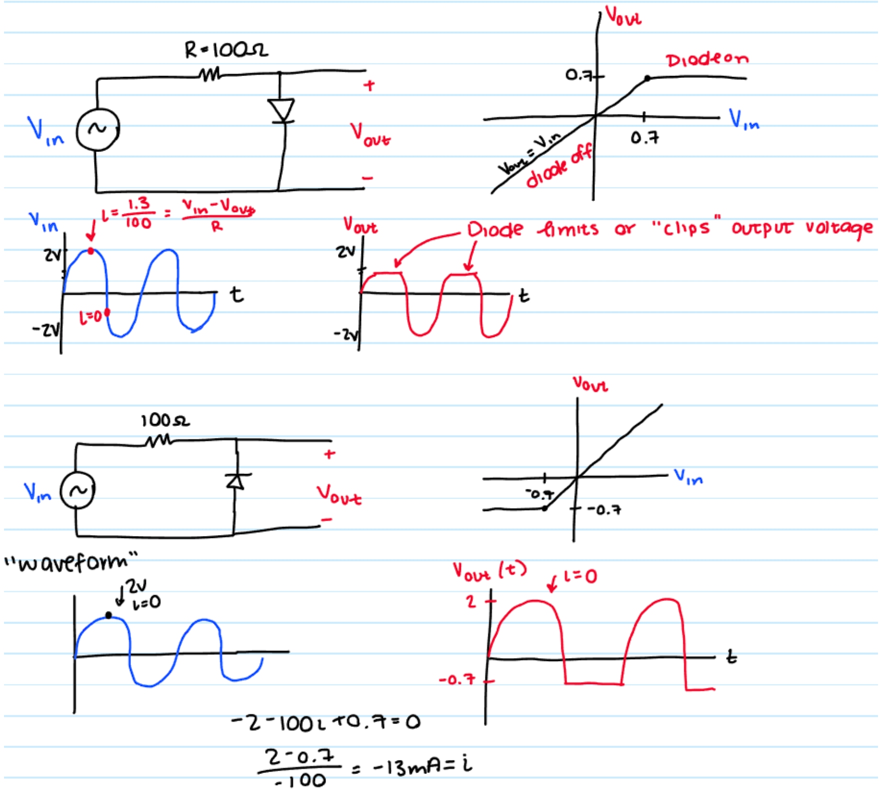
$$I_x = \frac{V_{out}}{33} + \frac{V_{out}-2}{47}$$

need to apply KCL|KVL, nodal or mesh

Diodes So Far

- When a certain amount of voltage is sent across a diode, it turns on and allows current to flow
- If current is unable to flow, all components that normally eat up voltage can't, so all will be across the diode
- Also important to note is that voltage is the same across parallel components. This means that if a diode is in parallel with any other component, the voltage across it will be 0.7V. This can lead to determining other behaviors of a circuit
    - In relation to this, the easier branch is always the one that should be calculated for values since they are the same
- Diodes can be considered loosely as a 0.7V DC battery with the line on the diode representing the negative terminal
    - This leads to regular KVL analysis when considering it very simply as just another battery
- FINAL THING:
    - Since diodes limit the voltage effectively, they can be specifically configured for limiting or clipping capabilities
    - In other words, prior theory is important to understand function and how to calculate, but the same theory can be used for different specific functions which will be covered after this

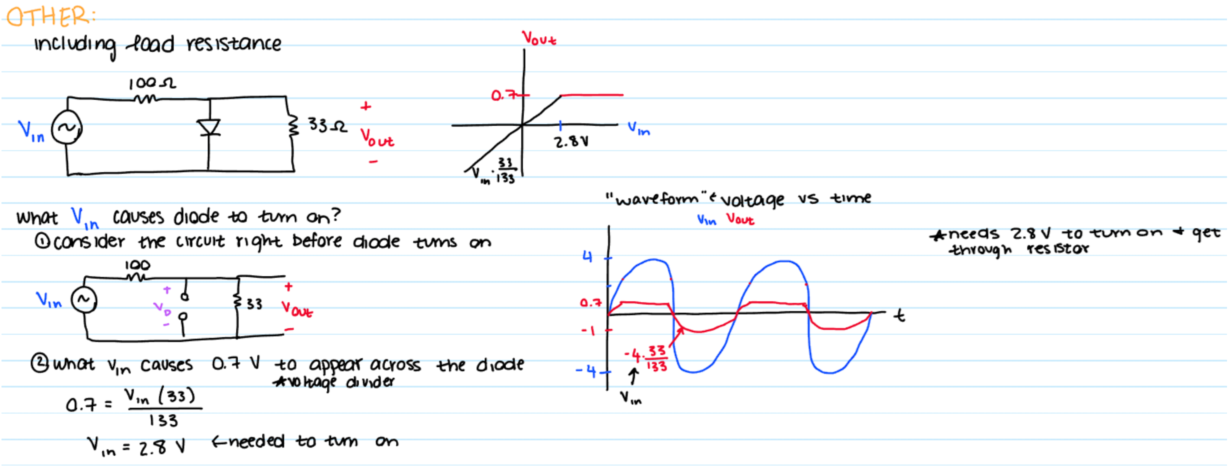
Intro to Limiters

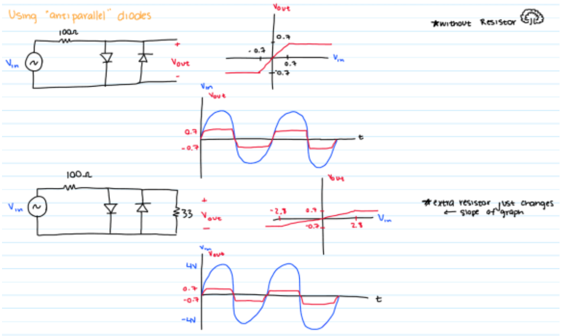


Notice how this isn't anything new:

- First circuit, its a simple diode circuit that we've already evaluated
  - Always use practical unless specified otherwise
  - It's a simple 0.7V drop and that's all that will ever be observed on the output
  - That means that V_o will evaluate the input wave as normal except always limit the top to 0.7V output
- Second circuit, if we flip the circuit around it will instead limit the negative output to -0.7V which makes sense
  - Current can be deduced from recognizing how much voltage is seen across the components at that point and the resistive elements that are at play as well

Slightly more complicated limiter

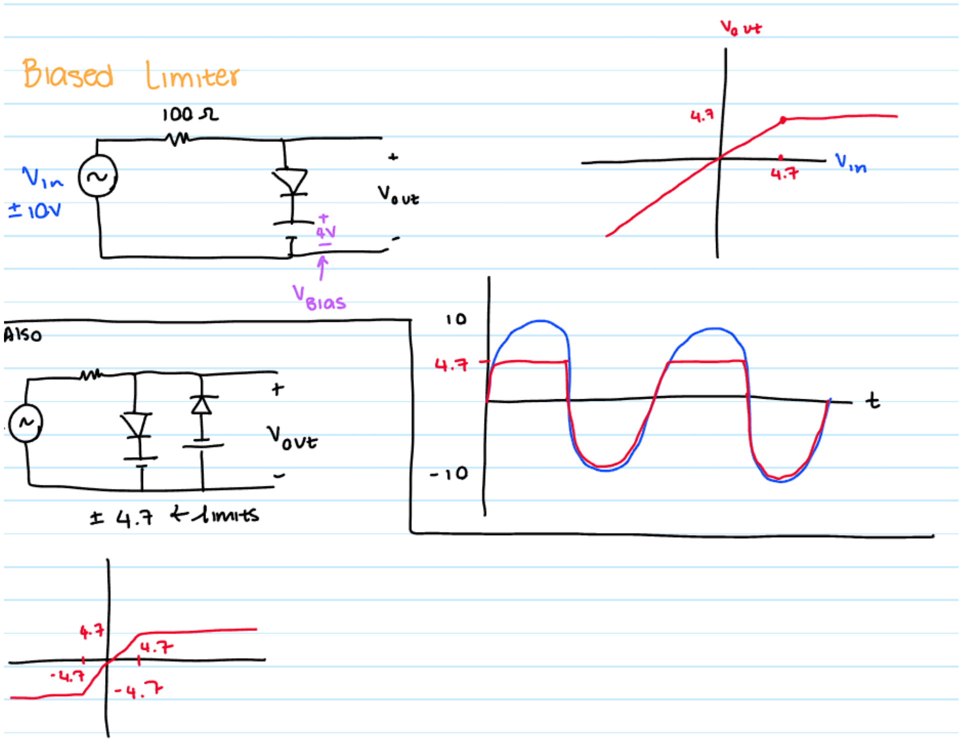




Quick note (bc simple): Using 2 diodes causes double limiting, adding an additional resistor just changes the slope since different V_{in} will be required

- The voltage across the 33 ohm is 0.7V
- Additionally, it can be seen which V_{in} causes exactly 0.7V across the diode by using a voltage divider for the 33 ohm resistor
- If 2.8V (calculated above) wasn't used, and instead 4V was, 0.7V would clip the positive voltage, and the negative would result in -1V, which can be calculated as voltage in the 33 ohm by a voltage divider (volt divider solves a lot of these relationships)

Biased Limiter - Uses an additional source and considers diode as a source as well

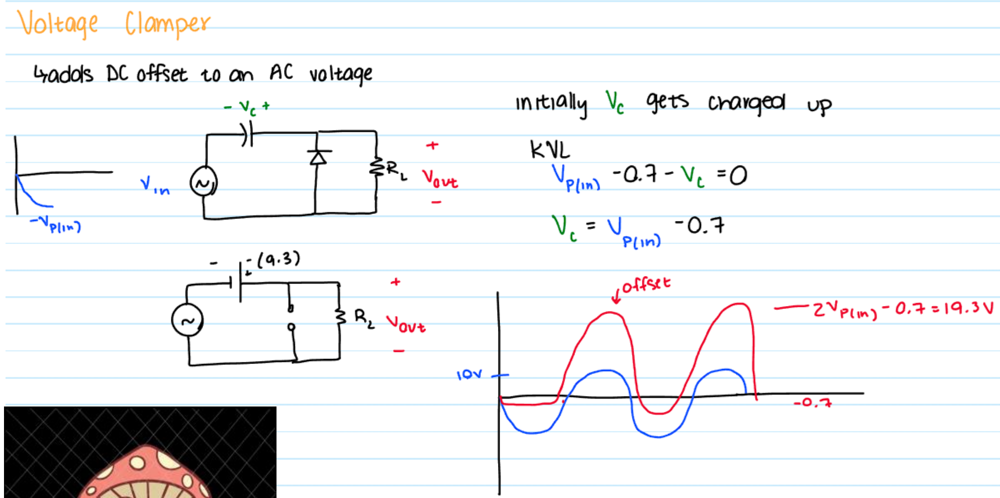


Since the polarities are the same, they add, so 4.7V is required in order to allow current flow

This also means that the voltage will be clipped for the output

Similar to the double reversed diodes, it will be clipped positively and negatively at the same p

Voltage Clampers



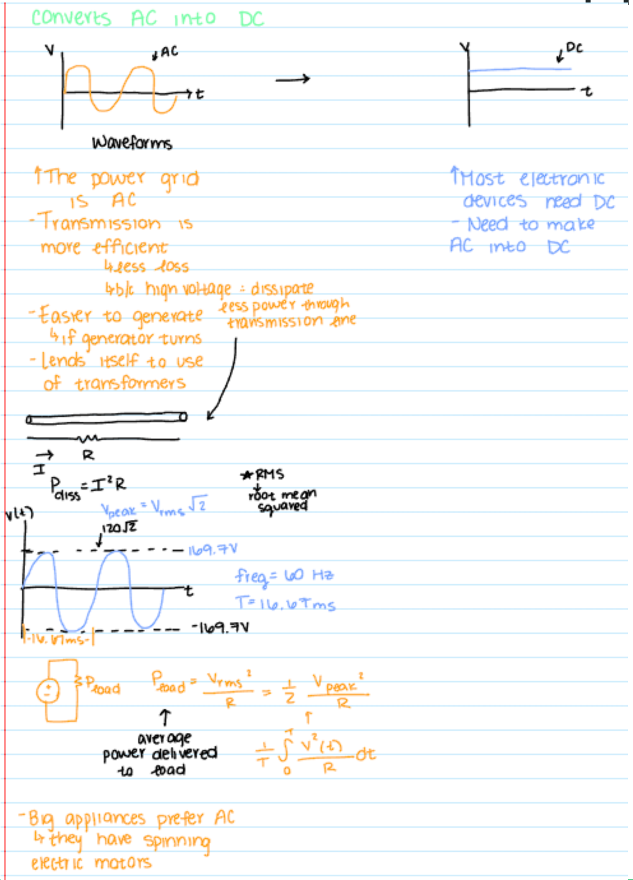
Background is a lil ugly but the froggy is cutie so its fine <3

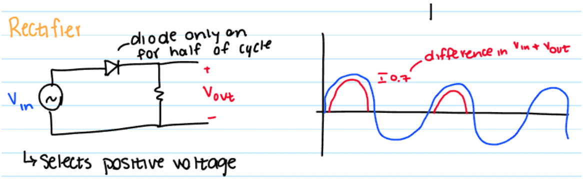
The way to think about clampers is that they will charge up initially and then function as a battery of the voltage that was able to be seen across it

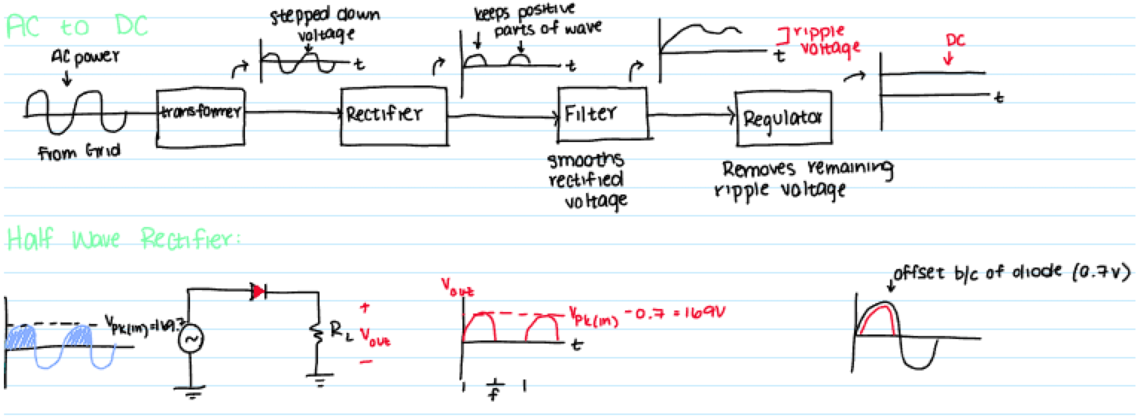
This results in doubling the voltage minus the amount seen by any diodes in the circuit

This is the simplest case of it, isn't super utilized but will be seen in some test scenarios

Basic DC Power Supply Background



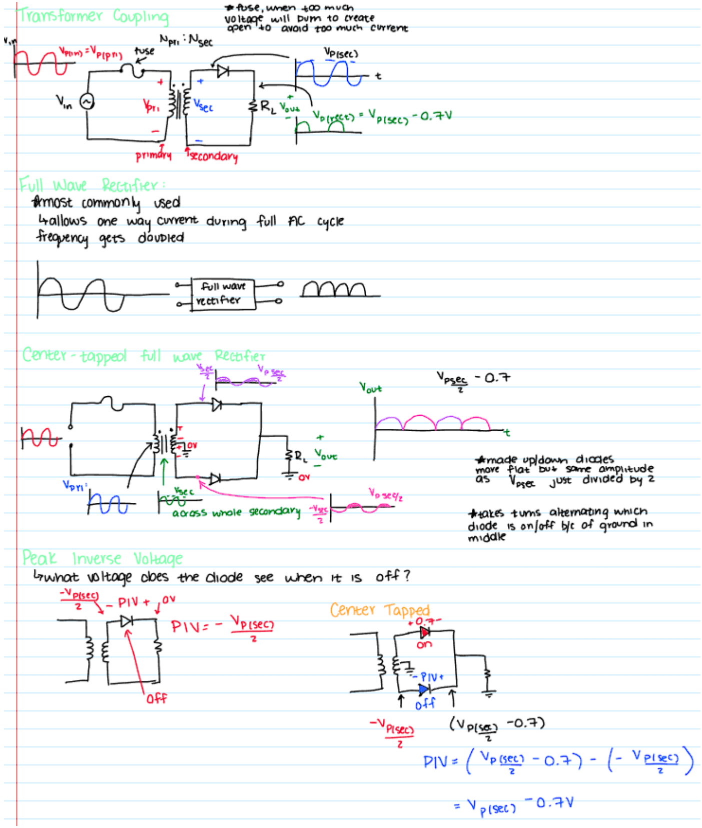




This is the basics of the upcoming topic and what a significant amount of it is based upon

Things are never tested in this context and in general are pretty known already, but here just in case are needed for context

Transformers and Power Supply Supplements



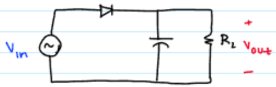
There's a problem that normally stems from all of this, but it needs context in order to be solved:

- Transformer Coupling:
  - First, all transformers need a fuse
  - It prevents failure if there's a blowout
  - Voltage from source is V_{in} or V_{RMS}
  - It's V_{peak} or V_{pri} if mult by sqrt2
  - Volt on the right after trans is V_{sec}
  - After it gets rectified by a diode (typically -0.7V), it becomes V_{rec}
- Full Wave Rectifier
  - Allows a one way current to flow both ways
  - There is center tapped and bridge
- Center Tapped:
  - The inductor involved in trans gets grounded and splits the voltage into negative and positive components
  - This results in $V_{sec}/2$. It will only be subtracted by 0.7V though since only one of the branches will flow at once
- Peak Inverse voltage
  - Since there are two diodes and it lows on different polarities, it sees negative V_{sec} - 0.7V
  - If it sees too much it can be really bad

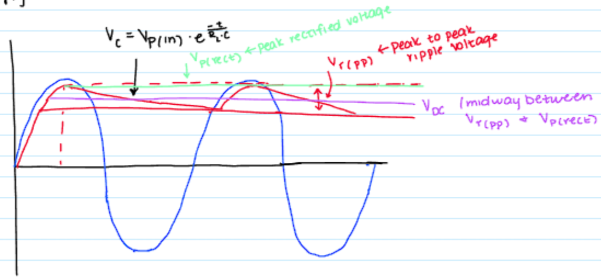
Full Wave Bridge Rectifier

Add filtering to build a DC power supply

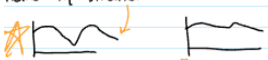
"capacitor input filter"



$V_{p(rect)} = V_{p(in)} - 0.7V$



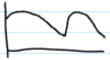
make R_L smaller?

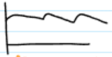


R_L very large

more resistance, C cant discharge

make C larger





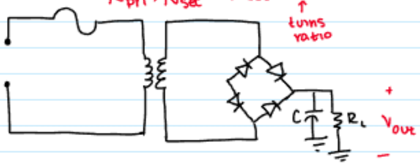
★ larger C = more smooth

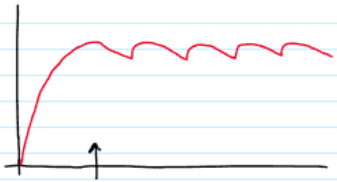
full wave?

$N_{pri} : N_{sec}$

$\frac{N_{pri}}{N_{sec}} = N = \frac{V_{pri}}{V_{sec}}$

↑ turns ratio

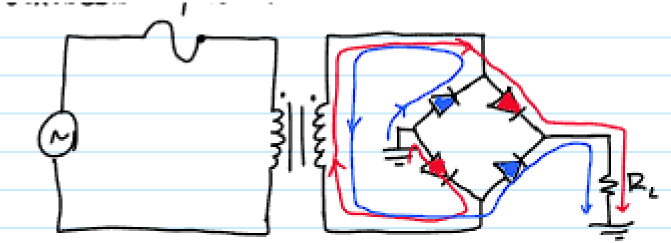




frequency is doubled
120 Hz or 100 Hz

All things being equal, would a full or half wave power supply need a bigger C for the same ripple voltage?

↳ half wave needs larger C to maintain smoother voltage b/c frequency 2x as fast



Formulas:

$$V_{r(pp)} = V_{p(rect)} \left(\frac{1}{f \cdot R_L \cdot C} \right)$$

$$V_{DC} = V_{p(rect)} - \frac{1}{2} V_{r(pp)}$$

$$= V_{p(rect)} - \frac{1}{2} V_{p(rect)} \left(\frac{1}{f \cdot R_L \cdot C} \right)$$

$$= V_{p(rect)} \left(1 - \frac{1}{2f \cdot R_L \cdot C} \right)$$

↑ f is the frequency

US − 60 Hz
EU − 50 Hz
> typically

Ripple factor

$$r = \frac{V_{r(pp)}}{V_{DC}}$$

$V_{r(pp)}$: V ripple peak to peak

V_{DC}: The DC equiv of the secondary after all things configured

Ripple Factor: How much it ripples, its a proportion kind of

Full Wave Bridge Rectifier Worked Out:

$V_{DC} = 11$

$RF = 0.07$

$f = 60\ Hz\ (\cdot 2)$

$V_{r(pp)} = 0.77$

$V_{p(rect)} = 11.385$

$V_{p(sec)} = 12.785$

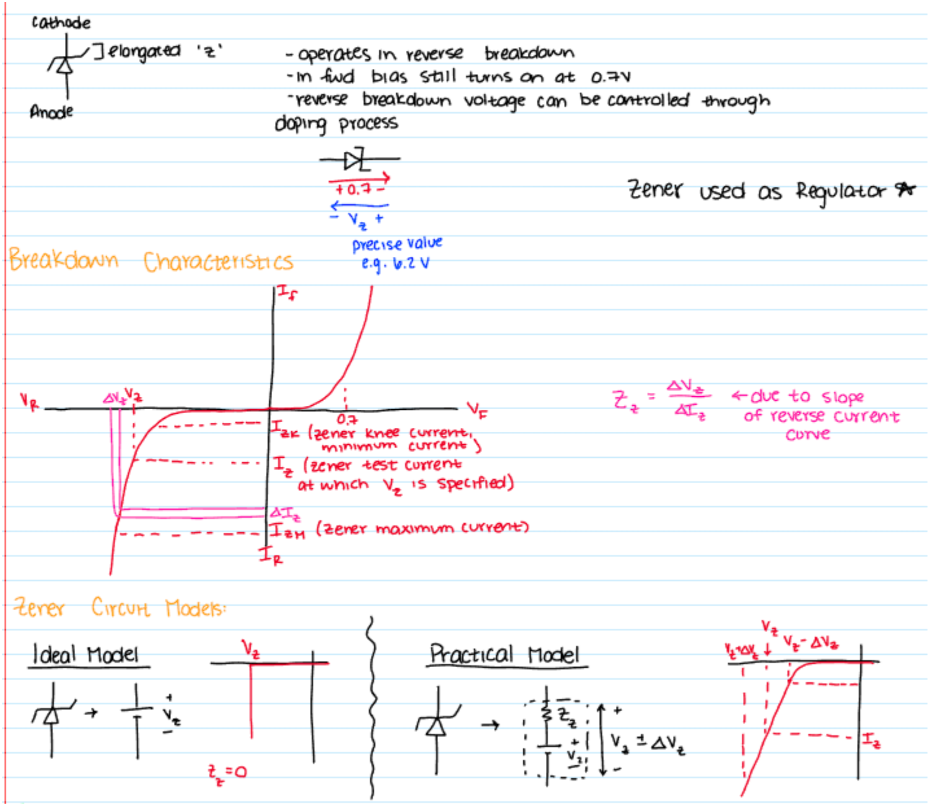
$V_{rms} \cdot \sqrt{2} = V_{primary}$

$N = \frac{120(1.414)}{12.785}$

$0.77 = 11.385 \left(\frac{1}{120 \cdot 124 \cdot C} \right)$

When practicing will add more

Zener Diode Intro

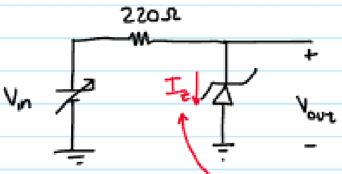


- Consider a diode that operates in reverse breakdown as well as regular states
- Has 3 states
  - I_{ZK} - The knee current (minimum)
  - I_Z - Zener test current(aka want)
  - I_{ZM} - Zener max current
- Has multiple applications that will be seen soon

Zener Diode Practice Problems/Application

Example:

Zener Regulation with Variable input Voltage



$I_{ZK} \le I_Z \le I_{ZM}$

$V_Z = 10V$ $\quad I_Z = 25\ mA$

$I_{ZK} = 0.25\ mA$

Max Power $= 1W$ $\quad I_{ZM} = \frac{1W}{V_Z} = \frac{1W}{10V} = 100\ mA$

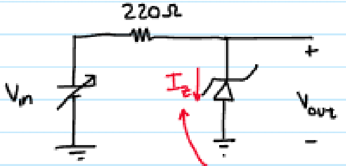
Assume Ideal $Z_Z = 0$

For what range of input voltages can a constant output of 10V be maintained?

Zener Diode Practice Problems/Application

Example:

Zener Regulation with Variable input Voltage



$I_{ZK} \leq I_Z \leq I_{ZM}$

$V_Z = 10V$ $I_Z = 25\ mA$

$I_{ZK} = 0.25\ mA$

Max Power = 1W $I_{ZM} = \frac{1W}{V_Z} = \frac{1W}{10V} = 100\ mA$

Assume Ideal $Z_Z = 0$

For what range of input voltages can a constant output of 10V be maintained?

Apply KVL:

$V_{in,min} - 220 I_{ZK} - V_Z = 0$

$V_{in,min} - 220(0.25\ mA) - 10V = 0$

$V_{in,min} = 10.055\ V$

$V_{in,max} - 220 I_{ZM} - V_Z = 0$

$V_{in,max} - 220(100mA) - 10V = 0$

$V_{in,max} = 32\ V$

$\underbrace{10.055 \leq V_{in} \leq 32.0V}$

In this range diode happy + on

more than 32 : blow up

less than, no reverse breakdown

Reminder:

I_{ZK} - knee

I_Z - asked for

I_{ZM} - max

Looking for maximum and minimum voltage, use maximum and minimum current

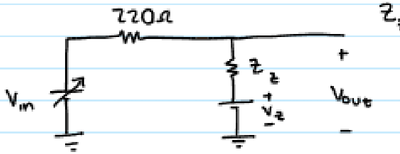
If not given I_{ZK} or I_{ZM} not entirely sure what to do :)

Zener Diode Practice Problems/Application

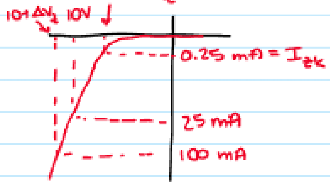
$V_z = 10V$ $I_z = 25$ mA
$I_{zk} = 0.25$ mA
Max Power = 1W $I_{zM} = \frac{1W}{V_z} = \frac{1W}{10V} = 100$ mA

Assume ideal $Z_z = 0$

Accounting for Zener Impedance (practical) same values

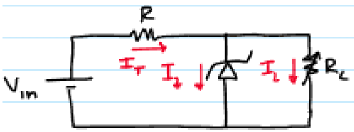


$Z_z = 7\Omega$



Zener Diode with Variable Loads

R_L: like potentiometer



Zener Diode Practice Problems/Application

$V_z = 10V$ $I_z = 25\ mA$
$I_{zk} = 0.25\ mA$
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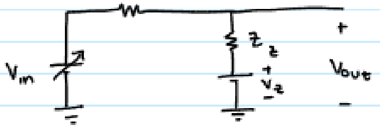
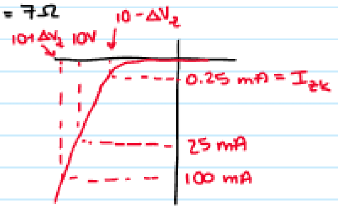
Assume Ideal $Z_z = 0$

Accounting for Zener Impedance (practical) — same values



$Z_z = 7\Omega$



$10 - \Delta V_z = 10 - 7\Omega(25\ mA - 0.25\ mA)$
$= 9.83\ V$ ← $V_{out\,min} = V_{z\,min}$

$10 + \Delta V_z = 10 + 7\Omega(100\ mA - 25\ mA)$
$= 10.525\ V$ ← $V_{out\,max} = V_{z\,max}$

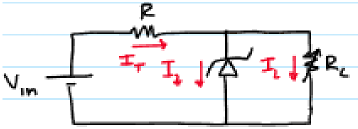
$V_{in,min} - 220 I_{zk} - 9.83 = 0$
$V_{in\,min} - 220(0.25\ mA) - 9.83 = 0$
$V_{in\,min} = 9.885\ V$

$$\boxed{9.885 \le V_{in} \le 32.525}$$

$V_{in\,max} - 220 I_{zM} - 10.525\ V = 0$
$V_{in\,max} = 32.525\ V$

Zener Diode with Variable Loads

R_L: like potentiometer



KCL: $I_{tot} = I_z + I_L$

$I_{zk} \le I_z \le I_{zm}$

If no load, $R_L = \infty$ (open circuit)
↳ All current must go through zener diode

As $R_L\downarrow$ $I_L\uparrow$, $I_z\downarrow$ I_{tot} (constant)

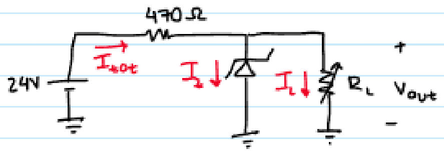
Zener Diode Practice Problems/Application

Determine Minimum, + maximum load currents for which the zener diode will maintain regulation. What is the minimum R_L that can be used?

$V_z = 12V$ $\qquad$ $Z_z = 0\Omega$ (ideal)

$I_{zk} = 1mA$

$I_{zm} = 50\ mA$ $\qquad$ Find range of R_L



Zener Diode Practice Problems/Application

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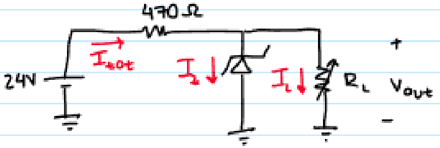
Find range of R_L



$24 - I_t(470) - 12 = 0$

$I_t = 25.5\ mA$

$I_{tot} = I_z + I_L$

$R_L > 489\Omega$

$25.5mA = I_{zK} + I_{Lmax}$

$I_{Lmax} = 24.5\ mA$

remember inverse relationship

$R_{Lmin} = \frac{12\ V}{24.5\ mA} = 489\Omega$

$R_{Lmax} = \infty$

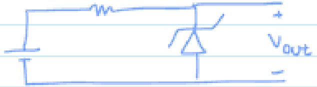
$25.5\ mA = I_{zM} + I_{Lmin}$

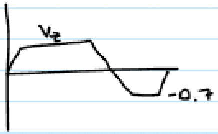
$I_{Lmin} = 25.5mA - 50\ mA$

$= \underbrace{\text{negative}}_{\text{can't have negative}}$

$\therefore I_{Lmin} = 0$

Zener Limiter is also possible





LED Applications/Questions

LED Examples:

NOTE: LED fwd V drop is not 0.7 V

→ V_f varies w/material/light color

example:

An LED is rated at $I_f = 20mA$ for $V_f = 1.3V$

if 5V power supply is used, what R_{Limit} resistor should be used?

max current

If we want to use a 12V, 1A power supply to power as many of these LEDs as possible, design circuit.

how many branches in parallel?

LED Applications/Questions

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An LED is rated at $I_f = 20\,mA$ for $V_f = 1.3V$
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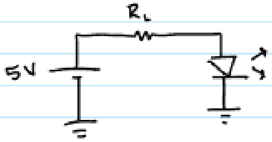
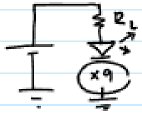


$$\frac{5V - 1.3}{20mA} = 185\,\Omega$$

If we want to use a 12V, 1 A (max current) power supply to power as many of these LEDs as possible, design circuit.



$$\frac{12V}{1.3V} = 9.23 \text{ LEDs in series}$$

$\therefore 9$

$$12 - 1.3(9) - R_L \cdot 20mA = 0$$

$$R_L = 15\,\Omega$$

how many branches in parallel?

$\frac{1A}{20mA} = 50$ branches $\boxed{50 \cdot 9 = 450 \text{ LED}}$

Operational Amplifiers - We stan the cute little bug

Operational Amplifiers

Ideal vs practical characteristics

- Open loop gain
- Input impedance
- Output impedance
- Bandwidth

Terminals and connections
Differential mode
Common mode
Common mode rejection ratio
Maximum output voltage (saturation)
Input offset voltage
Input bias current
Slew rate
Op amp bandwidth limitations
Negative feedback
Closed loop gain
Non-inverting amplifier
Inverting amplifier
Voltage follower / Unity gain buffer
Comparator
Node-voltage analysis method

- Finding all voltages and currents in an op amp circuit

Summing amplifier
Difference amplifier
Integrating amplifier

Filters

- Ideal vs real filter responses
- Passive RC filter circuits
- Active filter circuits
- Why use active filters?
- Analysis
- Transfer functions
- Poles
- Magnitude, phase response
- 3-dB cutoff frequency
- Lowpass filters
- Highpass filter
- Cascaded filters
- Bandpass filters
- Band-reject / notch filters
- Sallen-key topology
- Butterworth filters
- Chebyshev filters
- Passband ripple
- Bode plot
- / Gain bandwidth product

Additional Content

Exam 2

Labs

Homeworks

Pertinent Clicker Questions

Characteristics of Operational Amplifiers

Ideal:

- Infinite Loop Gain (AV)
  - Give 1, get ∞, “when there is no feedback”
- Infinite Input Impedance
  - Infinite resistance against input
- Zero Output Impedance
  - 0 resistance to reduce output
- Infinite Bandwidth
  - Wave with power

Practical:

- 100K - 1M Open Loop Gain
  - Gain without feedback
- “Big” Input Impedance
  - Resistance between input
- Low Output Z
- Approx. 3 MHZ Bandwidth

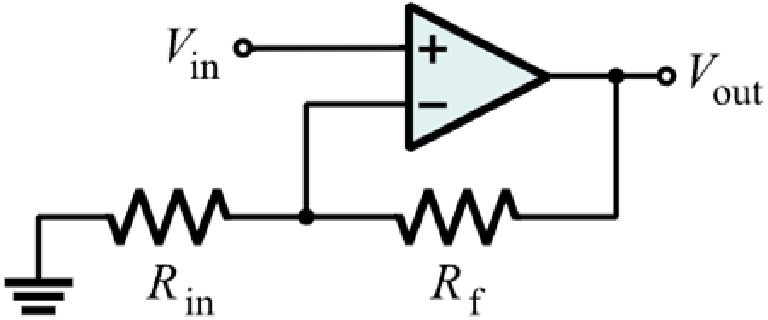
	Voltage Gain	Input Z	Output Z	Bandwidth
Without negative feedback	A_{ol} is too high for linear amplifier applications	Relatively high	Relatively low	Relatively narrow (because gain is so high)

	Voltage Gain	Input Z	Output Z	Bandwidth
With negative feedback	A_{cl} is set to desired value by the feedback circuit	Can be set as desired	Can be set as desired	Significantly wider

Additional Definitions and Equations

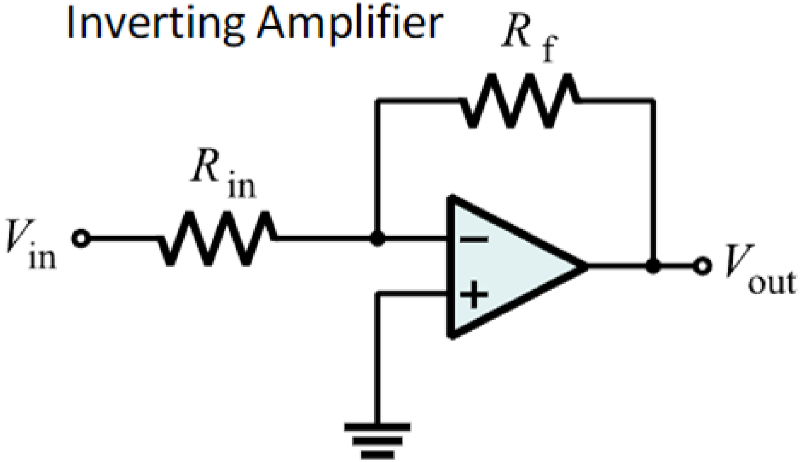
- Input Signal Modes
  - Common Mode: Signals applied to both sides with same phase (calculated with 0) $V_a = V_b = V_{in}$
  - Differential Mode: Amplifiers difference btwn the two terminals $V_a = -V_{in}/2$; $V_b = +V_{in}/2$; $V_a = -V_b$
  - Noise Calculation: $dB = 20 \log (V_2/V_1)$
  - Common Mode Rejection Ratio: $|A_{dm}/A_{cm}|$ CMRR: ability to amplify signals and reject common mode
- Slew Rate: How quickly Op Amp Can respond to changing output: $\Delta V_o/\Delta t$
- Negative Feedback: Taking output and feeding back to input (all op amp)
  - 180^o out of phase
- Max Output Voltage Swing: Limited to less than $+-V_{cc}$ depends on load resistance
- Input Offset Voltage: When both V_- and V_+ are 0, V_o ideally 0,but offset volt @input causing output offset
- Input Bias Current: Small current needed at input for operation, assumed 0 $I_{Bias} = I_1+I_2/2$

Non-Inverting Amplifier



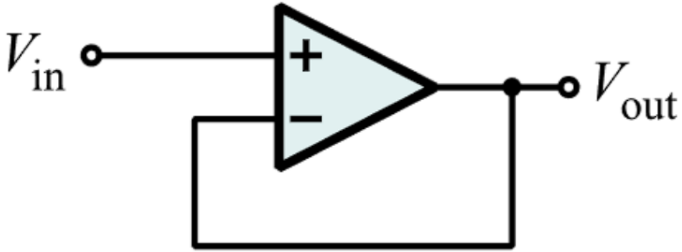
$V_o/V_i = A_{cl} = 1+R_f/R_i$

Inverting Amplifier



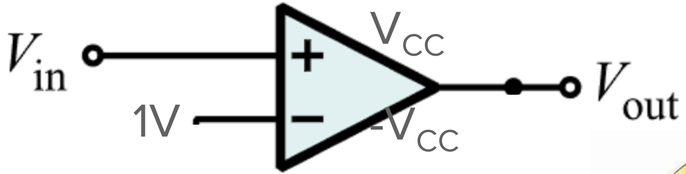
$V_o/V_i = A_{cl} = -R_f/R_i$

Voltage Follower / Unity Gain Buffer



$V_o/V_i = A_{cl} = 1$

Comparator

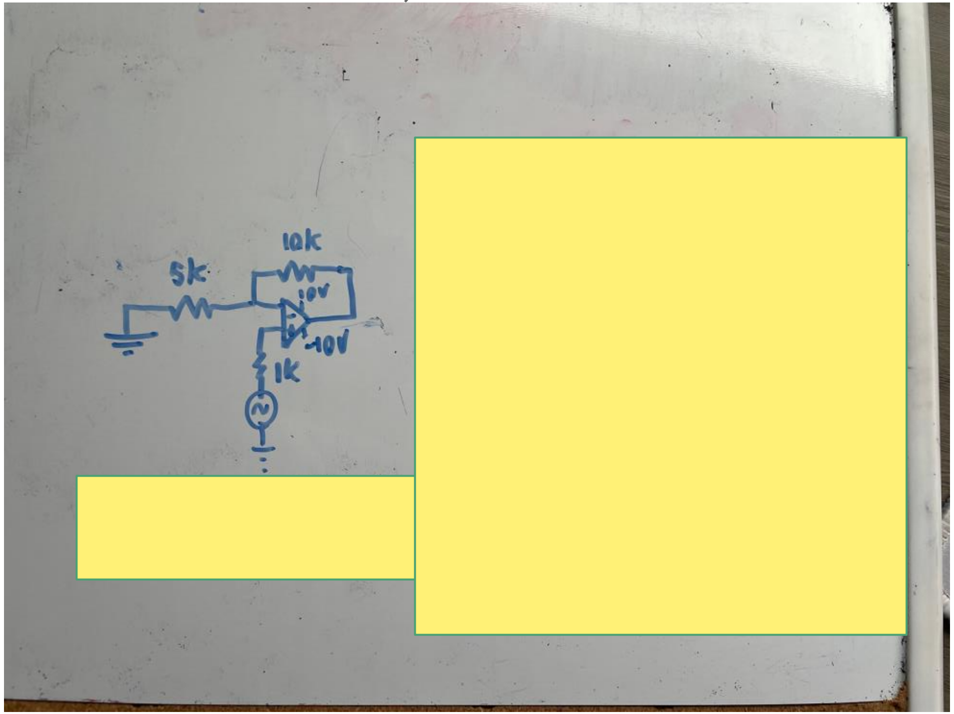


$V_{in} < 1V : V_o = -V_{cc}$

$V_{in} > 1V : V_o = +V_{cc}$

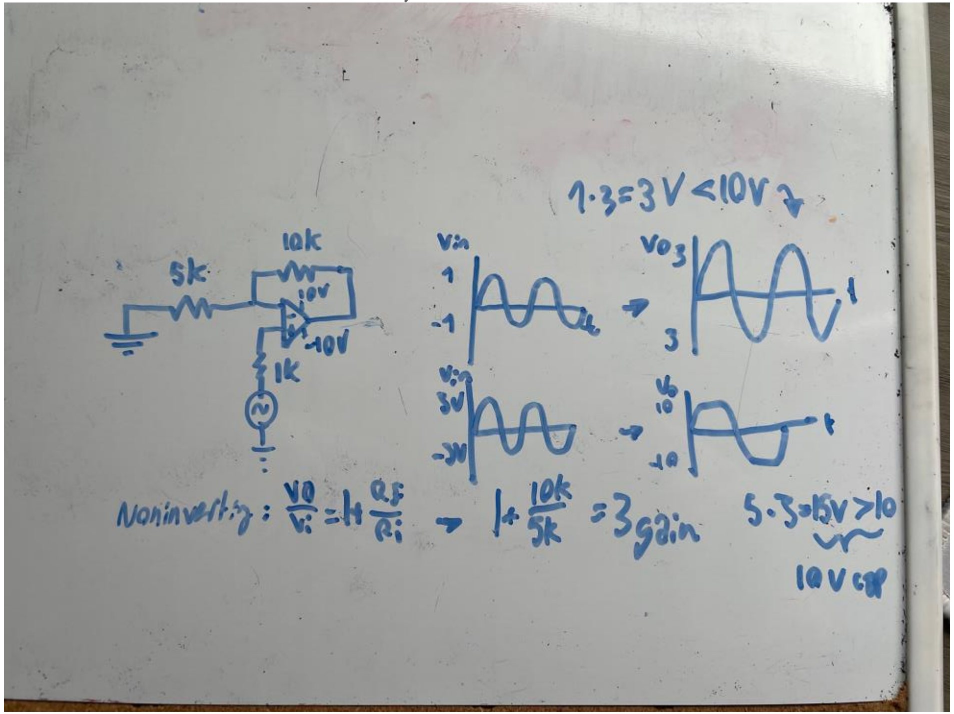
Voltage Saturation

These don't need to be the same either, each end determines its individual limit

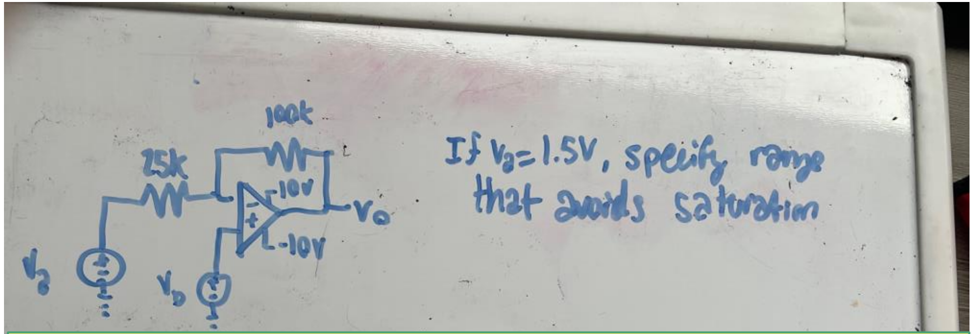


Voltage Saturation

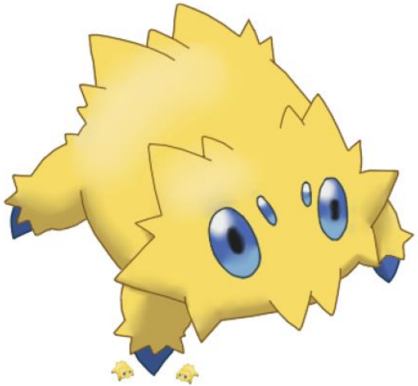
These don’t need to be the same either, each end determines its individual limit

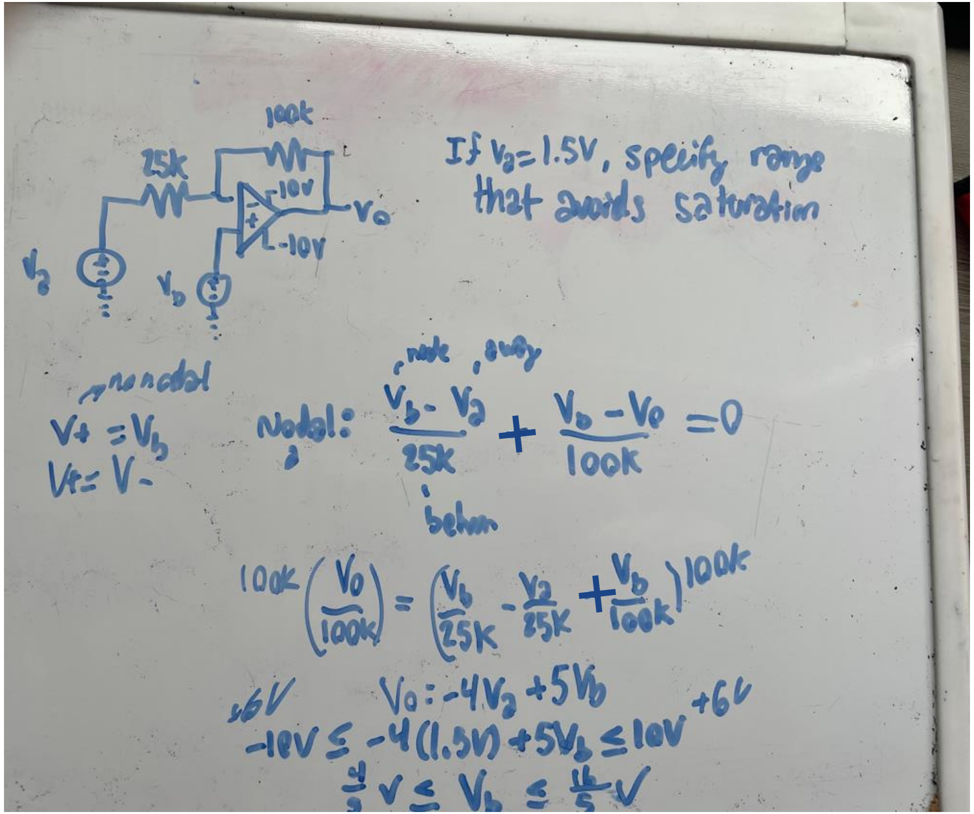


Ranges that avoid saturation

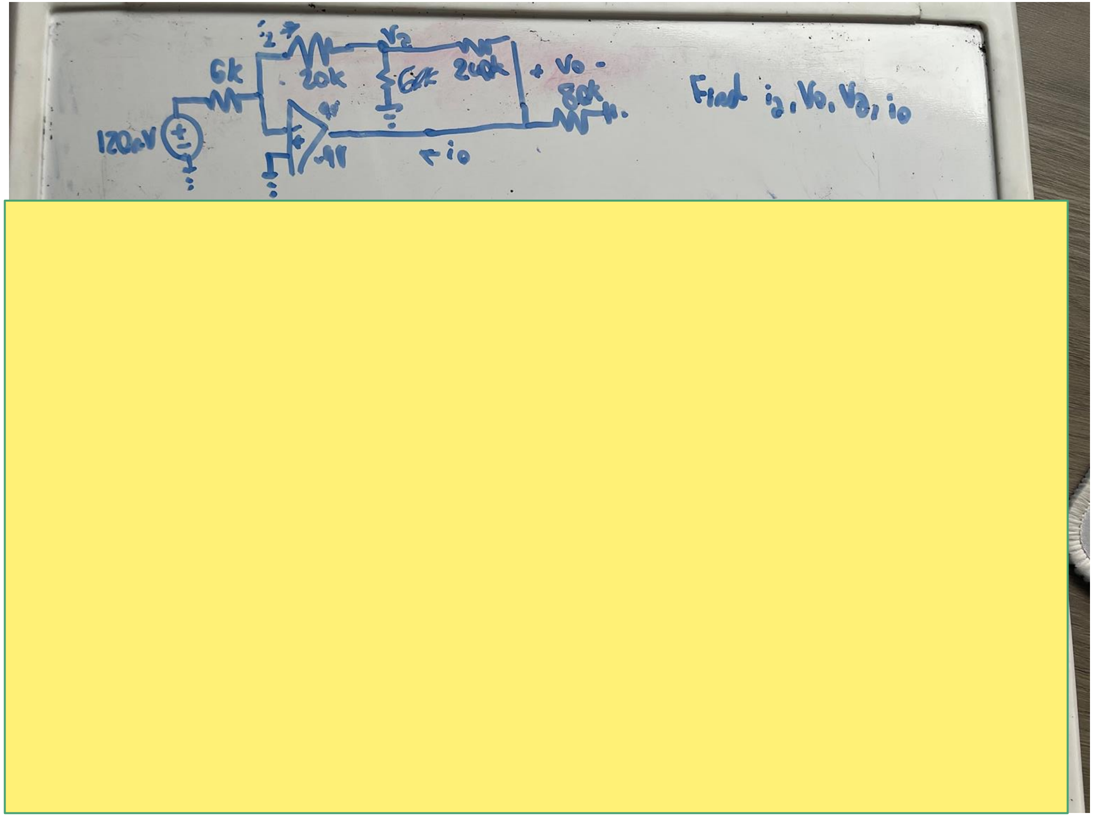


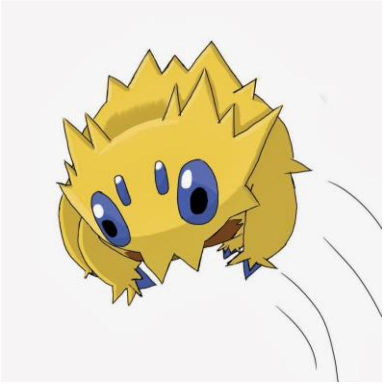
Ranges that avoid saturation



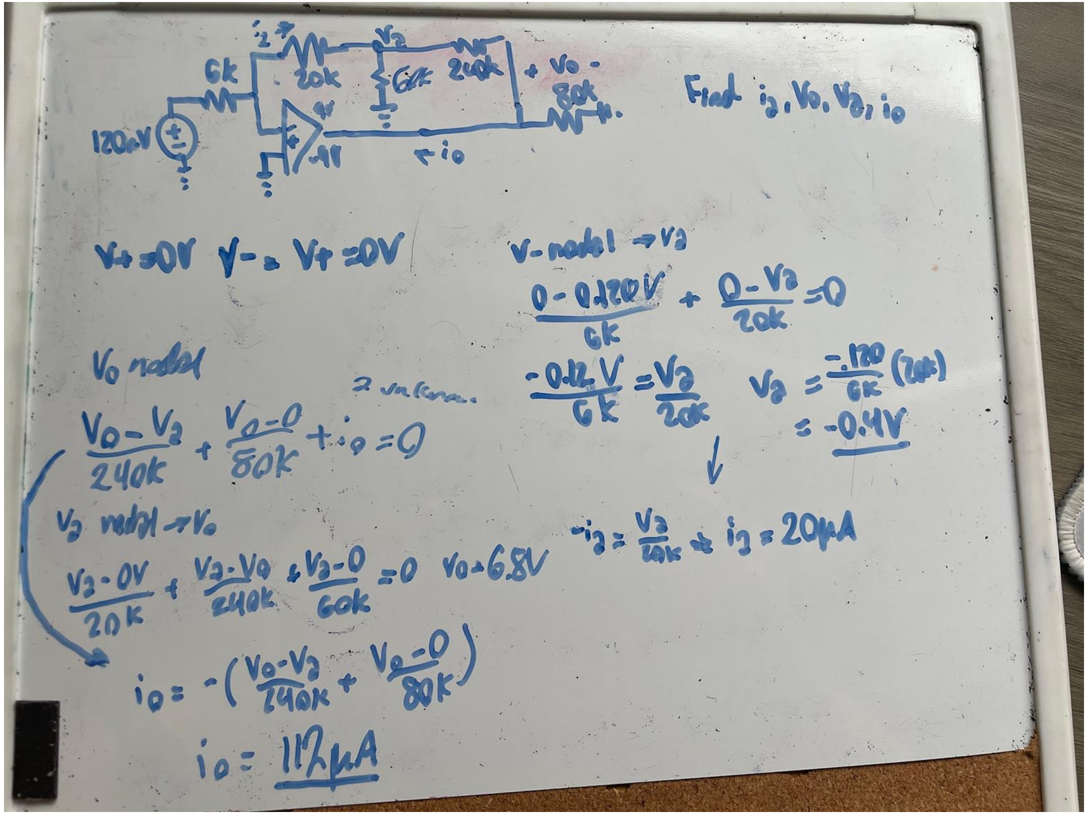


Fully Evaluated Operational Amplifier Circuit

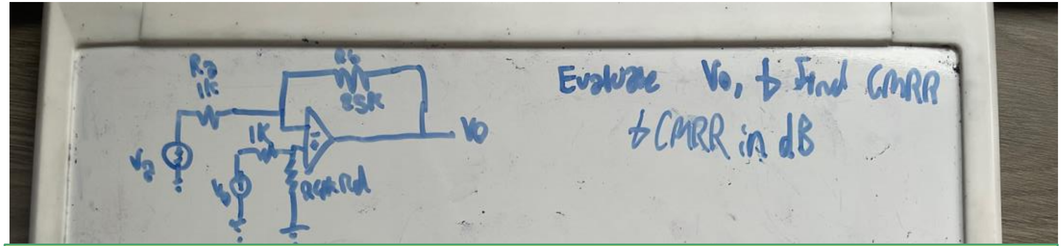




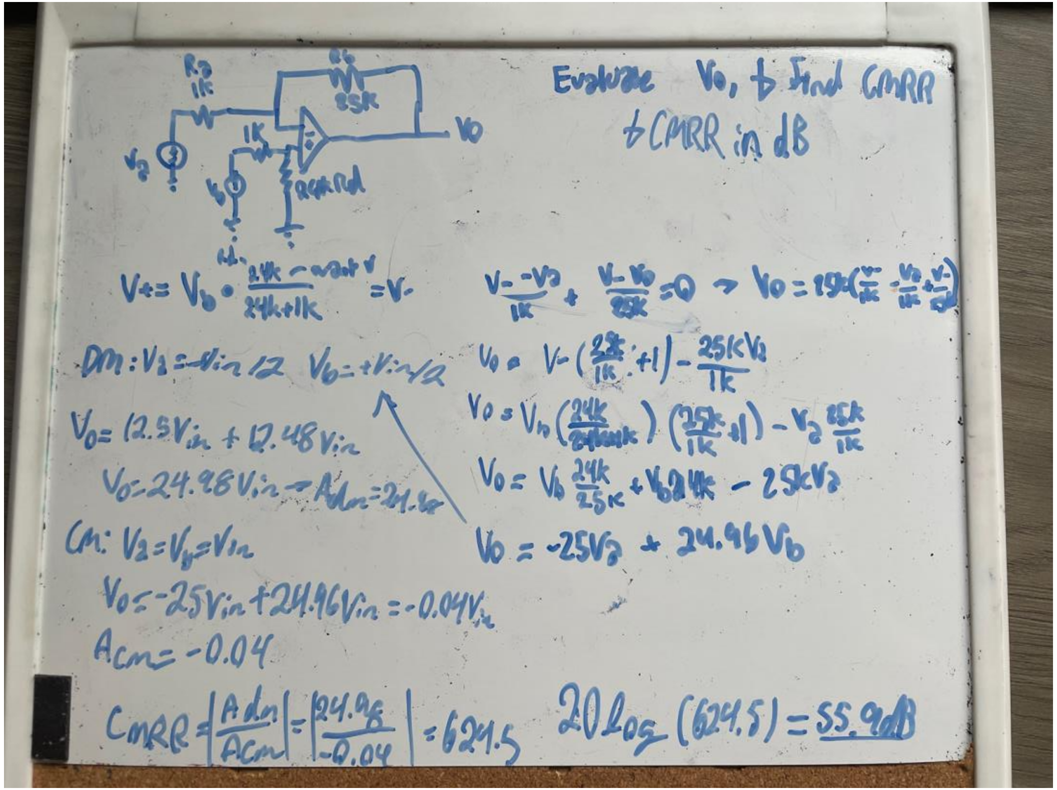
Fully Evaluated Operational Amplifier Circuit



Common Mode Rejection Ratio

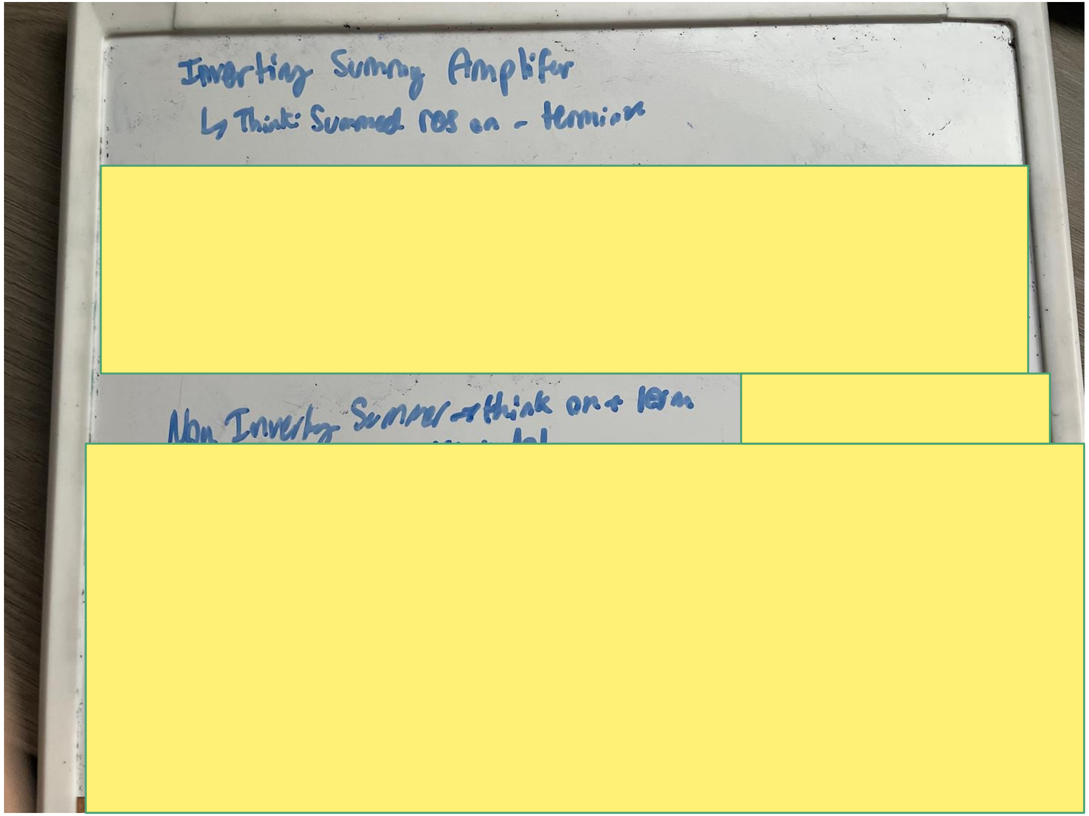


Common Mode Rejection Ratio



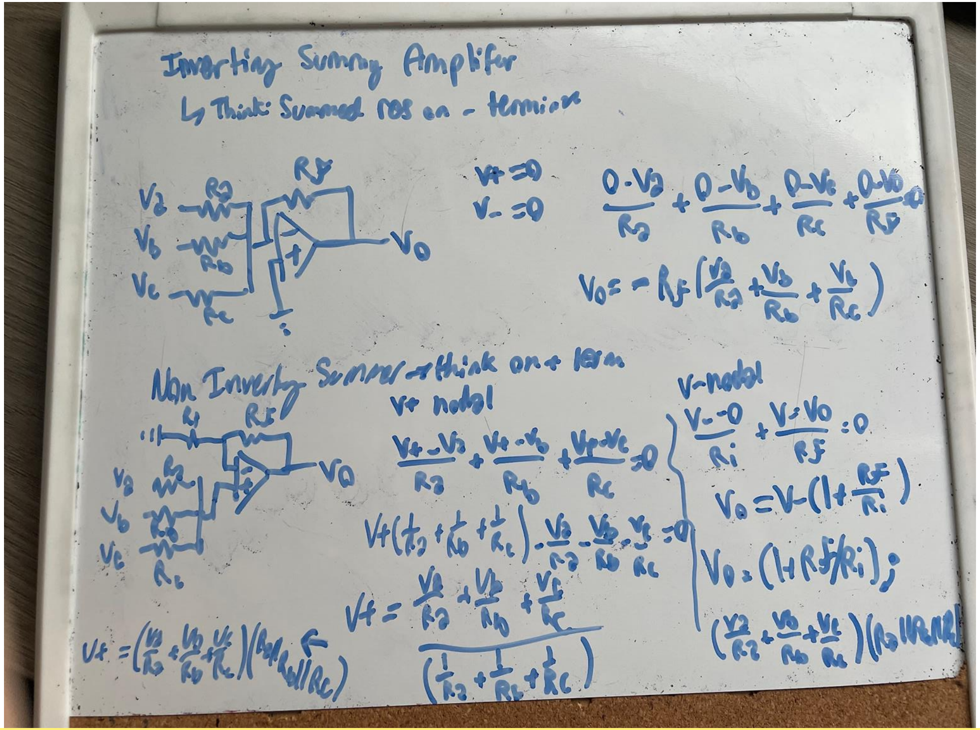
595

Inverting and Non-Inverting Summing Amplifier

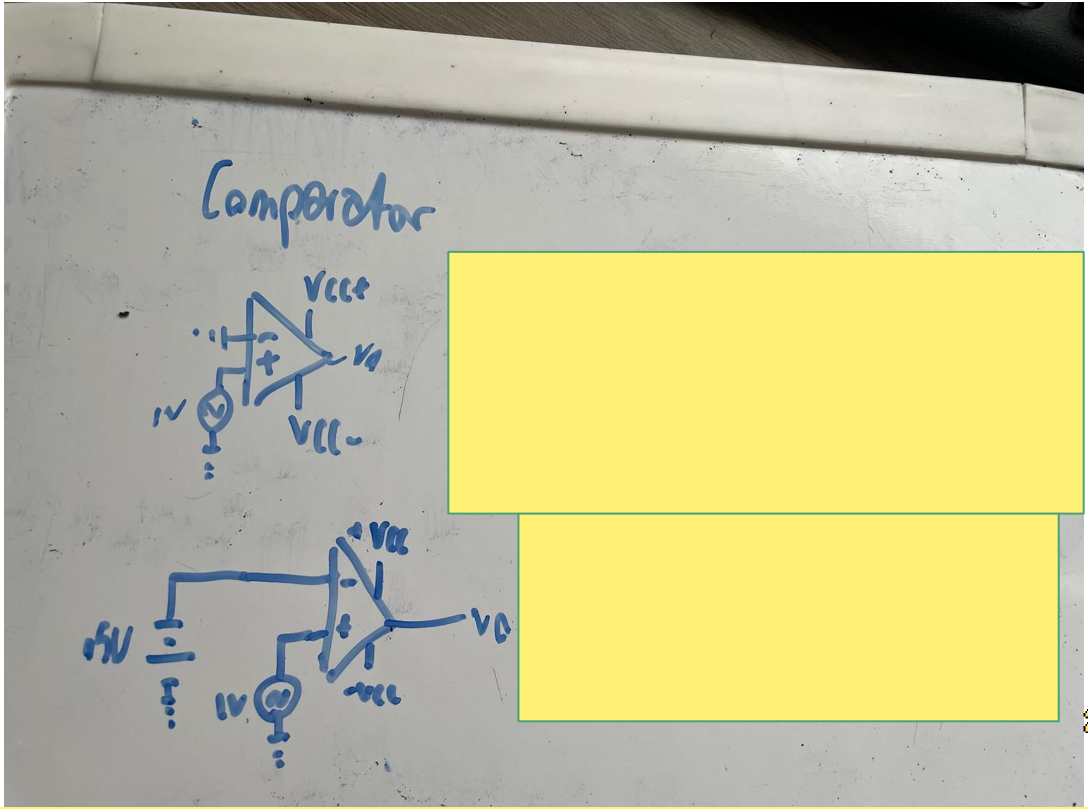


Inverting and Non-Inverting Summing Amplifier

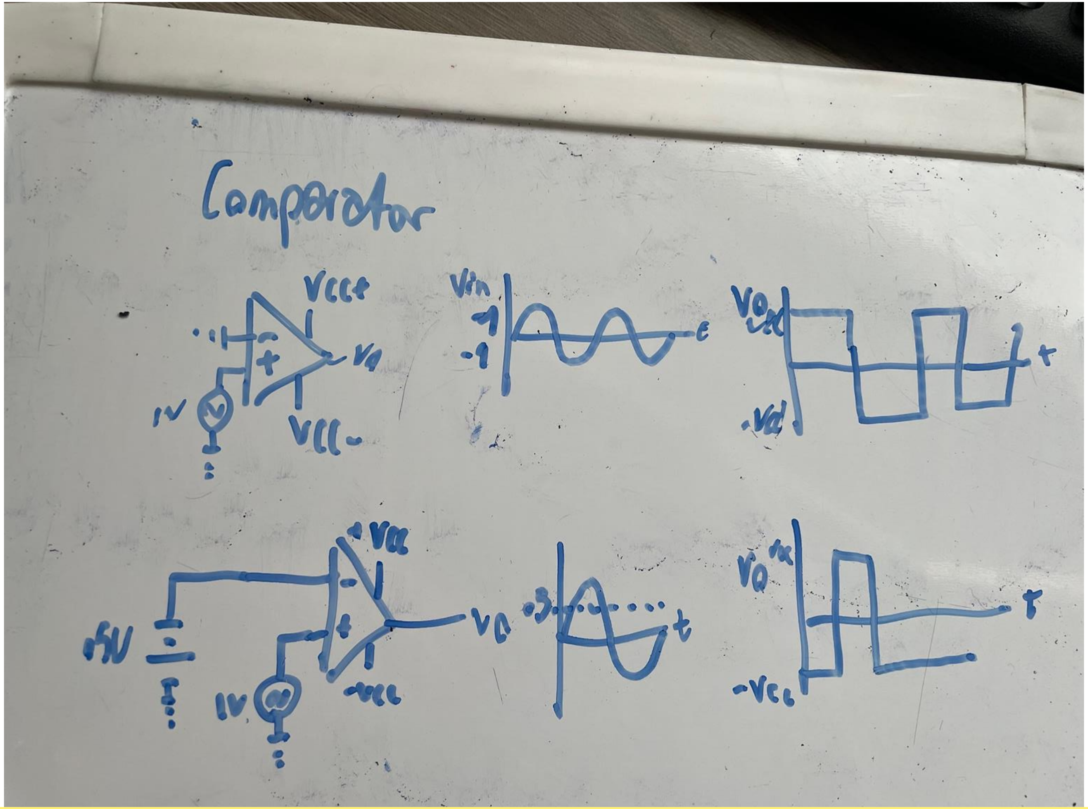
Worth Noting: Superposition allows eval by setting all but $V_a = 0$ and repeat



Comparators

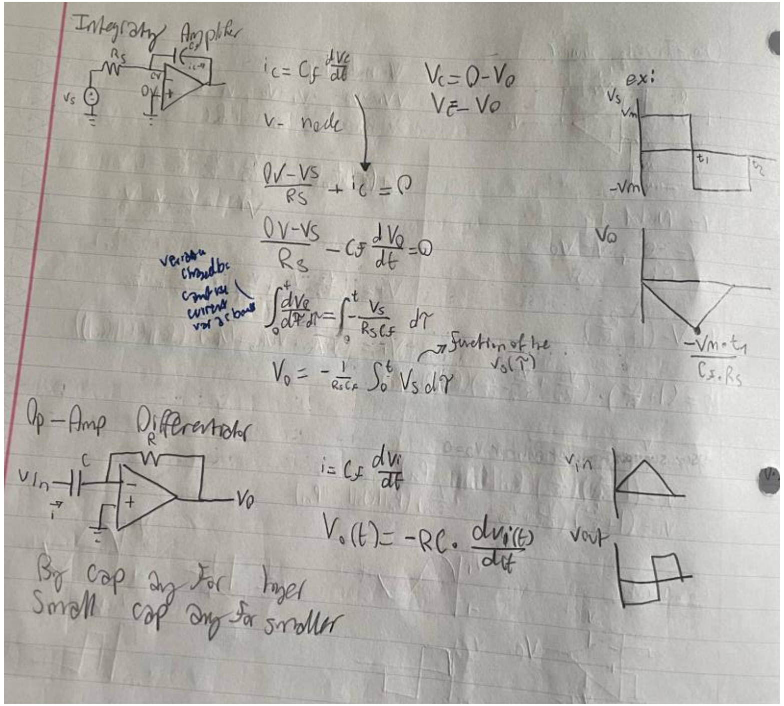


Comparators



Integrating and Differentiating Amplifier

- We've never worked out one so thoughts
- Integrating Amplifier
  - Uses $I_c = C_f * dV_c/dt$
  - Integrating has cap above op amp
    - Think I has the bar above
  - Integrating turns square to triangle
    - Think direction it will drop or rise as tri dir
- Differentiating Amplifier
  - Uses $i = C_f * dV_i/dt$
  - Differentiator has cap before, know by compare
  - Goes from triangle to square
    - Think where it will be abt to go



Dumb Nomenclature :(

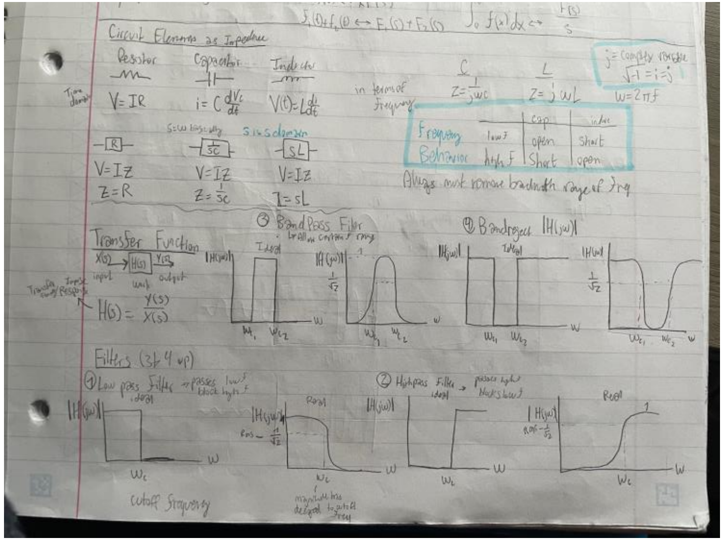
- Integrating draw a cap I, starts at cap (block) goes to tri
- Diff draw a lowercase d, start at tri go to block

LaPlace Transformations HEAVILY Simplified

Laplace Transforms are fancy function conversions that simply the differential equations and allow for easy usage of capacitors and inductors in algebraic form

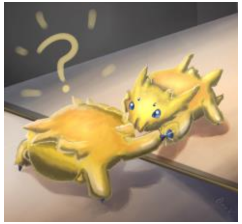
Changes from F(s) (laplace domain) to F(t) (time domain) or freq domain if s=jw

- Allows for active filters
- They follow these formations
- IMPORTANT NOTE:
  - Going forward know filters super well
  - Everything can be simplified as
    - Passive Filter
    - Simple 3 Op Amp Circuits

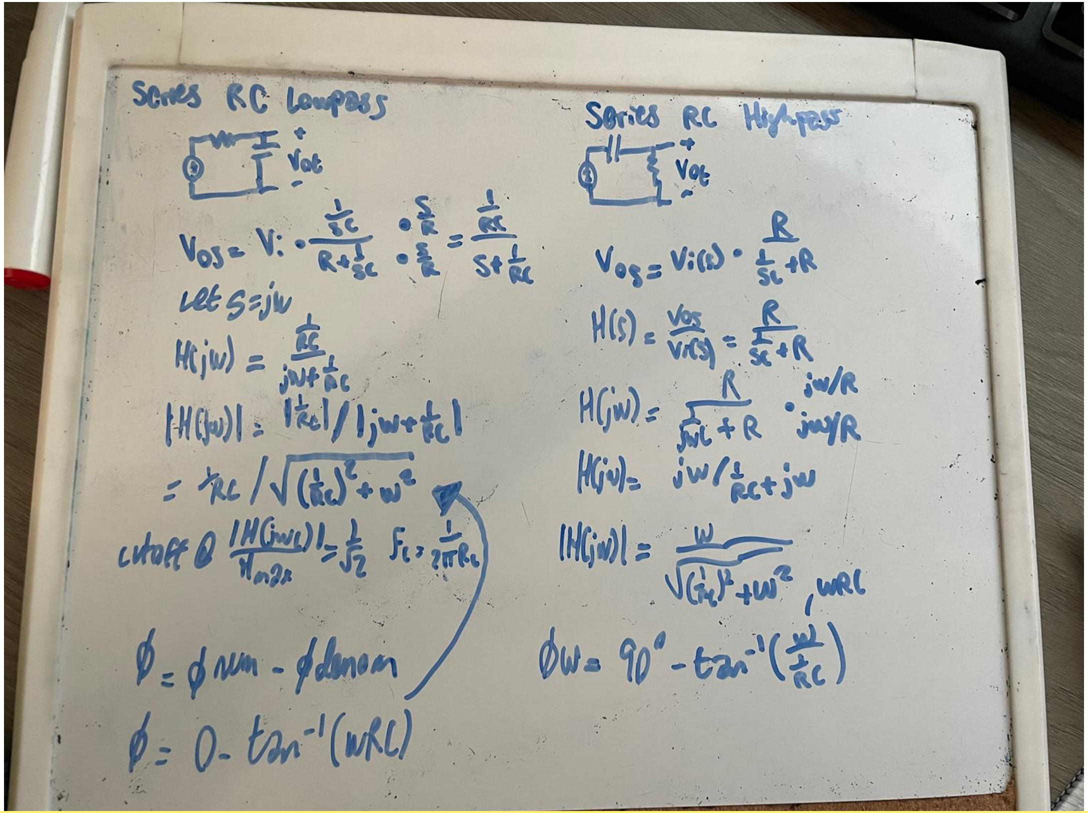


Passive RC Filter Circuits

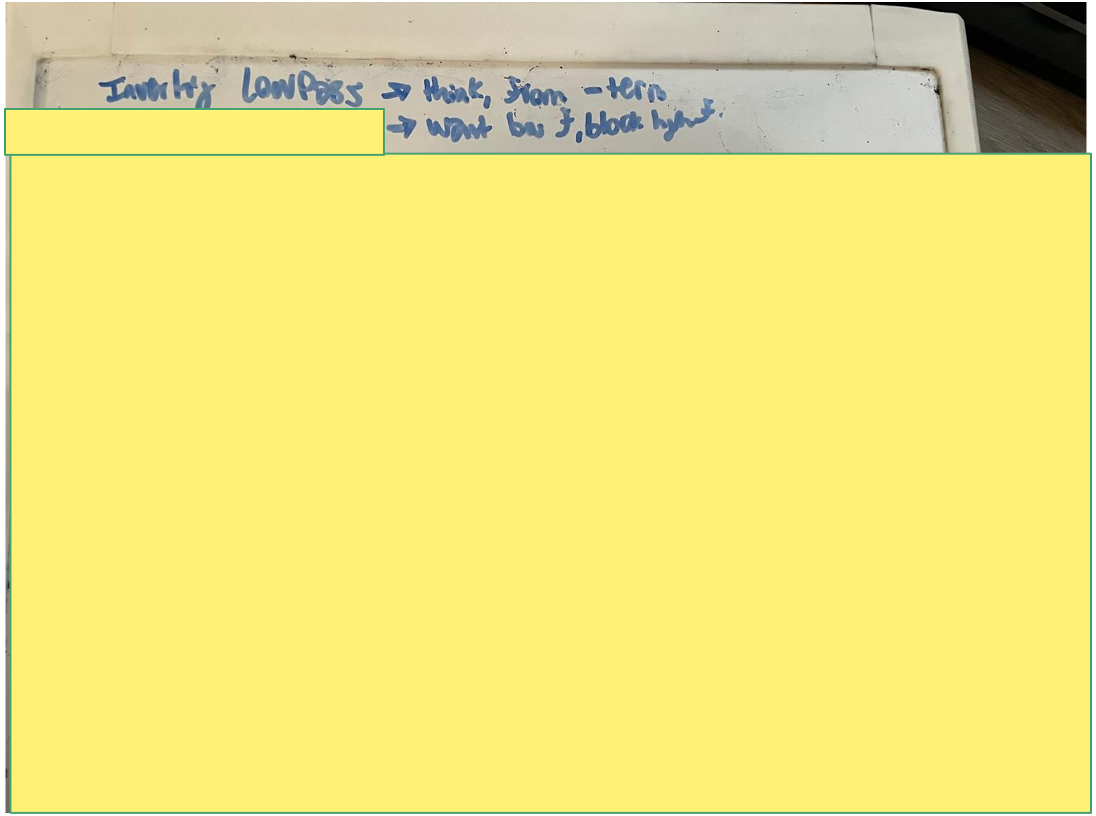
Passive RC Filter Circuits



- Each are used to allow f
- LP:
  - Allow low f (short)
  - Open at high f
- HP:
  - Allow high f (short)
  - Open at low f
- Use caps in spots to make
- Certain passes (will show)
- ALSO! Intro to transforms



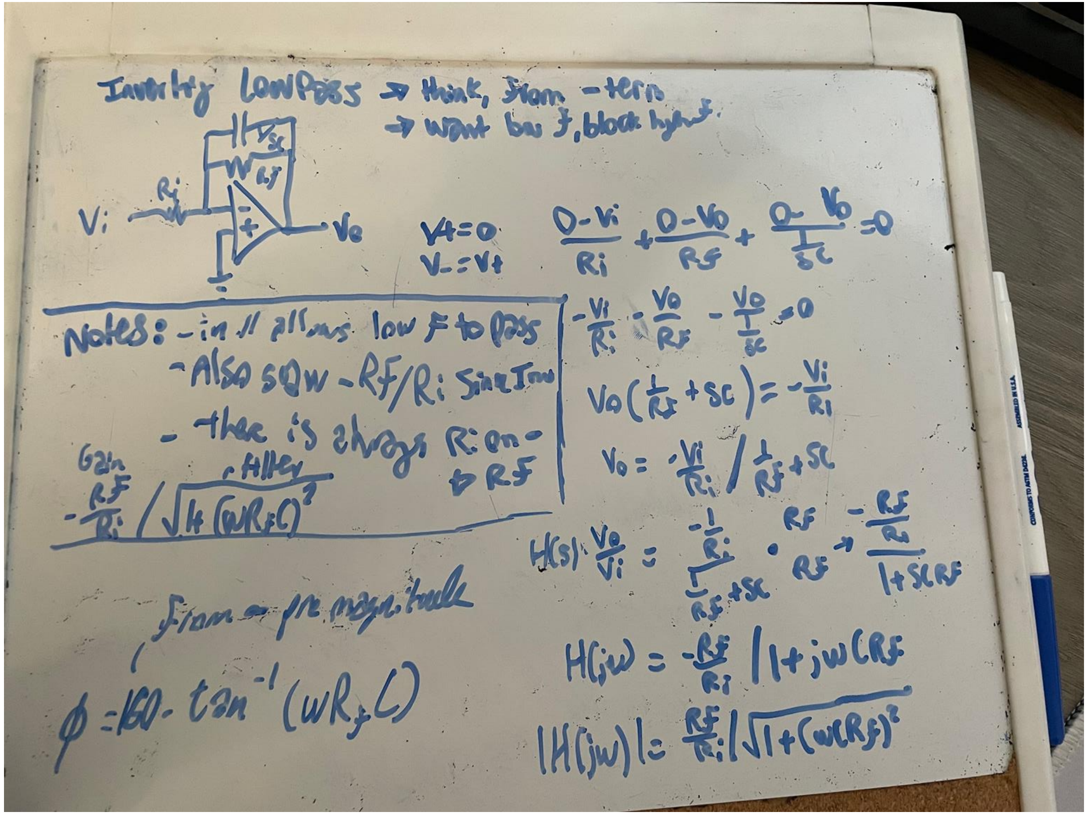
Inverting Low Pass



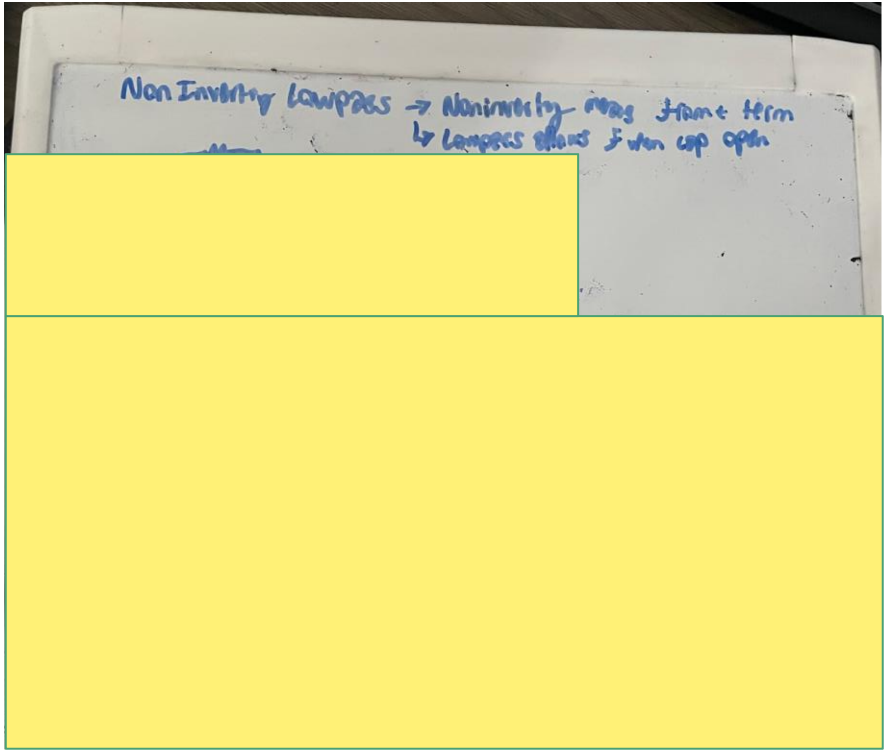
Inverting Low Pass

She knows

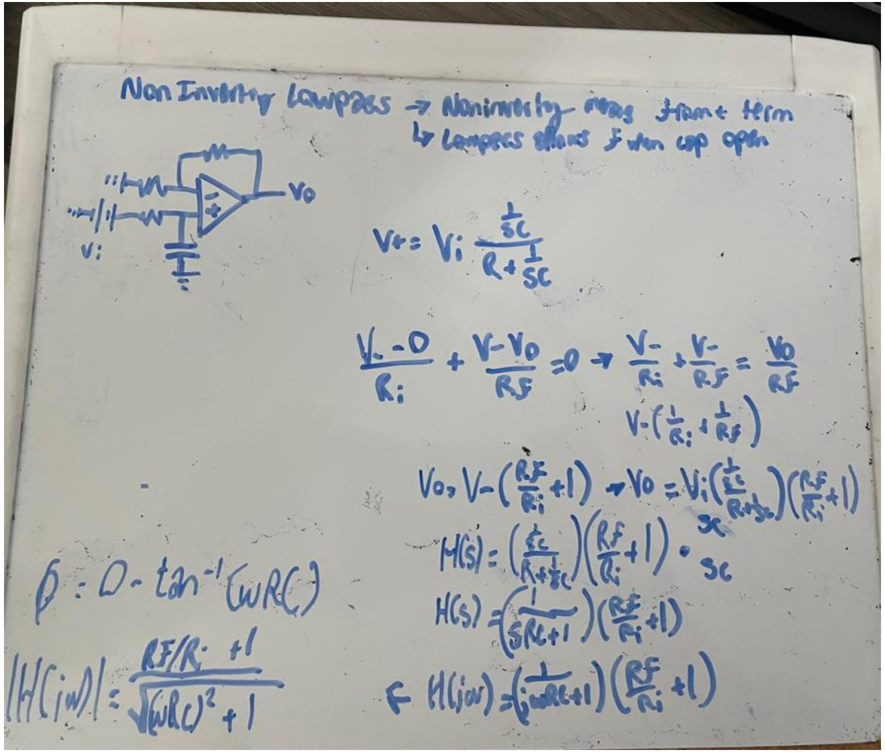
whats up ->



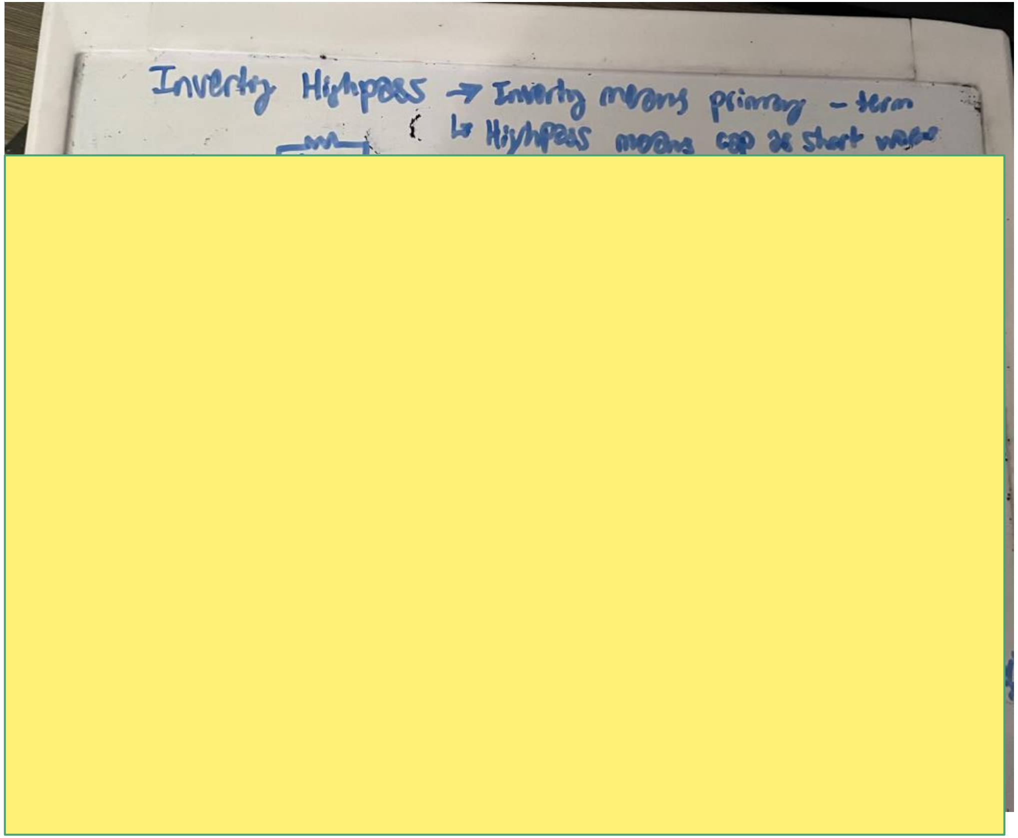
Non-Inverting Low Pass



Non-Inverting Low Pass



Inverting High Pass



She’s so proud of you !

Inverting High Pass

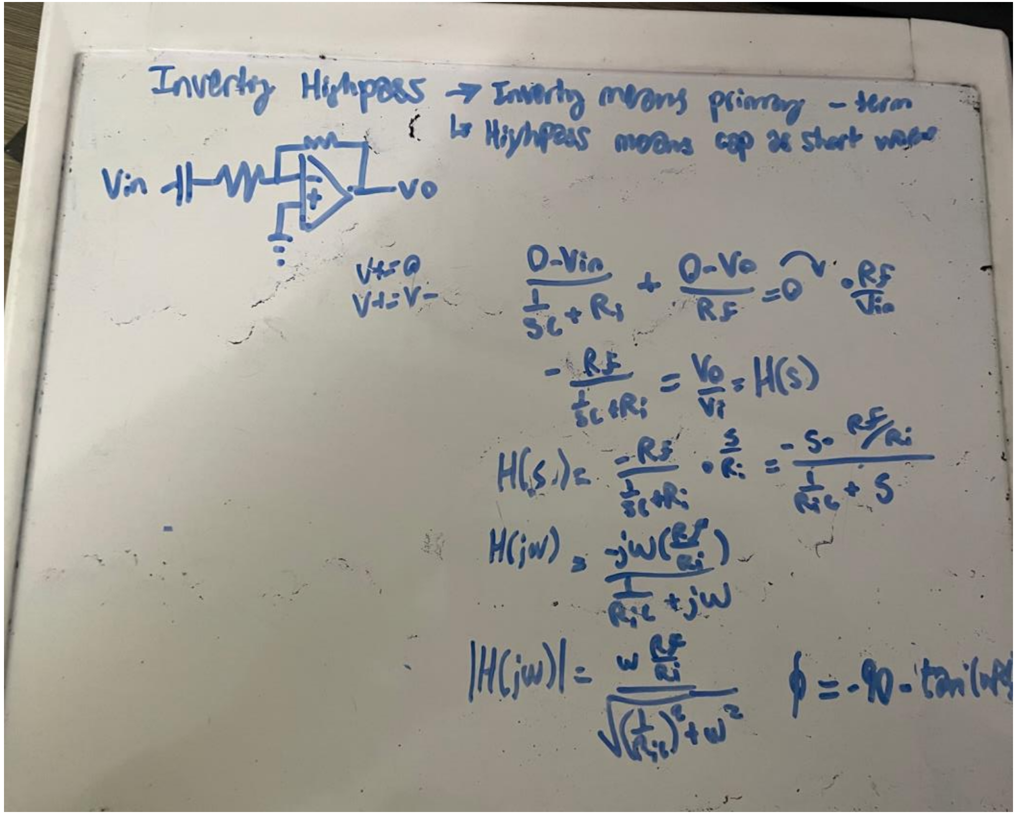
-Use multiplying term

To make fit what

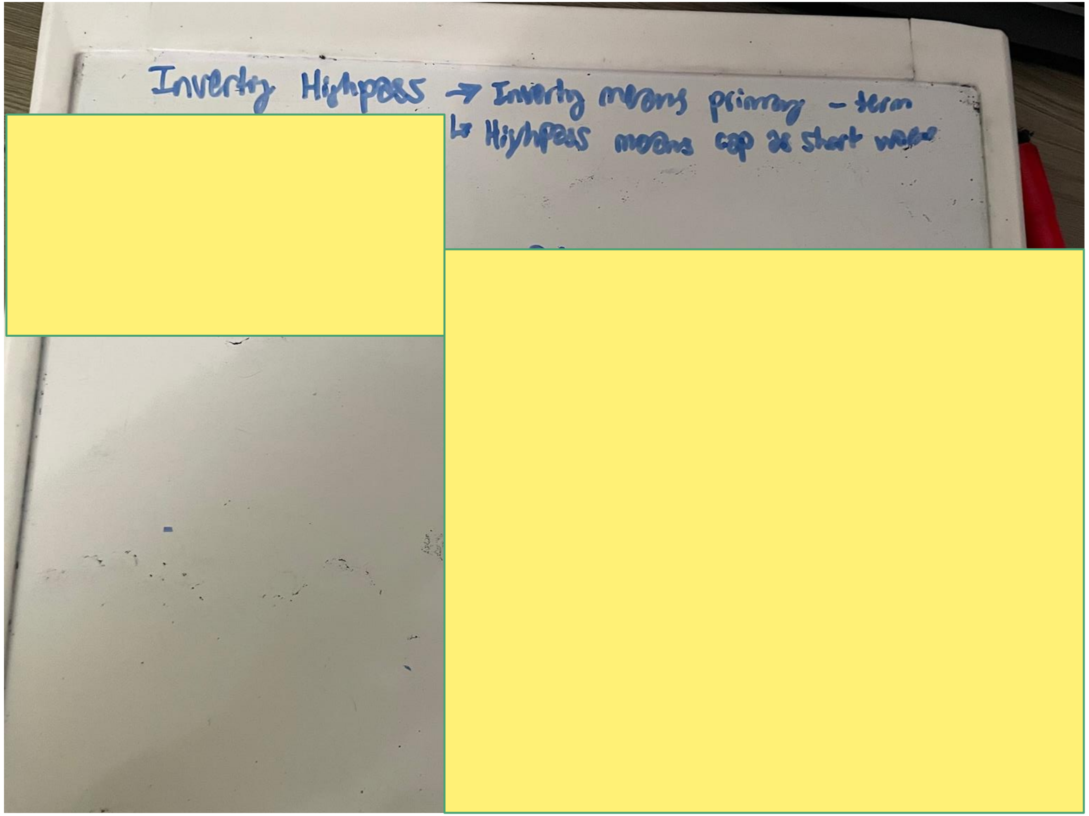
The theorhetical/

Easy gain should be

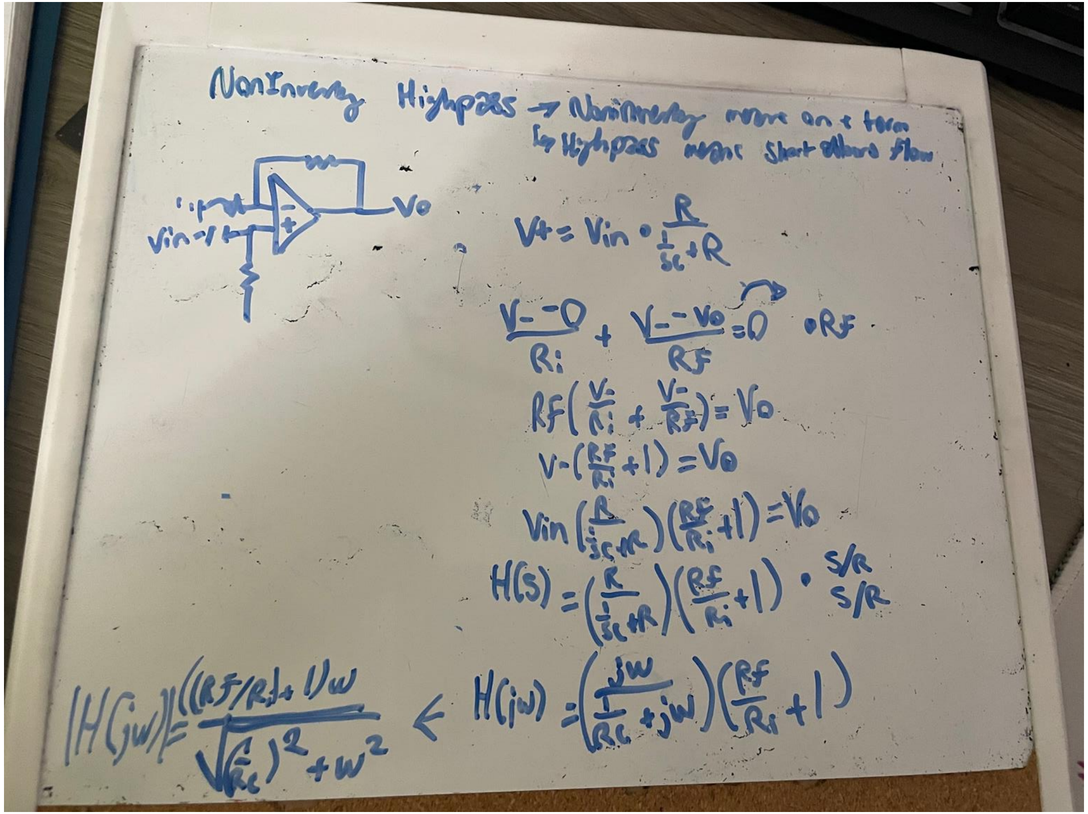
Ex: Made the -Rf/Ri for top



Non-Inverting High Pass

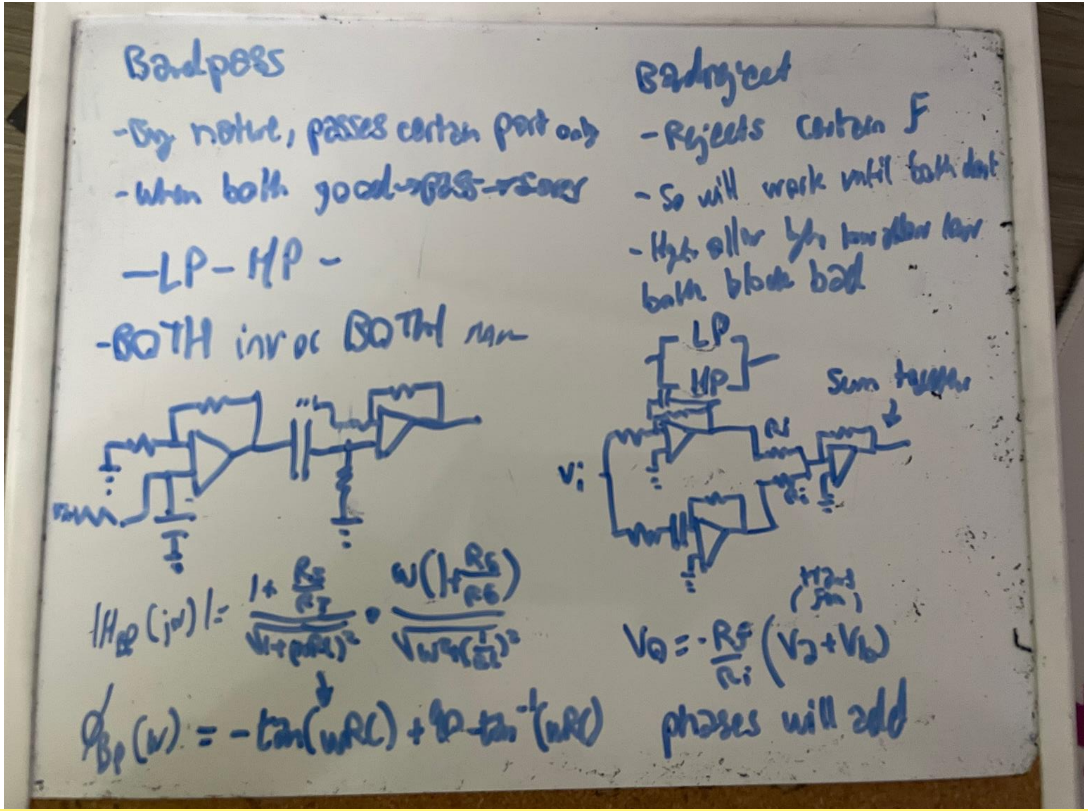


Non-Inverting High Pass



Bandpass + Band Reject

Bandpass + Band Reject



Sallen-Key Topology

Higher Order Filters allow for sharper cutoffs or more rejection of the frequency range

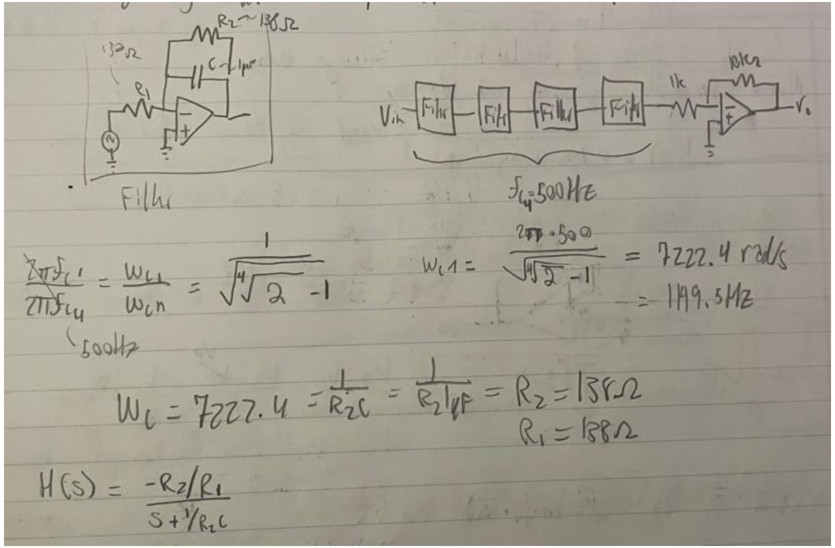
Bode Plots indicate different orders, higher orders have steeper cutoffs

20log(0.707) = -3dB@ cutoff as .707 is where the cutoff is

When filters are cascaded: w_{c1}/w_{cn}: 1/sqrt(nthroot(2)-1) lowpass w_{c1}/w_{cn}: sqrt(nthroot(2)-1) highpass

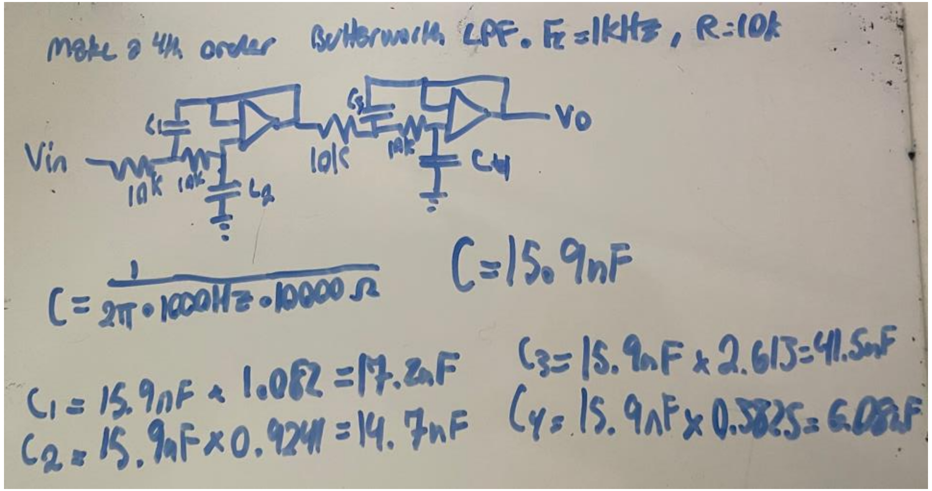
Higher Order Filter Example

Design a 4th order low pass filter with a cutoff frequency of 500 Hz and a passband resistor gain of 10. Use 4 identical unit gain active filter stages followed by gain stage. Use 1 microfarad capacitor.



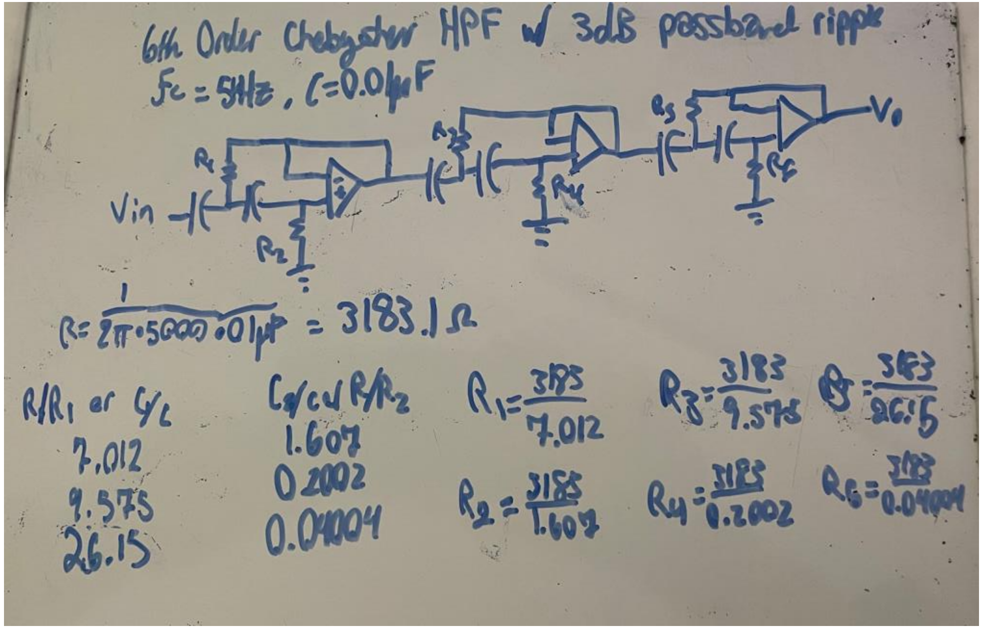
Butterworth Filter: Example and Basis (think smooth like butter (cringe is fine if = an A))

- Designed to have frequency response as flat as possible
- Steps:
  - Determine whether C values or R values need to be determined
  - Then, use the variables given in order to determine the values to scale the tables by
  - Then multiply the values by each and plug them back in



Chebyshev Filter: Example and Basis

- Designed to have ripple and a more steep slope



Passband Ripple & Gain Bandwidth Product

Why use active filters: No gain with passive filters, load impacts filters typically but the op amp protect this

Gain Bandwidth Product = gain x bandwidth (its a constant for an amplifier)

Passband ripple is involved with Bode plots, anyways have a Joltik

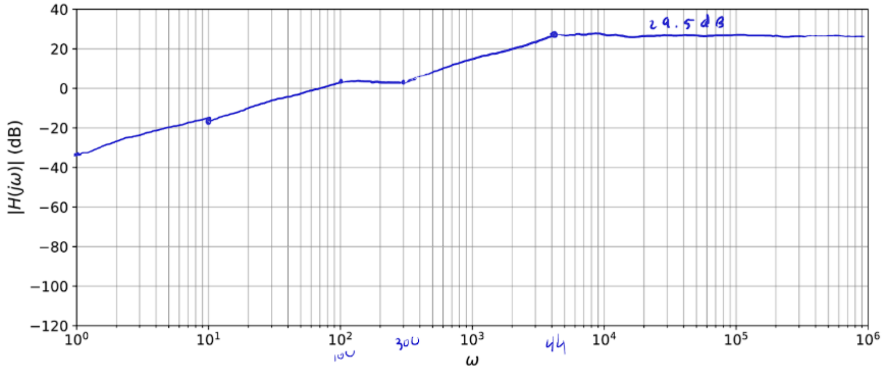
Bode Plot

Steps:

- Pull out the values:
  - See what values for numerator and denominator would make the non-s term as 0. Deal with the consequences
- Identify zeros and poles
  - Z: Up in slope, when num = 0
  - Z: $S^1 = 20$, $s^2 = 40$, $s^3 = 60$
  - P: Down in slope, dem = 0
- Determine the base value
  - 20log(factored value)
  - This is where you start, horizontal line all the way across
- Add slopes of each segment
  - Starting at base value, add the slopes of each portion
- Determine what type of filter
  - Highpass ends high
  - Lowpass ends low
  - Bandpass is a hump
  - Bandstop is a valley

4. $H(s) = \dfrac{30s(s+300)}{(s+100)(s+4000)} = \dfrac{9}{400} \dfrac{s\left(\frac{s}{300}+1\right)}{\left(\frac{s}{100}+1\right)\left(\frac{s}{4000}+1\right)}$

$20\log\left(\frac{9}{400}\right) = -32.96\,dB$



b) high pass

c) $w_c = 100, 300, 4000$

d) $s \to \infty$ $\quad H(s) \to 30$

$20\log(30) = 29.5\,dB$

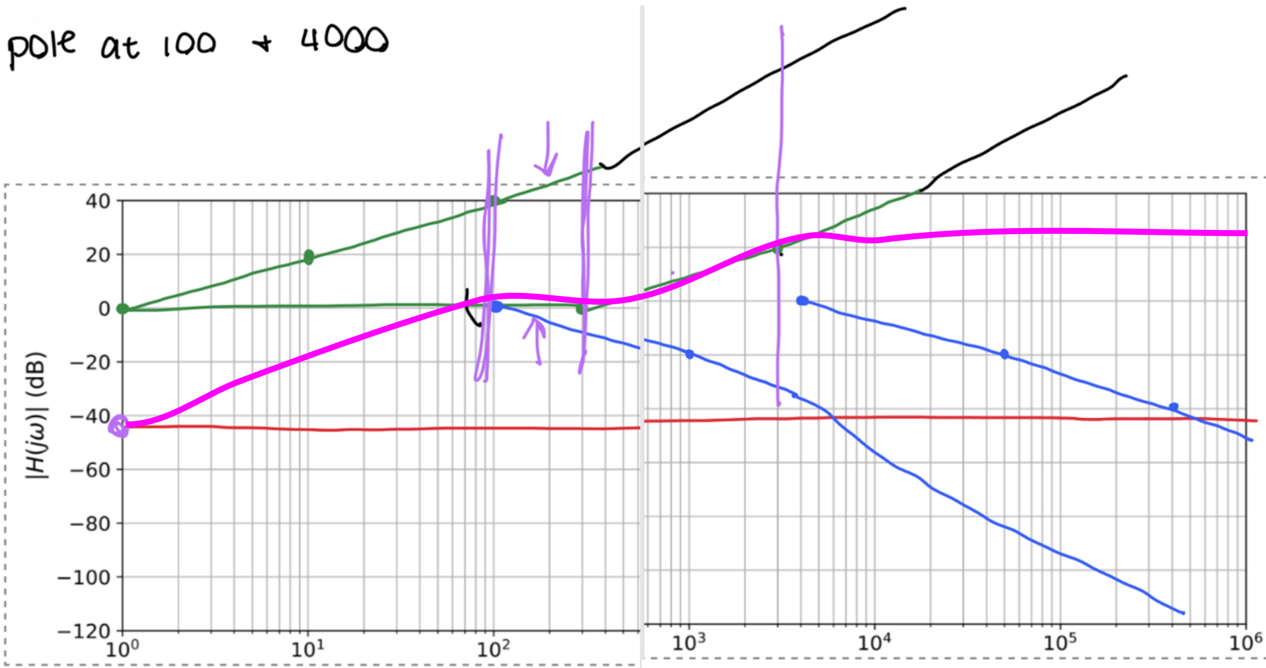
Examples of Previous steps

- 30s: 0 would make 0, so its a zero
- S+300: 300 would make 0, so its a zero
- S+100: 100 would make 0, so its a pole
- S + 4000: 4000 would make 0 so its a pole

4) $H(s) = \frac{30s(s+300)}{(s+100)(s+4000)}$

zero at 0 + 300 Hz

pole at 100 + 4000



Transistors - Spidermancore bc getting back up when you’re down

Transistors
Bipolar Junction Transistors (BJTs)

- npn and pnp transistors
- Base
- Collector
- Emitter
- Currents
- Current controlled current source

Biasing

- Forward-active
- Saturation
- Cutoff
- Reverse-active

DC Beta βDC
Collector Characteristic Curves
DC Load Line
Voltage amplification (e.g., common emitter amplifier)
BJT as a switch
DC operating point or quiescent point (Q-point)
Voltage divider bias

- Thevenin’s theorem

Linear operation

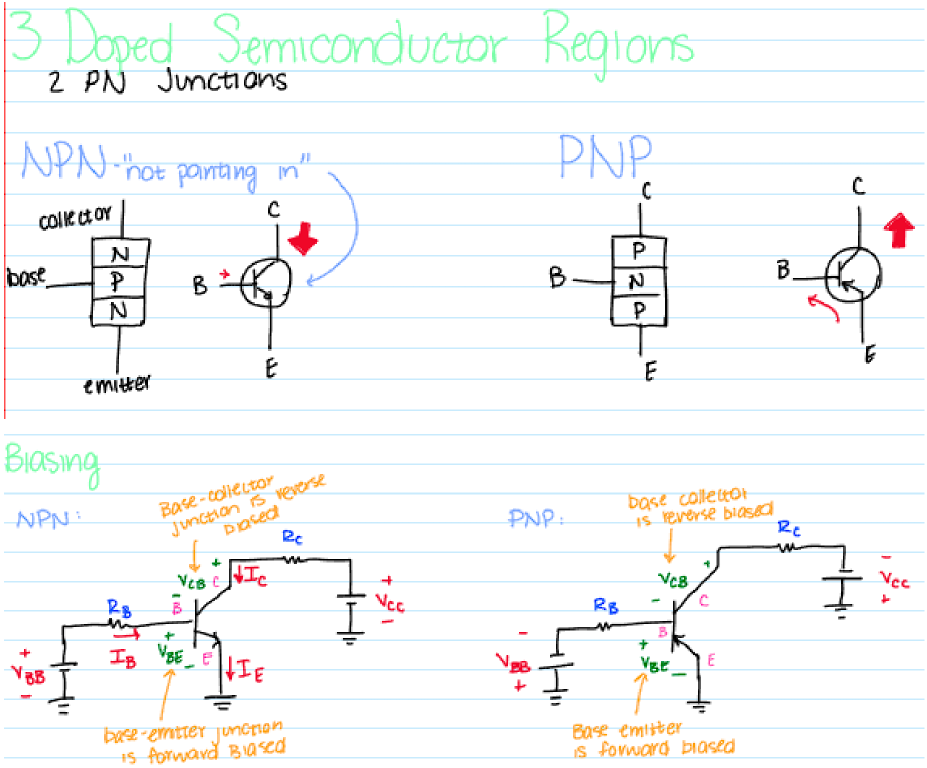
- Waveform distortion (saturation, cutoff)
- Coupling capacitors

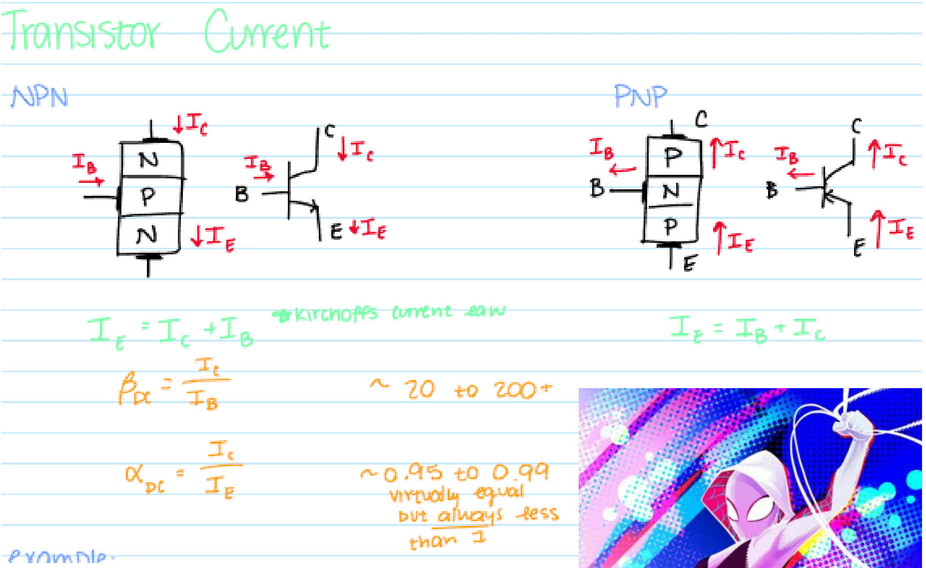
Field Effect Transistor (see slides)

- Voltage controlled current source
- High input resistance
- Non-linear
- JFET – p-channel and n-channel
    - Vp, VGS(off) and IDSS
    - ID as a function of VGS
    - Universal Transfer Characteristic
- MOSFET
    - E-MOSFET (no structural conduction channel)
    - D-MOSFET (has a channel)
    - Basic MOSFET switching circuit (refer to last lab)
    - Basic idea behind CMOS (logic)
- FET transistor switch
- Amplifier Classes (A, B, AB, C, D)

I kind of don’t care that this is a little lame, I feel like this and I just don’t wanna BUT that’s when it’s important to do

Basics about Transistors





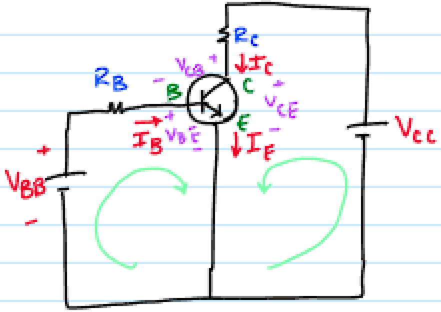
Takeaways:

- Always have a takeaway textbox
- The beta and alpha eq is very important
- The biasing works very similar but this is known at this point
- IMPORTANT: Transistors are voltage controlled devices that permit current flow

Example of Transistor Analysis

[N]

Analysis:



I_B - DC base current
I_C - collector current
I_E - DC emitter current

$V_{BE} = V_B - V_E$ voltage at base w/respect to emitter

$V_{CB} = V_C - V_B$ voltage at collector w/respect to base

$V_{CE} = V_C - V_E$ voltage at collector w/respect to emitter

base bias (V_{BB})
↳ forward biases base-emitter junction
collector bias (V_{CC})
↳ reverse biases base-collector junction

★ rule: when the base emitter junction is forward biased, $V_{BE} \approx 0.7V$

KVL:

$V_{BB} - I_B \cdot R_B - V_{BE} = 0$

$I_B = \frac{V_{BB} - V_{BE}}{R_B}$

$V_{CC} - I_C R_C - V_{CE} = 0$

$V_{CE} = V_{CC} - I_C R_C$

$I_C = \beta_{DC} I_B$

$V_{CB} = V_{CE} - V_{BE}$

Takeaways:

- Voltages are always determined this way
- Forward bias until later noted always has V_{BE} as 0.7V, this is a rule
- These types of problems are standard atp

Example 4-2/The Most Basic Problem Possible:

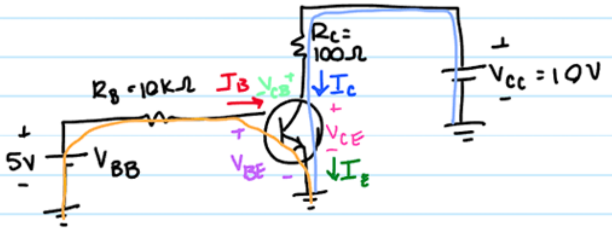
Example 4-2:

base emitter junction is forward biased, ∴ $V_{BE} = 0.7V$

↳ b/c positive 5V

determine I_B I_C I_E V_{BE} V_{CE} V_{CB}

$\beta_{DC} = 150$



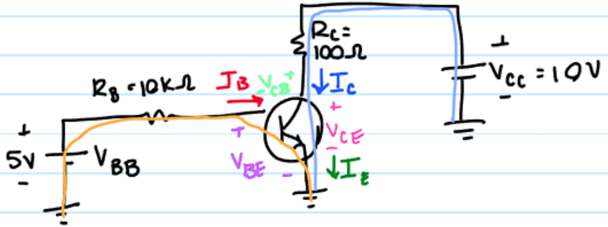
Example 4-2/The Most Basic Problem Possible:

Example 4-2:

base emitter junction is forward biased, ∴ $V_{BE} = 0.7V$
b/c positive 5V

determine I_B I_C I_E V_{BE} V_{CE} V_{CB}

$\beta_{DC} = 150$



KVL Base-emitter

$5V - I_B \cdot 10K - V_{BE} = 0$

$5V - I_B \cdot 10K - 0.7V = 0$

$I_B = \frac{5-0.7}{10K} = 0.43\ mA$

$V_{BE} = 0.7V$

$6.9 - I_B \cdot 11K - 0.7 = 0$

$I_B = 0.56\ mA$

$I_C = I_B \cdot 87$

$\frac{I_C}{I_B} = \beta_{DC}$ ∴ $I_C = 150 \cdot 0.43mA$

$I_C = 64.5\ mA$

KVL - Collector Emitter:

$10V - I_C \cdot 100\Omega - V_{CE} = 0$

$10 - 64.5mA \cdot 100 = V_{CE}$

$V_{CE} = 3.55V$

$I_E = I_C + I_B = 64.93\ mA$

$V_{CB} = V_C - V_B$

$V_{BE} = V_B - V_E$

$V_{CE} = V_C - V_E$

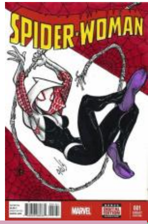
$\boxed{V_{CB} = V_{CE} - V_{BE}}$ ★

$V_{CB} = 3.55 - 0.7$

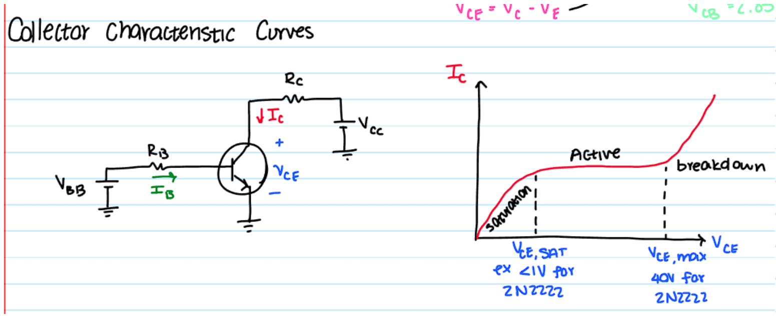
$V_{CB} = 2.85\ V$

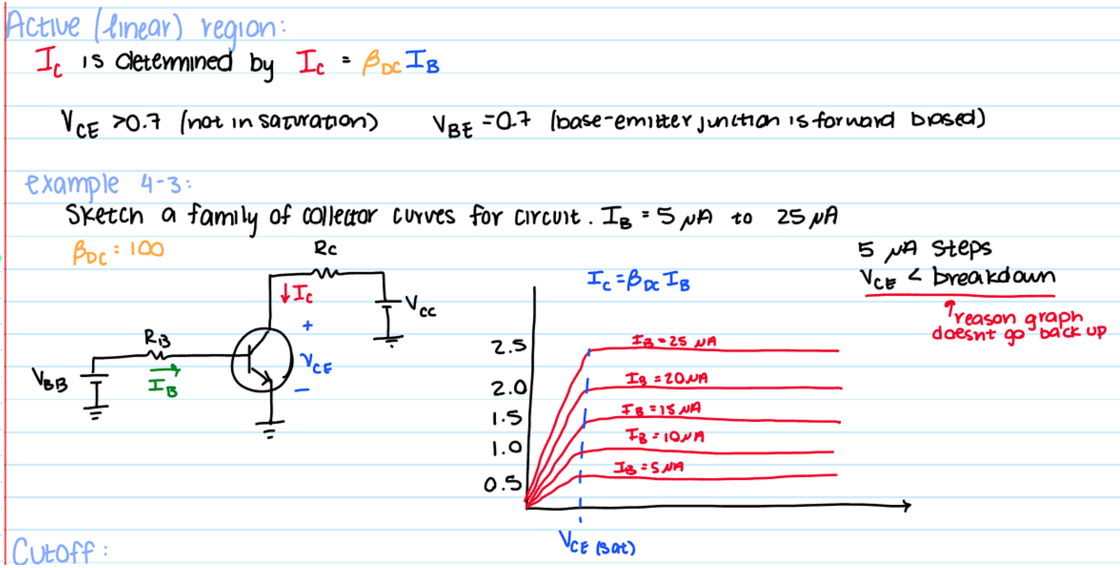
Takeaways:

- V_{CE} always accounts for 0.7V from V_{BE}
- In fact, V_{CB} is just V_{CE} - V_{BE} which is 0.7 V lower



Regions of Collector Current + Active

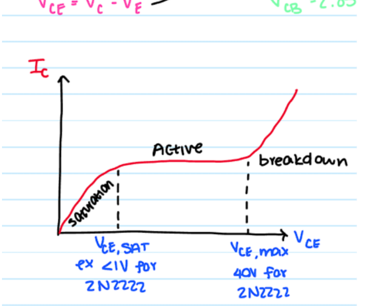


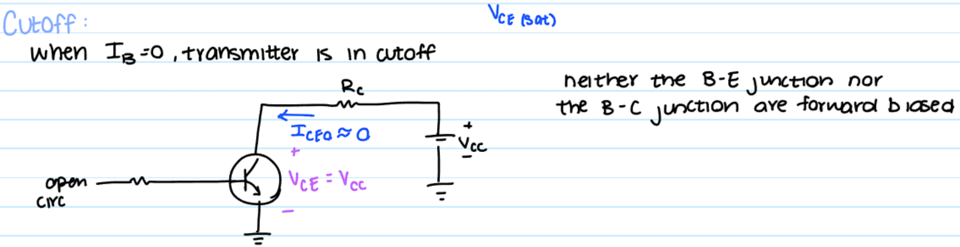


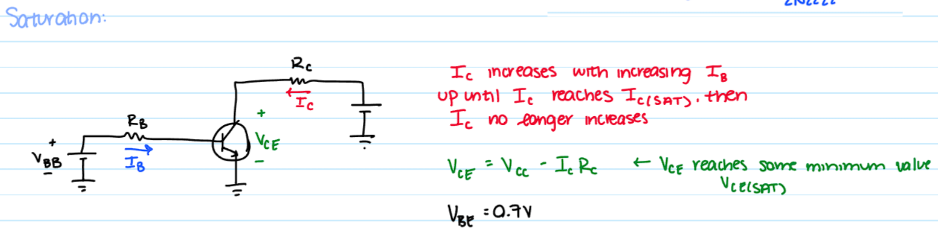
Takeaways:

- Most of the operations happen in the active region
- When V_{CE} > 0.7 V it is not in saturation, which makes sense because 0.7 V has to come from V_{BE}
- This is when B_{DC} applies
- Family of collector currents is covered in homework for what it applies to

Cutoff + Saturation of Collector Current







Cutoff: This is whenever the junctions are not forward biased. No current flow is possible.

V_{CE} will equal V_{CC} due to all of the voltage drop being across the transistor

Ignore the state of the chart for this one

Saturation: Saturation occurs prior to the active region. Think of it as the point that I_C can no longer increase as being bound by the circuit. It is bound by the line above.

V_{CE} can be considered V_{CC} - $I_C R_C$ as the voltage drop across is full dependent on the individual I_C value that is in an incomplete state

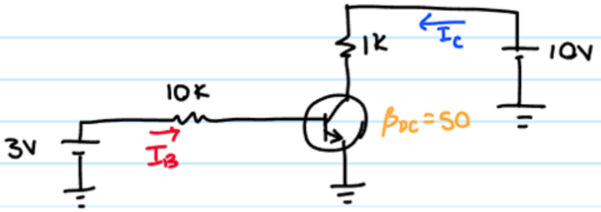
V_{BE} is 0.7V as I_C trying to flow more than it can is evidence of a near active state

Example of Saturation

Look for I_C vs $I_{C(sat)}$ and minimum I_B that causes saturation

Example 4-4:

determine if the following circuit is in saturation. Assume $V_{CE(SAT)} = 0.2V$



Example of Saturation

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example 4-4:

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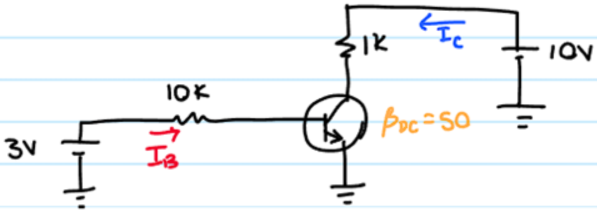


$I_B = \frac{3-0.7}{10K} = 0.23\ mA$

$I_C = \beta_{DC} \cdot I_B = 50 \cdot 0.23mA = 11.5\ mA$

*reality: if transistor replaced w/ ground then $I_c = 10\,mA$

I_c more than 10 mA ∴ saturated

Find $I_{C(SAT)}$

$10V - I_{C(SAT)} \cdot 1K - V_{CE(SAT)} = 0$

$\frac{10V - 0.2V}{1K} = I_{C(SAT)}$

I_c cannot exceed 9.8 mA

minimum I_B causing saturation

$I_{B(min)} = \frac{I_{C(SAT)}}{\beta_{DC}} = \frac{9.8mA}{50}$

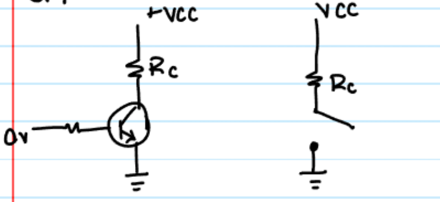
$\underbrace{I_{B(min)} = 0.196\ mA}_{\text{more current than this results in saturation}}$

BJT as a Switch Intro

BJT:

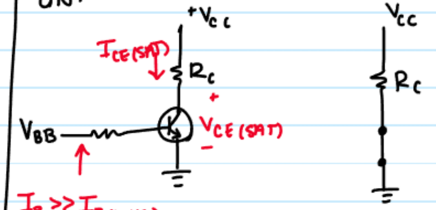
as a switch

OFF:



cutoff region, open switch

ON:



$I_B >> I_{B(min)}$
↳ double
triple
5x

Saturation region,
closed switch

In cutoff B-E junction, not forward biased
All currents are zero

In saturation B-E is forward biased
I_c is reached its maximum

$$I_{C(SAT)} = \frac{V_{CC} - V_{CE(SAT)}}{R_C}$$

$$I_{B(min)} = \frac{I_{C(SAT)}}{\beta_{DC}}$$

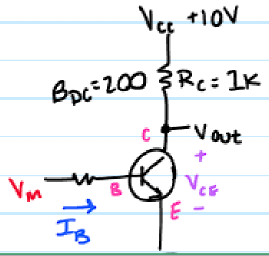
$I_B >> I_{B(min)}$ to ensure saturation

Fundamentally:

- Using a transistor as a switch when off doesn't allow current flow
- When On allows for current flow
- It's exactly what's been done so far but this is a more intentional version of it
- Cutoff = OFF
- Saturation + Active = ON
- EQUATION:
- In sat: $B_{DC} = I_{C(sat)}/I_{B(min)}$

BJT as Switch Example

example: 4-10



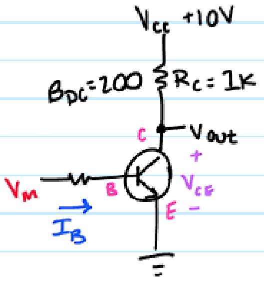
a) what is V_{CE} when $V_{in} = 0V$

b) what minimum value of I_B is needed to saturate this tranistor if $\beta_{DC} = 200$, and $V_{CE(SAT)} \approx 0V$.

c) Calculate maximum value of R_B that will put the transistor in saturation when $V_{in} = 5V$

BJT as Switch Example

example: 4-10



a) what is V_{CE} when $V_{in}=0V$

→ base-emitter junction is NOT forward biased

→ transistor is NOT on $\therefore I_c = 0$

↳ voltage drop through $R_c = 0$

$V_{CE} = V_{CC} = 10\ V$

b) what minimum value of I_B is needed to saturate this transistor if $\beta_{DC} = 200$, and $V_{CE(SAT)} \approx 0V$.

↳ can we find $I_{C(SAT)}$

KVL at SAT → $V_{cc} - I_{C(SAT)} \cdot R_c - \cancel{V_{CE(SAT)}} = 0$, $V_{CE(SAT)} \to 0$

$$\frac{10}{R_c} = I_{C(SAT)}$$

$$I_{C(SAT)} = 10\ mA$$

$$I_{B(min)} = \frac{I_{C(SAT)}}{\beta_{DC}} = \frac{10\ mA}{200} = 50\mu A$$

c) Calculate maximum value of R_B that will put the transistor in saturation when $V_{in} = 5V$

KVL through BE junction

$$V_{in} - I_B \cdot R_B - V_{BE} = 0$$

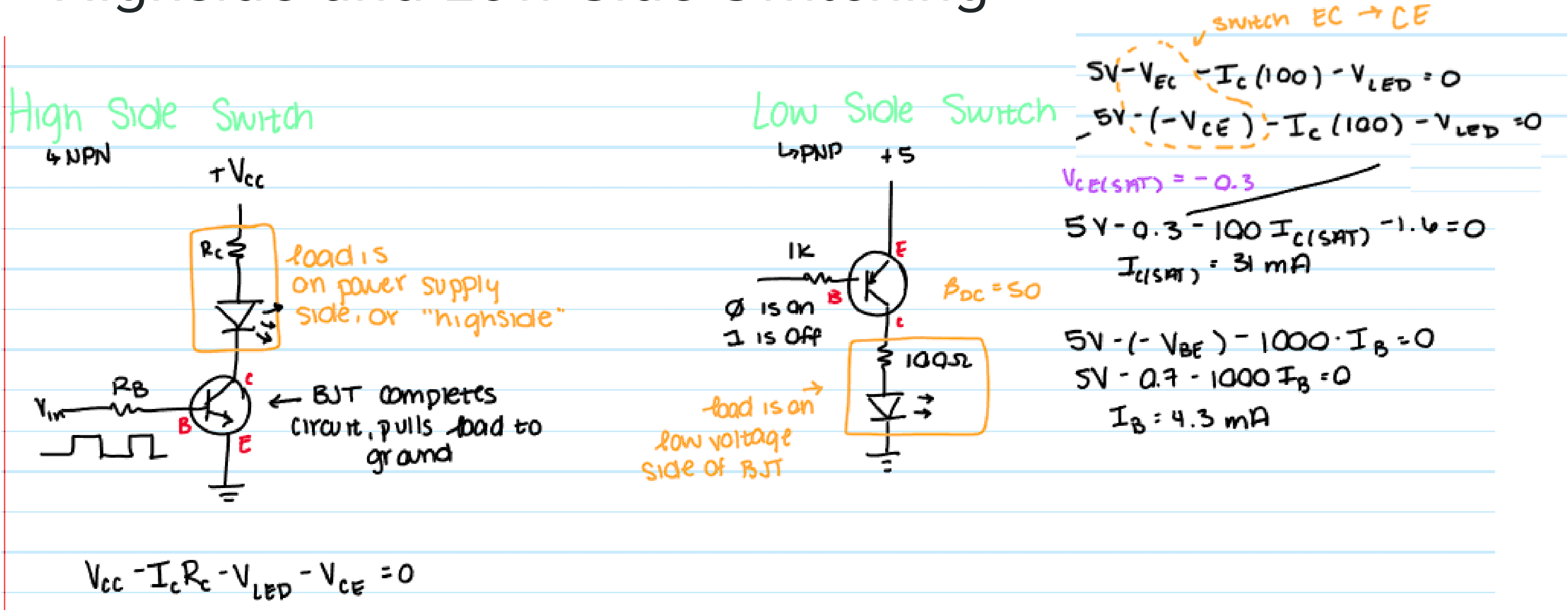
$$5 - (50\mu A) \cdot R_B - 0.7 = 0$$

$$R_B \leq 86\ k\Omega$$

This is easy built on these realizations:

- $I_{c(sat)}/I_{b(min)} = B_{DC}$
- $V_{CE(sat)}$ is set to whatever exists in the relationship
- KVL it with given values
- If I_C found by other given values is bigger, it's in saturation; $I_{C(sat)}$ is the max I_c can be

Highside and Low Side Switching



- Requires V_{CC} >= 0.7 V, load is on power supply/ high side

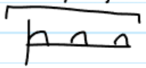
- After plugging values in and performing a KVL at saturation, can determine values
- Also uses a PNP transistor so less often seen

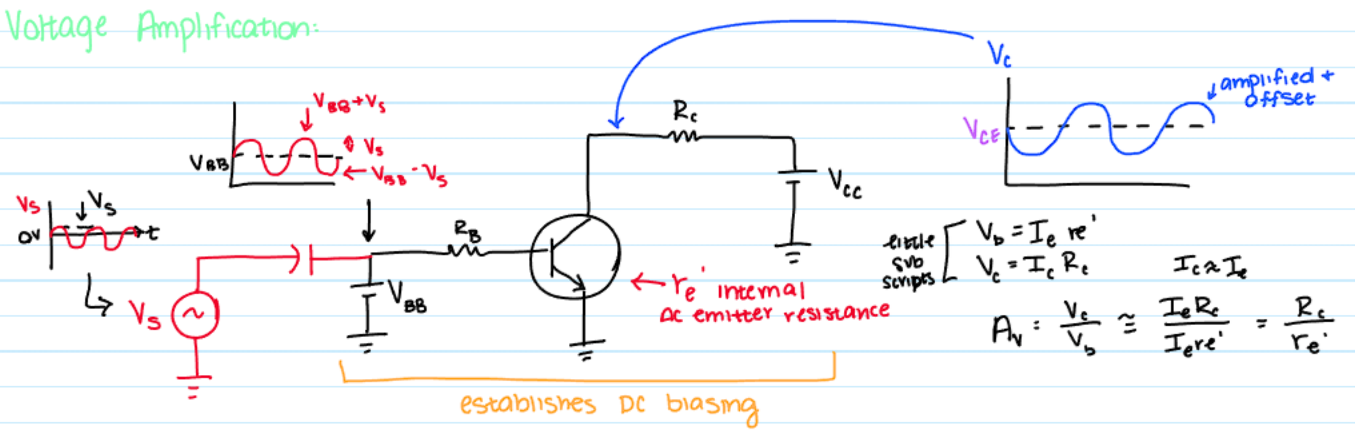
BJT As an Amplifier Intro

DC Quantities (uppercase)
I_B, I_C, I_E
V_{BE}, V_{CB}, V_{CE}
V_B, V_C, V_E

AC Quantities (lowercase)
I_b, I_c, I_e
V_{be}, V_{cb}, V_{ce}
V_b, V_c, V_e

w/ only AC source





DC Biasing is in blakc and is what we've done so far

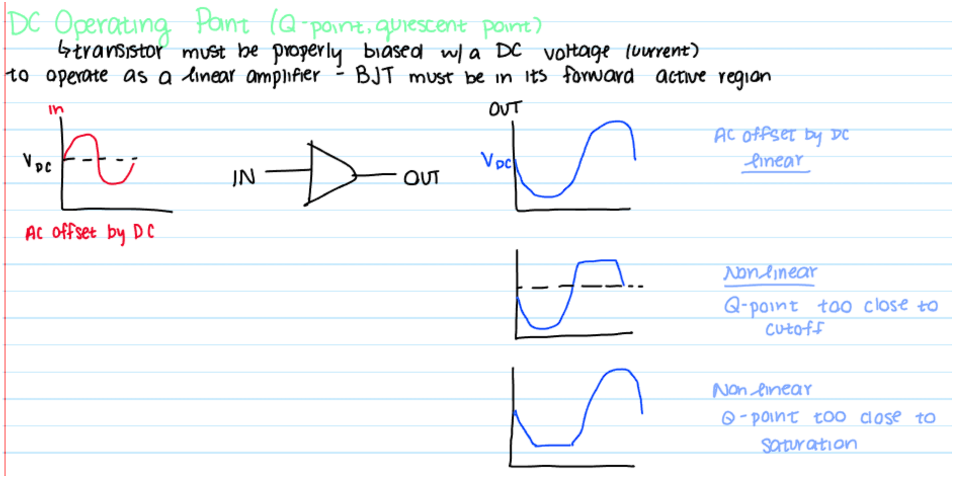
Voltage from V_S allows for $V_{BB} + V_S$

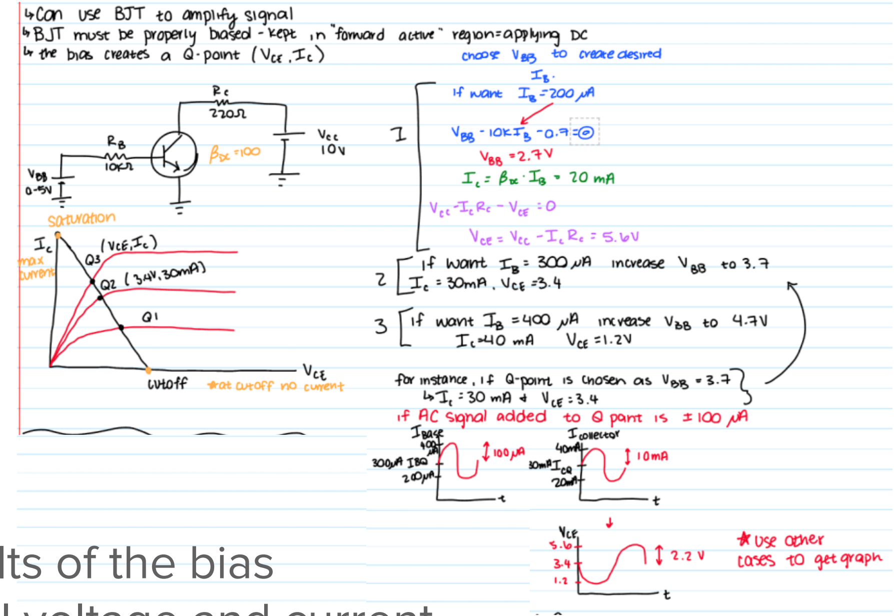
V_{BB} sets the initial offset as well

Gain involves the difference between V_C and V_b

There are a lot of equations and steps that will be involved in the flagship problem of this unit so stay tuned :)

Q Point + Motivation for Q Point + DC Load Line





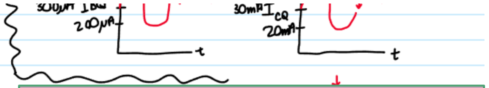
- Q Point functions as the point that results of the bias required for amplification and sets ideal voltage and current
- It is a point that can be determined by the things happening in the circuit and gives information about the current and voltage

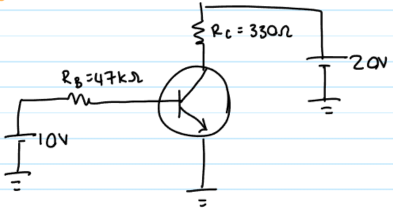
Q Point + DC Load Line Example

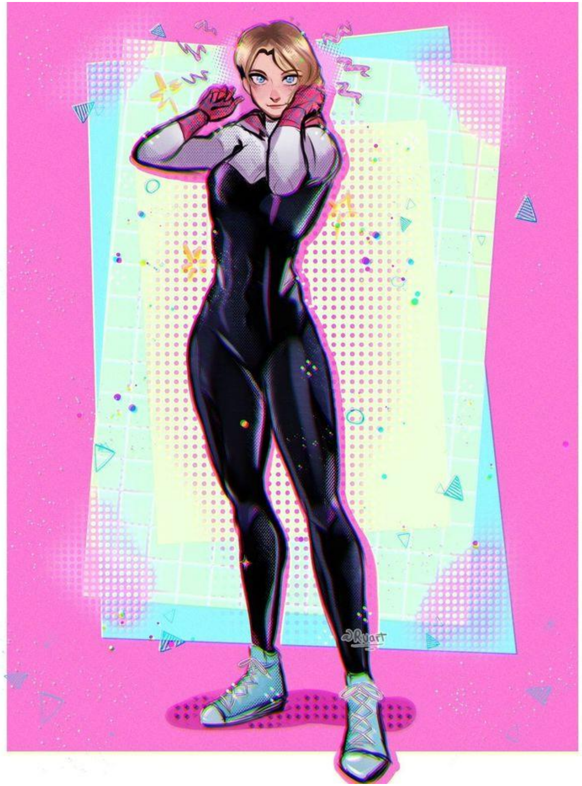
example 5-1:
determine Q-point, draw DC load line, find maximum peak voltage of base current for linear operation

Q poi

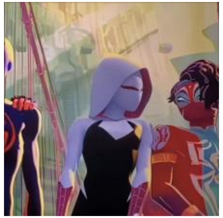
$\beta_{DC} = 200$

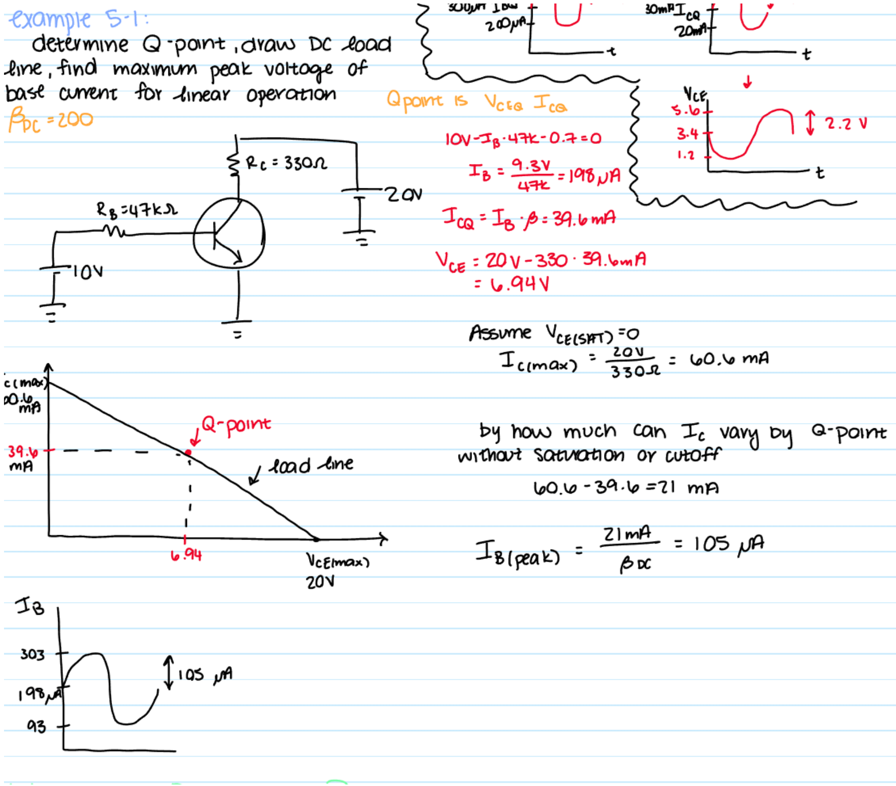






Q Point + DC Load Line Example

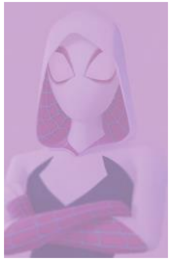




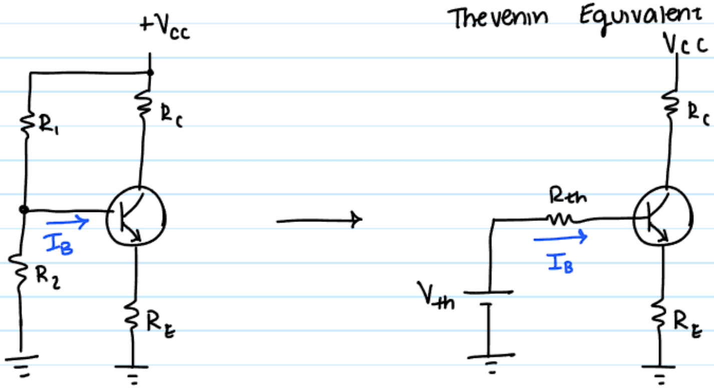
Important Notes:

- Looking for Q Point involves 4 steps that have already been examined:
- I_{CQ}
  - This invovles using the given values and KVL to find I_C. Will usually involve I_B which is needed for later
- V_{CEQ}
  - This is just V_{CE} using I_{CQ}. Since it is a scenario aiming to find the Q point, these points are normally found but are called the Q point of the relationship
- $V_{CE(SAT)}$
  - This is a value that's given, if it isn't ask but can assume 0
- $V_{CE(max)}$
  - $V_{CE(max)}$ is V_C as when there is no current, that's how much voltage will be in V_{CE}
- $I_{C(max)}$
  - This is I_C when you consider $V_{CE(max)}$
- $I_{B(peak)}$
  - Subtracting $I_{C(max)}$ by I_{CQ} grants the variance between saturationand cutoff
  - This current can be used with B_{DC} to get the I_B peak, which allows for a plot to be formed from I_B varying by this value

Voltage Divider Bias



Voltage Divider Bias



$$V_{th} = V_{cc} \cdot \frac{R_2}{R_1+R_2}$$

$$R_{th} = \frac{R_1 R_2}{R_1+R_2}$$

$$V_{th} - I_B \cdot R_{th} - V_{BE} - I_E \cdot R_E = 0$$

$$I_C = \beta_{DC} \cdot I_B$$

$$I_E \approx I_C \quad \Rightarrow I_E \approx \beta_{DC} \cdot I_B$$

$$V_{th} - \frac{I_E}{\beta_{DC}} \cdot R_{th} - V_{BE} - I_E R_E = 0$$

$$I_E = \frac{V_{th} - V_{BE}}{\frac{R_{th}}{\beta} + R_E}$$

- This is a portion of the flagship problem, it basically involves putting transistors in useful forms
- These two equations are ESSENTIAL to know and are not on the formula sheet
- They allow for simple conversion
- Voltage is just a divider using R_2
- R_{th} just considers them as \\
- $I_C = I_E$ for these problems

General Information

Voltage gain without swamping:

$A_v = R_C/r_e'$

(dependent on transistor parameters, not *stable*)

Swamping --- R_E replaced by R_{E1} and R_{E2} with emitter bypass connected between them:

$A_v = \frac{R_C}{r_e' + R_{E1}} \cong \frac{R_C}{R_{E1}}$

$R_c \| R_L$

depends on I_E, which depends on β_{DC}

(as long as R_{E1} is ten times greater than r_e')

Gain is largely independent of r_e' and thus does not depend on transistor parameters

r Parameter	Description
r_e'	ac emitter resistance
r_b'	ac base resistance
r_c'	ac collector resistance
α_{ac}	ac alpha (I_c/I_e)
β_{ac}	ac beta (I_c/I_b)

internal emitter resistance

$$r_e' = \frac{25\text{ mV}}{I_E}$$

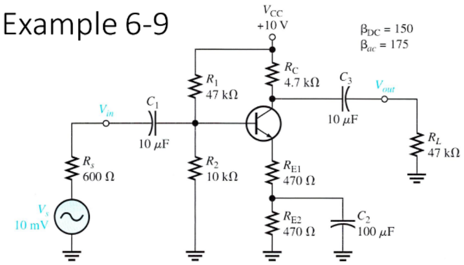
depends on β_{DC}

DC emitter

ere I_E is the DC

tter current

Example 6-9



Determine:

(a) dc collector voltage

(b) ac collector voltage

(c) Draw total collector voltage waveform and total output voltage waveform

TL: Voltage gain will be saught after for BR so these are two equations that are needed for the final step (covered later)

TR: These are explanations of parameters needed for flagship

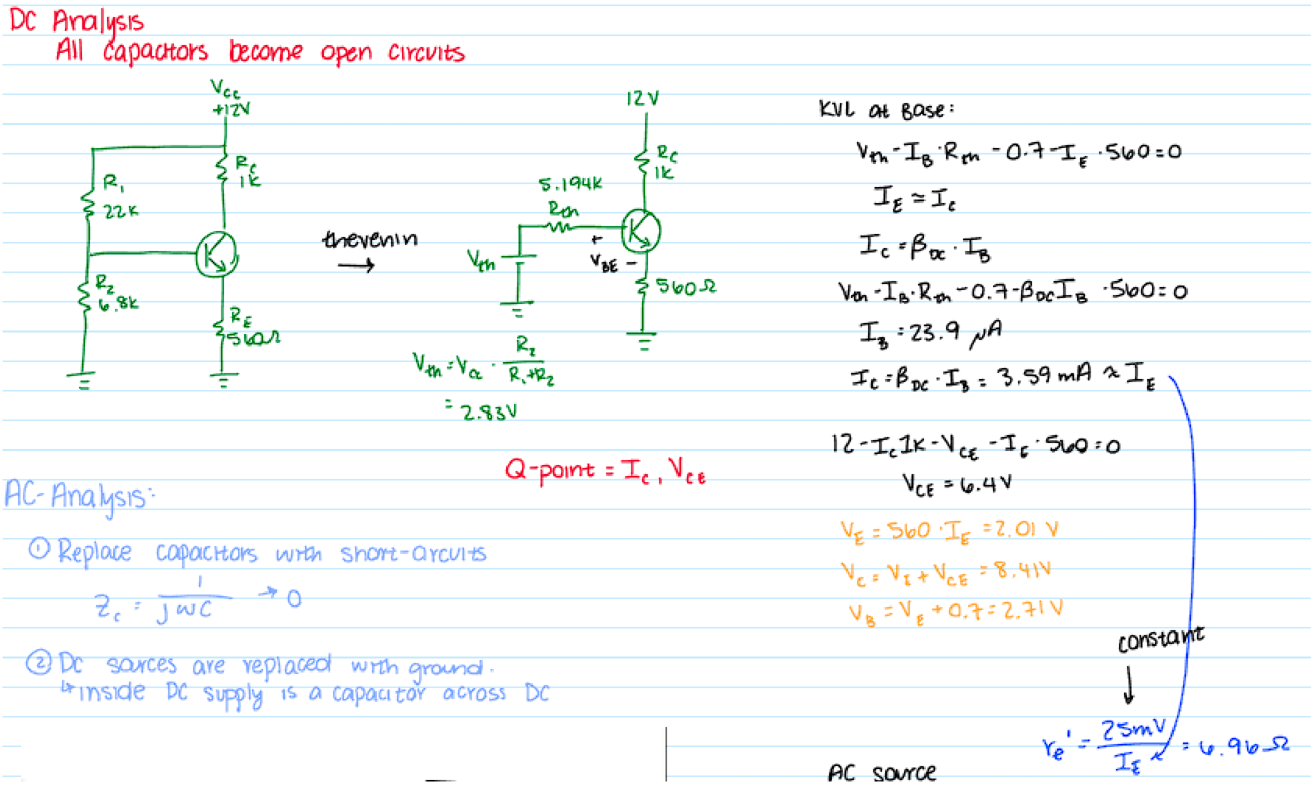
BL: This is an always relationship, so needed to be noted

BR: The flagship that we've been so afraid of thusfar

This is the big problem

Take a deep breath, you’ve got this

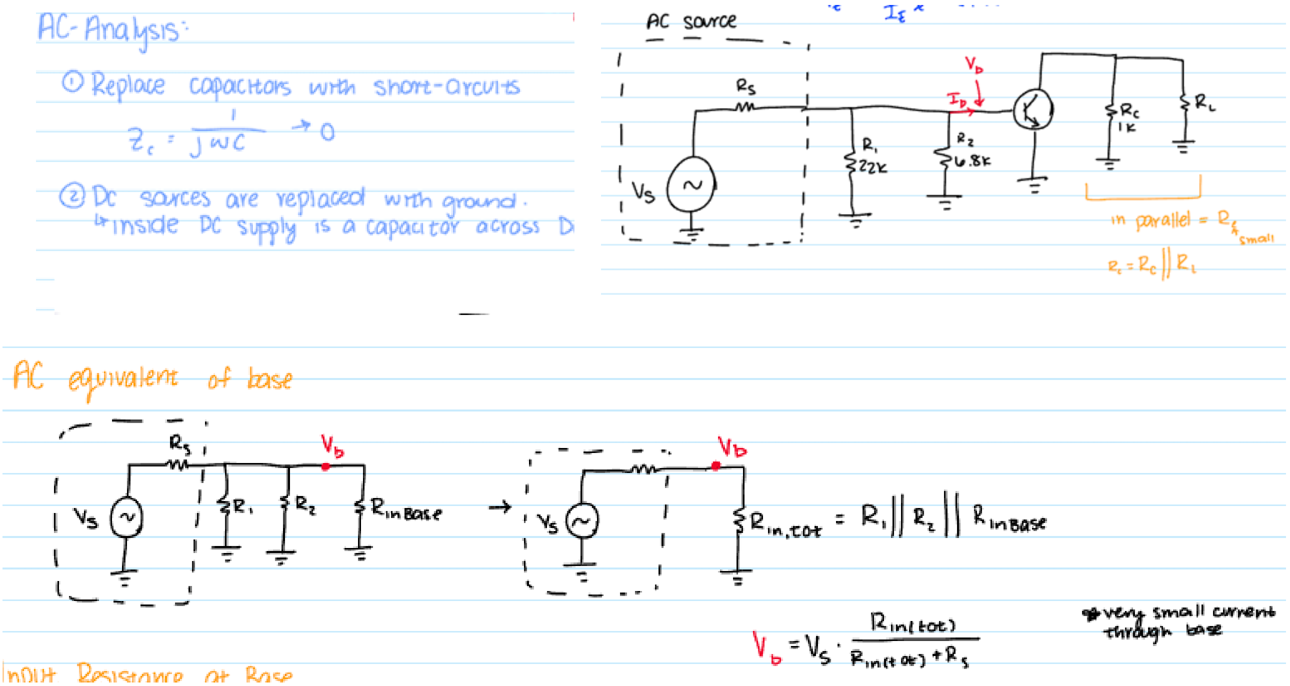
Flagship - Step 1: DC Analysis



This is the previously mentioned easy part:

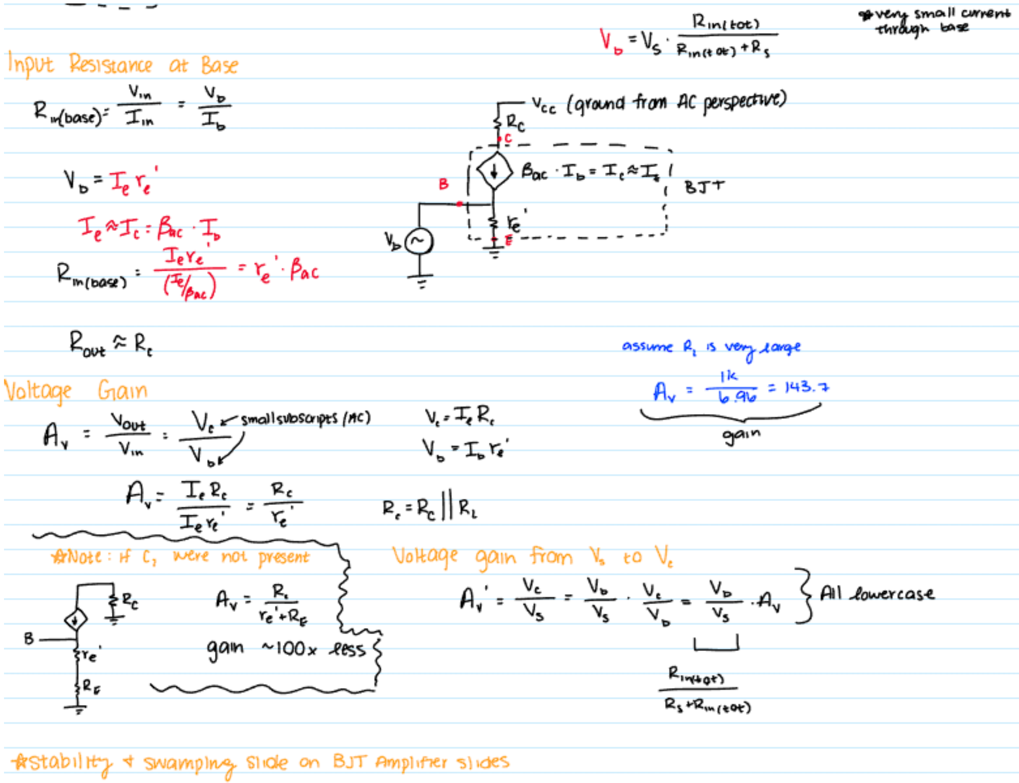
- Important to get I_C, I_E, and keep everything clear
- Must use thevenin
- Capacitors are opens for DC and then short for AC analysis, fully ignore
- $I_c = I_E$ when doing thevenin stuff

Flagship - Step 2: AC Analysis Reduction

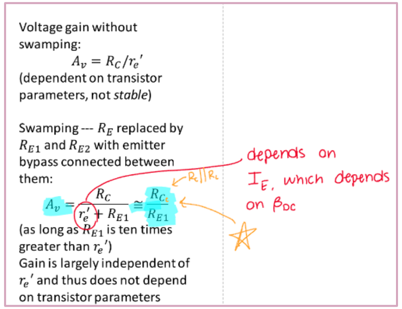


1. Replace capacitors with short circuits and DC sources are grounds
2. Create the form with the AC source and R_S (I think R_s is given)
3. Condense C by parallel (transistor will temporarily disappear)
4. Condense Base Resistors (not R_s)
5. Get your terms in line
    a. $R_{in\ Base}$ = All resistors that input to the base
    b. $R_{in,\ total} = R_1 \| R_2 \| R_{in\ base}$
    c. V_b = Voltage divider with $R_{in,\ total}$ and R_s using V_s

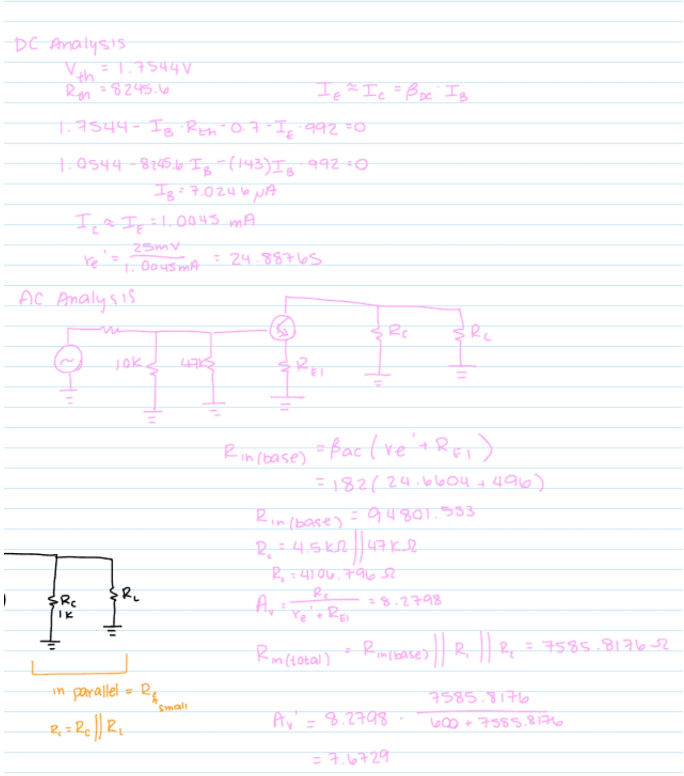
Flagship - Step 3: AC Analysis Application



1. (Input Res) From this point, identities can be used to calculate $R_{in(base)}$ by setting it equal to r_e'* B_{ac}. R_{out} is approximately R_c
2. (Voltage Gain) V_c/V_b can be considered voltage gain, but by subsitution so can R_c/r_e'
   a. This is because $V_c = I_eR_c$ and $V_b = I_bR_e$'
3. The additional cases are shown here invovling if C_2 was not present as well
4. Also below may be used for voltage gain in cases where more precision is wanted



Flagship worked out



DC Analysis

$V_{th} = 1.7544V$

$R_{th} = 8245.6$

$I_E \approx I_C = \beta_{DC} \cdot I_B$

$1.7544 - I_B \cdot R_{th} - 0.7 - I_E \cdot 992 = 0$

$1.0544 - 8245.6\, I_B - (143) I_B \cdot 992 = 0$

$I_B = 7.0246\ \mu A$

$I_C \approx I_E = 1.0045\ mA$

$r_e' = \frac{25mV}{1.0045mA} = 24.88765$

AC Analysis

$R_{in(base)} = \beta_{ac}(r_e' + R_{E1})$

$= 182(24.6604 + 496)$

$R_{in(base)} = 94801.553$

$R_c = 4.5k\Omega \| 47k\Omega$

$R_c = 4106.796\ \Omega$

$A_v = \frac{R_c}{r_e' + R_{E1}} = 8.2798$

$R_{in(total)} = R_{in(base)} \| R_1 \| R_2 = 7585.8176\ \Omega$

$A_v' = 8.2798 \cdot \frac{7585.8176}{600 + 7585.8176}$

$= 7.6729$

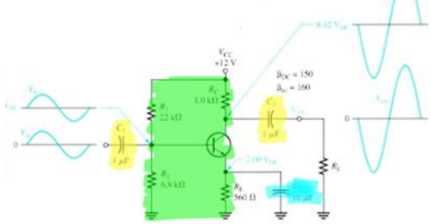
in parallel = $R_{c\,small}$

$R_c = R_C \| R_L$

[redacted]

C_1, C_3 - coupling capacitors b/c they couple AC signals into + out of BJT w/out disturbing DC bias

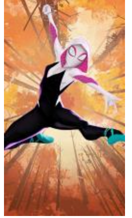
$C_2 \rightarrow$ bypass capacitor to keep emitter voltage constant. AC signals get shorted to ground



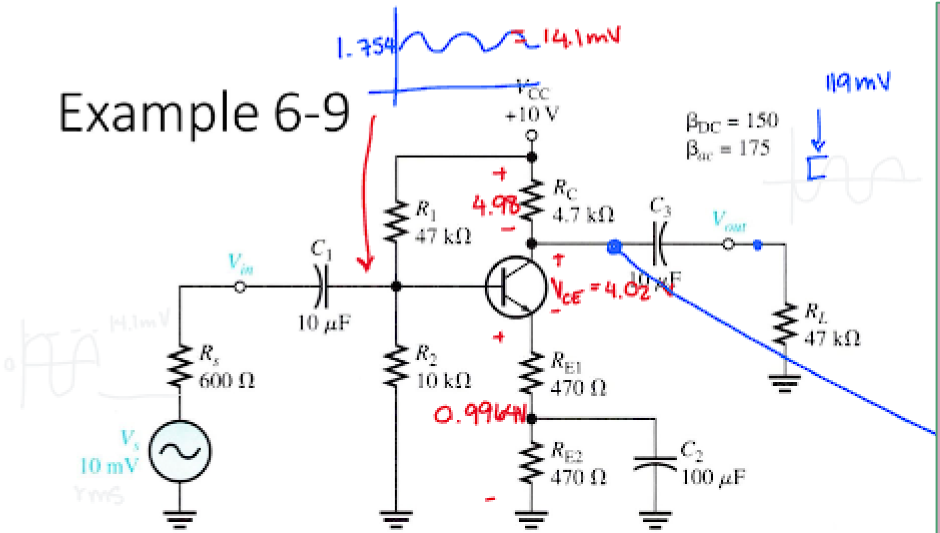
DC Analysis finds the values that we want
AC Analysis then goes into depth about
$R_{in(base)}$ which is calculated
R_c is then found by using R_s and 10k in ||
which is parallel with 47k as well
Then $A_v = R_c / r_e' + R_E$ can be used
$R_{in(total)}$ is found in order to find A_v'which
$Av' = Av * R_{in(total)}/R_s+R_{in(total)}$ i believe

These require separate workouts in order to be comfortable but these are the steps involved with regular soultions

Flagship practice - but like seriously do this please



Example 6-9



Determine:

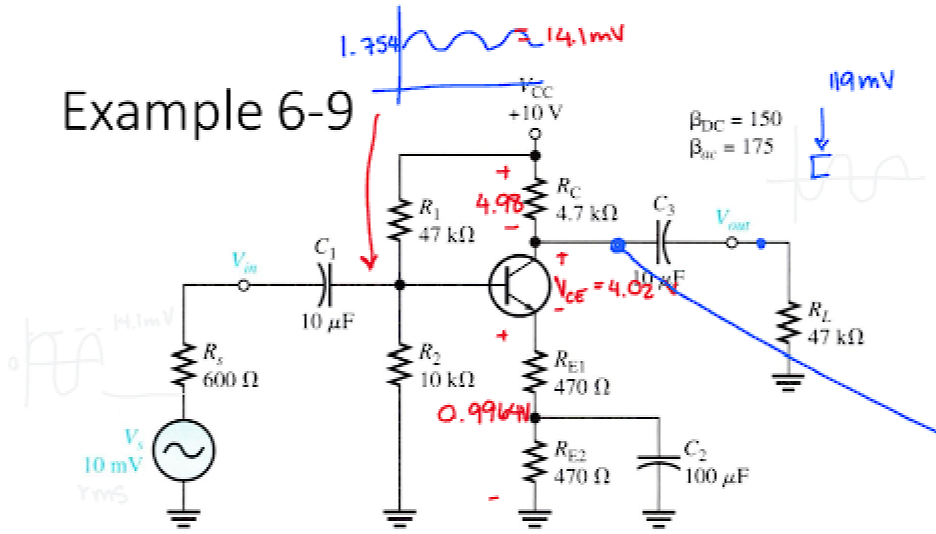
(a) dc collector voltage

(b) ac collector voltage

(c) Draw total collector voltage waveform and total output voltage waveform

Flagship practice - but like seriously do this please

Example 6-9



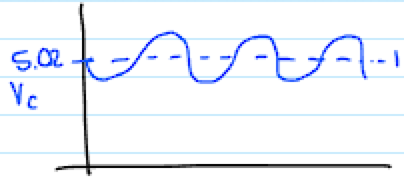
$V_{th} = 1.754$

$I_C = I_E \approx 1.00$ mA

$A_v \approx \frac{R_c}{R_{E_1}} = \frac{4.7k \| 47k}{470} = 9.09 \frac{V}{V}$

$A_v' = 8.42$

$14.1\text{mV} \cdot A_v' = 119$ mV



Determine:

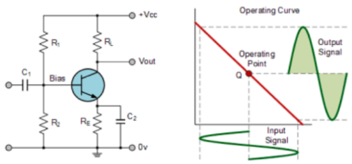
(a) dc collector voltage

(b) ac collector voltage

(c) Draw total collector voltage waveform and total output voltage waveform

Classes of Amplifier Transistors

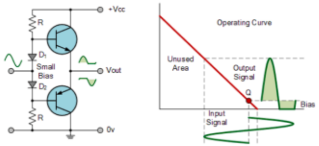
Class A



Least efficient (~30%): Amplifier is always biased, with current flowing to keep it at Q-point. Very liner.

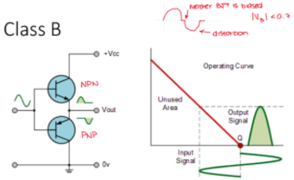
Class AB

Gets rid of dead band



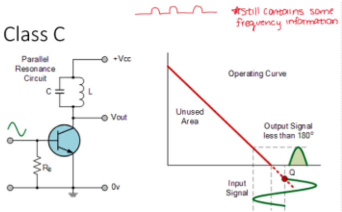
Like class B, but diodes added to overcome "dead band" when neither amplifier is on. Less distortion than class B.

Class B



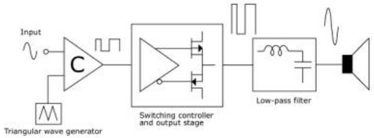
More efficient (~50%). Amplifier only conducts when it's amplifying the signal. Uses 2 transistors in a "push-pull" configuration. They take turns amplifying the positive and negative halves of the signal. Downside: distortion in dead band (-0.7 V to 0.7 V) when neither amplifier is on.

Class C



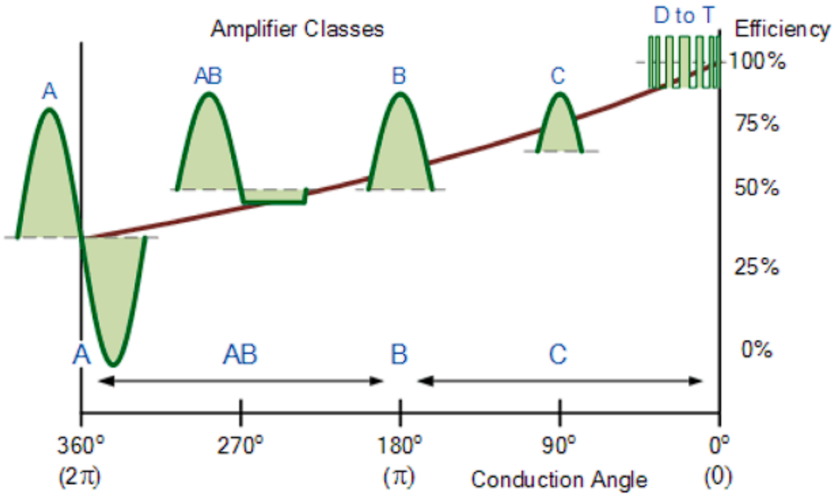
High efficiency (~80%) but highly non-linear. Heavy distortion (not suitable for audio amplification)

Class D



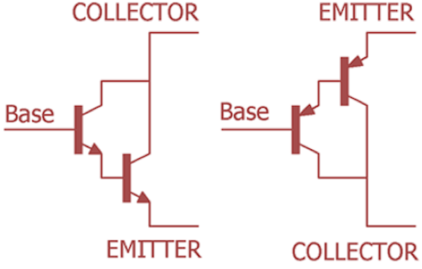
Highly efficient (>90%) pulse width modulation (PWM) switching amplifier. Low pass filter at output blocks pulses and only lets amplified version of input waveform pass.

Amplifier Classes



FETs Intro

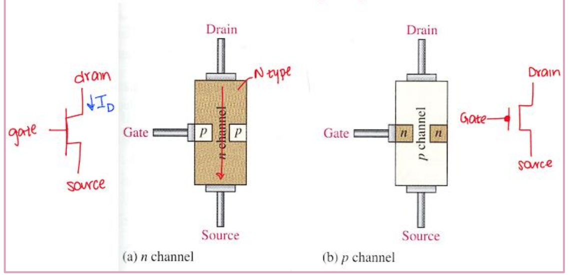
Darlington Pair – more gain



β multiply when cascade

Junction FET

P type - semiconductor
↳ doping - conduct better
↳ extra holes
N type
↳ extra electrons



- FETs are powered by voltages, have high input resistance, and do not have linear relations
- By charging or discharging the gate, a channel width is provided

Field Effect Transistor (FET)

- Unipolar (as opposed to bipolar BJT) – only one type of charge carrier
- Types:
  - JFET (Junction FET)
  - MOSFET (Metal oxide semiconductor FET)
- Voltage controlled current source (voltage between Gate and Source controls current)
- Very high input resistance
- Highly non-linear

Operation (Biasing)

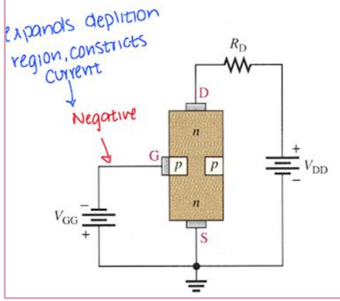


FIGURE 8–2

A biased *n*-channel JFET.

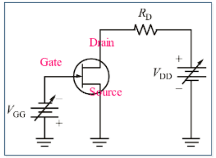
In a JFET, the gate-source pn junction is always reverse biase

Electronic Devices

The JFET

As in the base of bipolar transistors, there are two types of JFETs: *n*-channel and *p*-channel. The dc voltages are opposite polarities for each type.

The symbol for an *n*-channel JFET is shown, along with the proper polarities of the applied dc voltages. For an *n*-channel device, the gate is always operated with a negative (or zero) voltage with respect to the source.



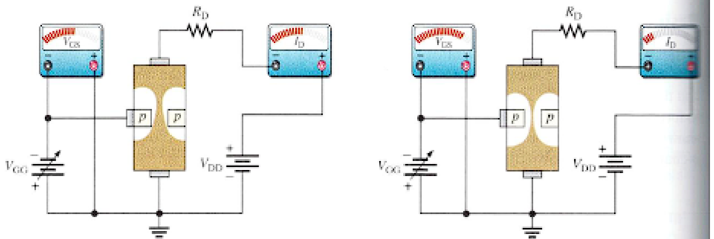
JFET States

E units = $\frac{V}{m}$

- Higher Voltage results in less current since the channel will decrease in size and vice versa
- There's a certain voltage where cutoff occurs and no current can be transmitted

Effect of V_{GS} on drain current

field effect
↳voltage creates an electric field



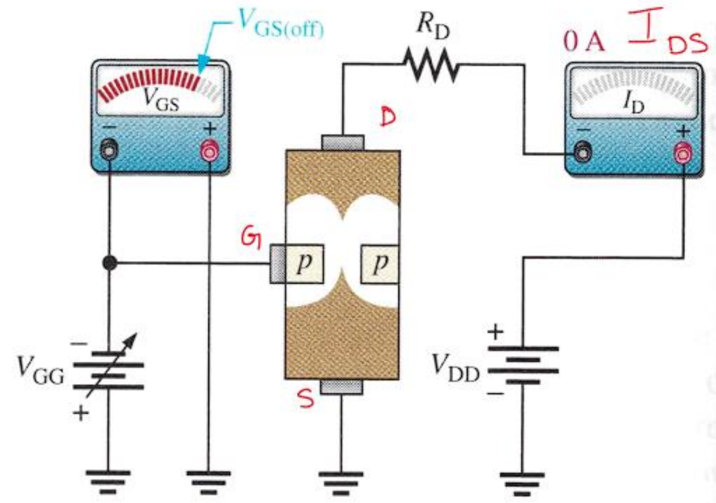
(a) JFET biased for conduction

(b) Greater V_{GG} narrows the channel (between the white areas) which increases the resistance of the channel and decreases I_D.

(c) Less V_{GG} widens the channel (between the white areas) which decreases the resistance of the channel and increases I_D.

- As V_{GS} is increased, the depletion region becomes larger, making the n-channel narrower
- Narrower channel restricts the current flow, due to increased resistance

JFET during cutoff



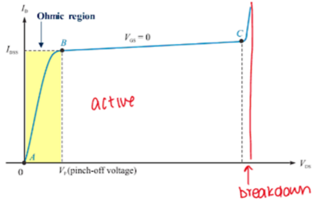
Regions for the JFET Characteristic Curve

There are three regions in the characteristic curve for a JFET as illustrated for the case when $V_{GS} = 0$ V.

Between A and B is the **Ohmic region**, where current and voltage are related by Ohm's law.

*ohmic means acts as resistor, more voltage = more current

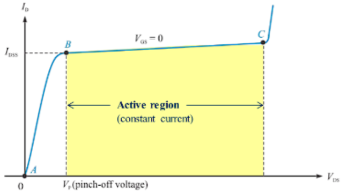
*active doesn't matter same amount of current



There are three regions in the characteristic curve for a JFET as illustrated for the case when $V_{GS} = 0$ V.

Between A and B is the **Ohmic region**, where current and voltage are related by Ohm's law.

From B to C is the **active** (or *constant-current*) **region** where current is essentially independent of V_{DS}.

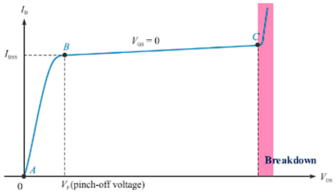


There are three regions in the characteristic curve for a JFET as illustrated for the case when $V_{GS} = 0$ V.

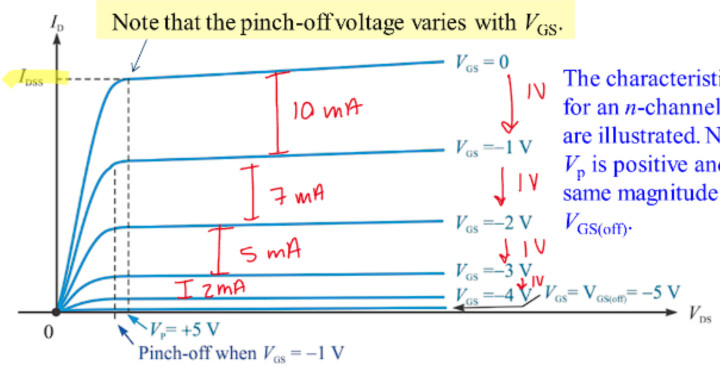
Between A and B is the **Ohmic region**, where current and voltage are related by Ohm's law.

From B to C is the **active** (or *constant-current*) **region** where current is essentially independent of V_{DS}.

Beyond C is the **breakdown region**. Operation here can damage the FET.

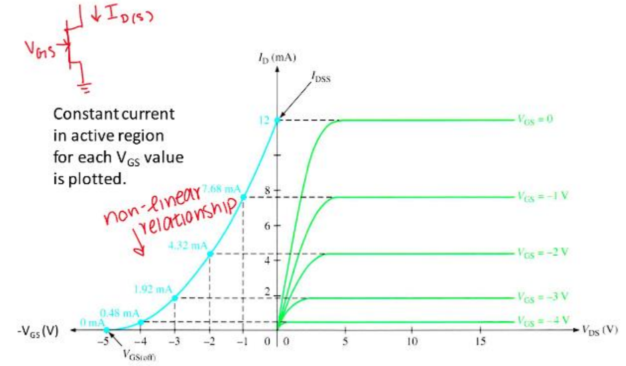


As V_{GS} is varied, a family of characteristic curves is developed that shows the relationship between V_{DS}, I_D and V_{GS}.



The characteristic curves for an n-channel device are illustrated. Notice that V_p is positive and has the same magnitude as $V_{GS(off)}$.

Universal Transfer Characteristic



Development of transfer characteristic curve (blue) from drain characteristic curves (green)

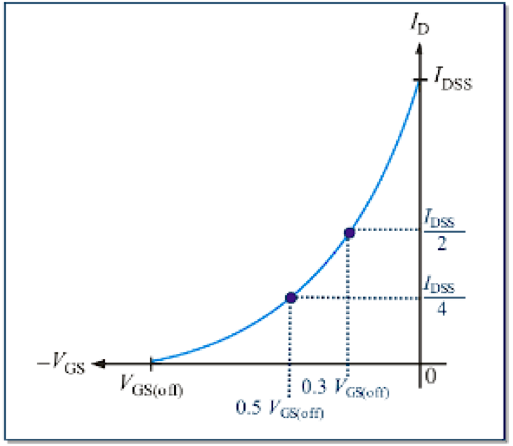
JFET Plots

A plot of V_{GS} vs I_D is called the transfer curve. The transfer curve is a is a plot of the output current (I_D) vs the input voltage (V_{GS}).

The transfer curve is the graphical representation of the equation:

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_{GS(off)}}\right)^2$$

By substitution, you can find other points on the curve for plotting the universal curve.



★ conductivity

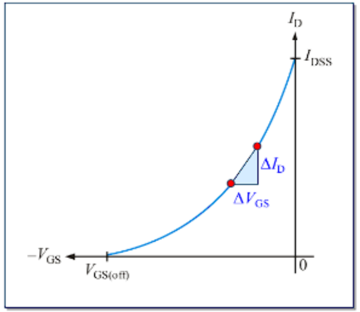
The transconductance is the ratio of a change in output current (ΔI_D) to a change in the input voltage (ΔV_{GS}).

This definition is $g_m = \frac{\Delta I_D}{\Delta V_{GS}}$

The following approximate formula is useful for calculating gm if you know g_{m0}.

$$g_m = g_{m0}\left(1 - \frac{V_{GS}}{V_{GS(off)}}\right)$$

The value of g_{m0} can be found from $g_{m0} = \frac{2I_{DSS}}{|V_{GS(off)}|}$



Transconductance is the slope of transfer curve

MOSFET Intro

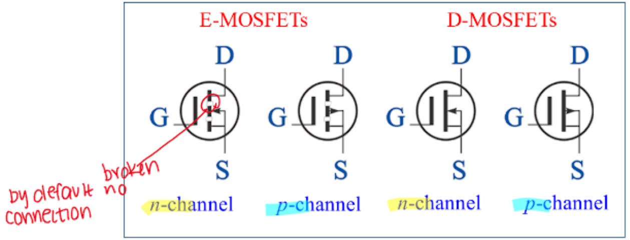
metal oxide semiconductor

MOSFETs

field effect transistor

- Like JFET – Small gate voltage controls drain current
- No *pn* junction structure, instead MOSFET gate is insulated from channel by silicon dioxide (SiO_2) layer
- Two types:
  - Enhancement (E-MOSFET, more widely used)
    - Only operated in enhancement mode (positive V_{GS})
  - Depletion (D-MOSFET)
    - Usually operated in depletion mode (negative V_{GS})

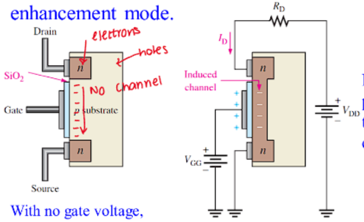
MOSFET symbols are shown. Notice the broken line representing the E-MOSFET that has an induced channel. The *n* channel has an i**n**ward pointing arrow.



E&D MOSFETs

enhancement mode ★until make channel no conduction

The E-MOSFET has no structural conduction channel. By default, it does not conduct. A channel is induced by the application of a gate voltage. E-MOSFETs can *only* operate in enhancement mode.



With no gate voltage, there is no channel.

Induced channel

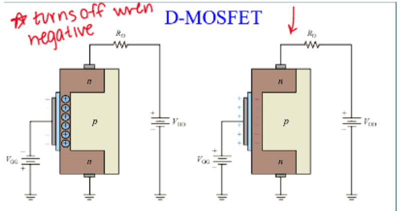
For an *n*-channel type, a positive voltage applied to the gate induces a conduction channel.

needed to make current flow

$V_{GS} > \boxed{V_{G(th)}}$

The D-MOSFET has a channel that is controlled by applying a gate voltage and comes as either an *n*-channel or a *p*-channel device.

better channel



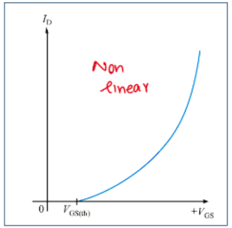
For an *n*-channel type, a negative voltage depletes the channel (D-mode); a positive voltage enhances the channel (E-mode).

D-MOSFETs can operate in either mode.

The transfer curve for a MOSFET is has the same parabolic shape as the JFET but the position is shifted along the *x*-axis. The transfer curve for an *n*-channel E-MOSFET is entirely in the first quadrant as shown.

The curve starts at $V_{GS(th)}$, which is a nonzero voltage that is required to have channel conduction. The equation for the drain current is

$$I_D = K\left(V_{GS} - V_{GS(th)}\right)^2$$

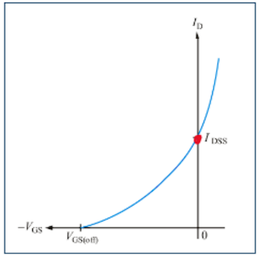


Transfer curve for E-MOSFET

Recall that the D-MOSFET can be operated in either mode. For the *n*-channel device illustrated, operation to the left of the *y*-axis means it is in depletion mode; operation to the right means is in enhancement mode.

As with the JFET, I_D is zero at $V_{GS(off)}$. When $V_{GS} = 0$, the drain current is I_{DSS}, which for this device is *not* the maximum current. The equation for drain current is

$$I_D \cong I_{DSS}\left(1 - \frac{V_{GS}}{V_{GS(off)}}\right)^2$$



Transfer curve for D-MOSFET

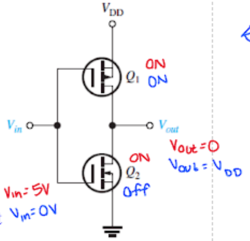
MOSFET Digital Logic

MOSFETs are widely used in digital logic circuits. **CMOS** (Complementary MOS) combines n-channel and p-channel **E-MOSFETs** in a series arrangement as shown for a basic logic inverter. The input voltage at the gates is either 0 V or V_{DD}. The term V_{DD} is used for the positive voltage, which is on the p-channel device's source terminal.

When $V_{in} = 0$ V, Q_1 is *on* and Q_2 is *off*. Because Q_1 is acting as a closed switch, the output is approximately V_{DD}, which is the inverse of the input.

When $V_{in} = V_{DD}$, V_{in} is *on* and Q_1 is *off*. Because Q_2 is acting as a closed switch, the output is essentially connected to ground (0 V). Again the output is the inverse of the input.

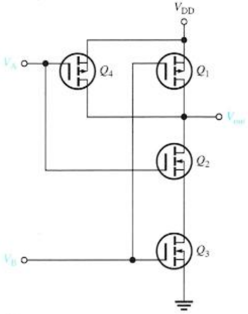
Because one of the two MOSFETs is always *off*, the drain current is kept near zero in both states, minimizing power dissipation.



← inverter

✱ based on N + P channels

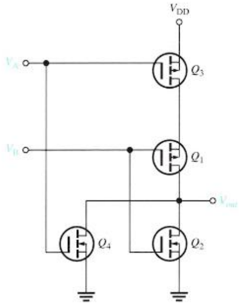
NAND gate



V_{DD} = 1

V_A	V_B	Q_1	Q_2	Q_3	Q_4	V_{out}
0	0	on	off	off	on	V_{DD}
0	V_{DD}	off	off	off	on	V_{DD}
V_{DD}	0	on	off	off	off	V_{DD}
V_{DD}	V_{DD}	off	on	on	off	0

NOR gate



V_A	V_B	Q_1	Q_2	Q_3	Q_4	V_{out}
0	0	on	off	on	off	V_{DD}
0	V_{DD}	off	on	on	off	0
V_{DD}	0	on	off	on	off	0
V_{DD}	V_{DD}	off	on	off	on	0

JFET Ex:

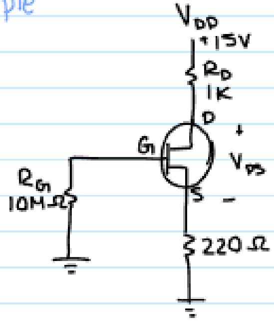
JFET example 8-3

$$I_D \approx I_{DSS}\left(1 - \frac{V_{GS}}{V_{GS(off)}}\right)^2$$

$I_{DSS} = 9mA$
$V_{GS(off)} = -8V$

find I_D if $V_{GS} = 0$ $9mA$
$= -1$ $6.89mA$
$= -4$ $2.25\ mA$

example



KVL

$15V - R_D I_D - V_{DS} - R_S I_S = 0$

$15V - 1k(5\ mA) - V_{DS} - 220(5mA) = 0$

$\boxed{V_{DS} = 8.9}$

$V_S = I_S \cdot 220 = I_D \cdot 220$
$= 1.1V$

$V_{GS} = V_G - V_S$
$0V - 1.1$

$\boxed{V_{GS} = -1.1V}$

MOSFET Ex:

Using a MOSFET as a switch

example:

we have a device with digital I/O Ports (computer, microcontroller, Arduino, Raspberry P.) which can source 1 mA. We want to use this to control an LED which needs 20 mA of current & has $V_f = 2.9V$

Assume there is a 5 V power supply

DIO logic level = 3.3V

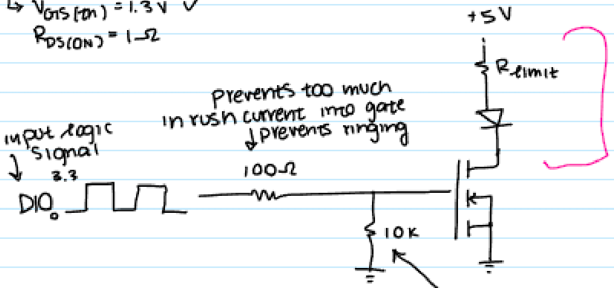
Design a circuit

- choose a device - "logic level" MOSFET. Enhancement - mode, n-channel
- BSS138
  - ↳ $V_{GS(on)} = 1.3V$ ✓
  - $R_{DS(ON)} = 1\Omega$

HW

$R_{limit} = R_{load}$

No LED



KVL:

$5V - I_D \cdot R_{limit} - 2.9V - R_{DS(ON)} \cdot I_D = 0$

$5V - 20mA(R_{limit} + 1) - 2.9V = 0$

$\boxed{R_{limit} = 104\,\Omega}$

Practice

Methods of Practice and Concept Strength

Diodes:
- HW2
- HW3
- HW4
- Exam 1
- Extra Exam 1

Confidence: 56/100
Explanation: Everything is fundamentally simple but it’s been a while
Biggest Worries: Zener Diodes and Transformers

Op Amps:
- HW5
- HW6
- HW7
- Exam 2

Confidence: 73/100
Explanation: Spent the most time studying for and remember things. Additionally, genuinely believe that Bode plots can be done
Biggest Worries: Exam 2 trauma and Bode Plots

Transistors:
- HW8
- HW9

Confidence: 40/100
Explanation: Haven’t really done anything for them, either really simple or confusing as hell
Biggest Worries: Saturation and literally any FET

Current Plan

Current Plan:
- Brief Break
- Once Over
- Exam 1
- Exam 2
- HW8
- Review how well this worked and figure what the next plan is
- Study for this :(

Unattempted:
- Exam 1
- Extra 1
- Exam 2
- HW2
- HW3
- HW4
- HW5
- HW6
- HW7
- HW8
- HW9
- Non 8&9 waste?

Worries:
- Zener Diodes
- Transformers
- Exam 2 Trauma
- Bode Plots
- FETs
- Saturation
- Overall Things that haven't been Seen

Done/Reward/Takeways:
- Tapped: Vpsec = (vprect+.7)*2
  - This is due to center tapped being half of it's thingy
- Bridge: Vpsec = vprect + 1.4
  - This is due to traversing through 2 diodes

Exam 1

Very Short Answer

(2) 1. The [redacted] diode model accounts for additional resistances.

(2) 2. A voltage doubler can be realized using two diodes and two [redacted]

(2) 3. A [redacted] region forms at a *pn* junction.

(2) 4. [redacted] The forward voltage drop of all LEDs is 0.7 V. [redacted]

(2) 5. For a specified load and ripple voltage, a DC power supply based on a half-wave rectifier will need a ([redacted] capacitor than one based on a full-wave rectifier.

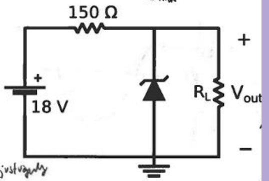
Short Answer

(8) 6. Sketch the *i*-*v* characteristic of an actual diode with voltage on the *x*-axis. Label the *turn-on* and *breakdown* voltages.

[redacted]

(8) 7. In the following circuit, assuming an ideal Zener diode with $V_Z = 7.2$ V, $I_{ZK} = 0.75$ mA, $I_Z = 30$ mA, and $I_{ZM} = 70$ mA, find the range of R_L for which regulation is maintained.

max



Full disclosure I dont remember this eq exactly just vaguely trying my best

[redacted]

Exam 1

Very Short Answer wrong

(2) 1. The "perfect"/complete diode model accounts for additional resistances.

(2) 2. A voltage doubler can be realized using two diodes and two capacitors.

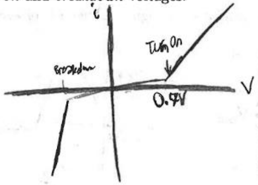
(2) 3. A depletion region forms at a *pn* junction.

(2) 4. (True / False) The forward voltage drop of all LEDs is 0.7 V.

(2) 5. For a specified load and ripple voltage, a DC power supply based on a half-wave rectifier will need a (larger / smaller) capacitor than one based on a full-wave rectifier. larger I think

Short Answer

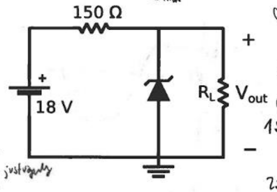
(8) 6. Sketch the *i-v* characteristic of an actual diode with voltage on the *x*-axis. Label the *turn-on* and *breakdown* voltages.



Curved

(8) 7. In the following circuit, assuming an ideal Zener diode with $V_Z = 7.2$ V, $I_{ZK} = 0.75$ mA, $I_Z = 30$ mA, and $I_{ZM} = 70$ mA, find the range of R_L for which regulation is maintained.

I_{ZM}: max



Full disclosure I dont remember this eq exactly just vaguely trying my best

$0 = 18V - 150\Omega i - 7.2V$ Node

$I_T = 72mA$

① $I_T = I_{ZK} + I_{Rmax}$ } makes sense, total current = max diode on + corresponding R current

② $I_T = I_{Zm} + I_{Rmin}$ } + min diode off + corresponding R current

1: $72mA = 0.75mA + I_{Rmax}$

$I_{Rmax} = 71.25mA$

2: $72mA = 70mA + I_{Rmin}$

$I_{Rmin} = 2mA$

Voltage in Z diode is 7.2V, so must be so in R_L too

$\frac{7.2V}{2mA} = 3600\Omega$

$\frac{7.2V}{71.25mA} = 101.05\Omega$

$\boxed{R_L \text{ Range: } 101.05\Omega \le R_L \le 3600\Omega}$

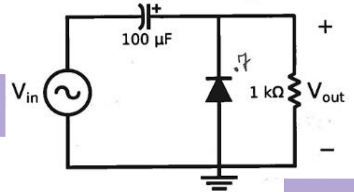
Exam 1

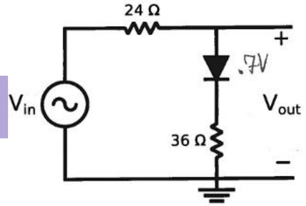
For all problems that follow, assume the practical / constant voltage drop diode model.

(12) 8. Design a circuit to limit voltages to a range of −3.6 V to +0.7 V, with diode current not exceeding 120 mA for a ±12 V input. Draw the circuit, labeling the input, output, and all component values.

To determine R, use the

(12) 9. Plot one period of $V_{out}(t)$ for a ±6 V sinusoidal input signal. Label all voltage values.



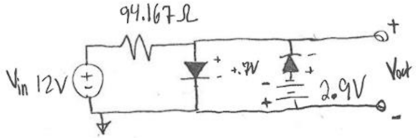


Exam 1

For all problems that follow, assume the practical / constant voltage drop diode model.

(12) 8. Design a circuit to limit voltages to a range of −3.6 V to +0.7 V, with diode current not exceeding 120 mA for a ±12 V input. Draw the circuit, labeling the input, output, and all component values.

To determine R, use the greater voltage

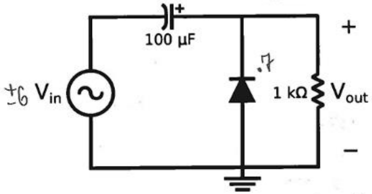


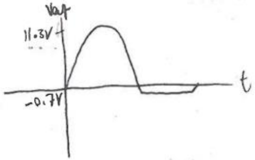
$12 - 0.7 = 11.3V$

$\frac{11.3V}{.12A} = \Omega$

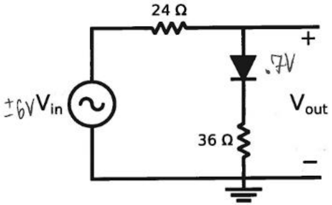
$= 94.167\Omega$

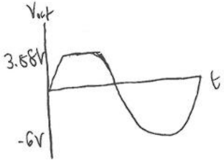
(12) 9. Plot one period of $V_{out}(t)$ for a ±6 V sinusoidal input signal. Label all voltage values.





$6 - .7 = 5.3 + 6 = 11.3V$





$V_T = 3.18 + 0.7V = 3.88V$ $\quad V_x = \frac{V_s R_x}{R_T}$

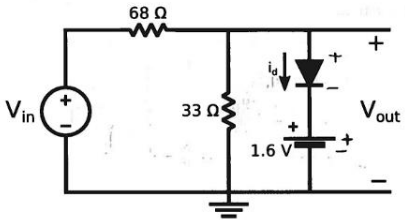
$V_x = \frac{(6V - .7V)(36\Omega)}{(24 + 36\Omega)}$

$V_x = 3.18V$ in 36Ω

Exam 1

Long Answer

10. Analyze the following diode circuit:



(20) (a) Complete the following table (show all work):

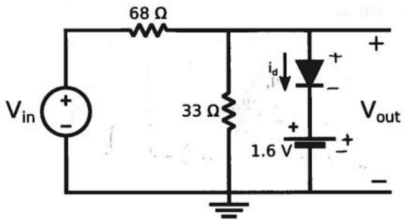
V_{in}	Diode Status	i_d	V_{out}
−5 V			
0 V			
	Just turning on		
10 V			

(4) (b) Plot V_{out} as a function of V_{in}

Exam 1

Long Answer

10. Analyze the following diode circuit:



Effective drop

$1.6V + 0.7V = 2.3V$

(20) (a) Complete the following table (show all work):

V_{in}	Diode Status	i_d	V_{out}
−5 V	Off	0A	−1.63V
0 V	Off	0A	0V
7.039V	Just turning on	0A	2.3V
10 V	On	43.53mA	2.3V

Does this make sense

The maximum V_{out} able to happen is 2.3V

V_{in}:

$2.3V = \frac{V_{in}(33)}{101}$

$\underline{V_{in} = 7.039}$

i_d: due to branch, id 0 til on

$\underline{7.039V}$

$7.039V - 2.3V = 4.739V$ in 68Ω

$4.739V / 68\Omega = 69.7mA$ total current

$2.3V/33\Omega = 69.7mA$ current in

0A in diode

V_{out}: $V_{-5V} = \frac{-5V(33\Omega)}{101\Omega} = \underline{-1.63V}$ Off

V_0: 0V

On

$\underline{10V}$

$10V - 2.3V = 7.7V$

$7.7V/68\Omega = 113.23mA$

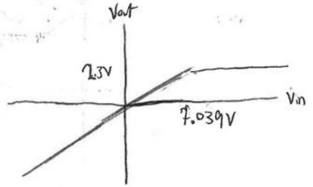
$113.23mA - 69.7mA$

$= \underline{43.53mA}$

No current thru diode

2.3V must be dissipated in 33Ω, that means its current remains constant. This means current through id will increase dramatically

(4) (b) Plot V_{out} as a function of V_{in}



Exam 1

11. Design a DC power supply using a center-tapped transformer and full-wave rectifier, having a DC output voltage of 15 V and a ripple factor of 0.09. The AC input power is 120 Vrms, 60 Hz. Assume a load of $R_L = 18\ \Omega$.

(10) (a) Draw the circuit.

(8) (b) Find the transformer turns ratio and filter capacitor value (show all work).

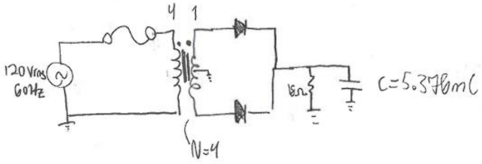
(8) (c) Plot several periods of $V_{out}(t)$ *with* and *without* the filter capacitor in the circuit. Indicate the *values* of $V_{r(pp)}$ and $V_{p(rect)}$.

Exam 1

N = 5.18

11. Design a DC power supply using a center-tapped transformer and full-wave rectifier, having a DC output voltage of 15 V and a ripple factor of 0.09. The AC input power is 120 Vrms, 60 Hz. Assume a load of $R_L = 18\ \Omega$.

(10) (a) Draw the circuit.



(8) (b) Find the transformer turns ratio and filter capacitor value (show all work).

$V_{p\,Rms} = 120\,V$

$V_p = 169.7\,V$

$f = 60\,Hz$

(act) $V_{DC} = 15\,V$

$r = 0.09$

$R_L = 18\,\Omega$

$r = \frac{V_{r(pp)}}{V_{DC}}$

$0.09 = \frac{V_{rpp}}{15V}$

$V_{rpp} = 1.35$

$V_{prect} = V_{DC} + \frac{1}{2} V_{(pp)}$

$V_{prect} = 15V + \frac{1}{2}(1.35)$

$V_{prect} = 15.675$

$V_{psec} = (15.675 + .7) \times 2$

$V_{psec} = 32.75V$

$V_{rpp} \cong \left(\frac{1}{fR_L C}\right) V_{prect}$

$1.35 = \frac{1}{(60Hz \cdot 2)(18\Omega) \cdot C}(15.675V)$

$\boxed{C = 5.376\ mC}$

$32.75V \times \sqrt{2} = 46.315$

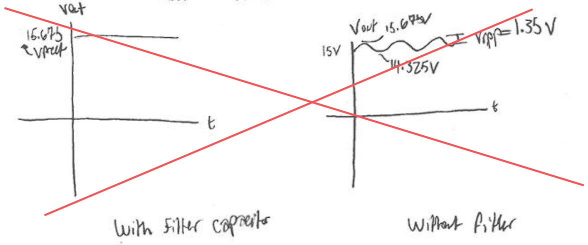
$\frac{V_P}{N_P} = \frac{V_S}{N_S} \rightarrow \frac{V_P}{V_S} = \frac{N_P}{N_S}$

$\frac{169.7V}{46.315V} = 3.66$

$\boxed{4\ turns}$

how do it in

(8) (c) Plot several periods of $V_{out}(t)$ *with* and *without* the filter capacitor in the circuit. Indicate the *values* of $V_{r(pp)}$ and $V_{p(rect)}$.



Exam 1

- Diodes require accounting very often
- Need work on zener
- Transformers are almost easy except when to account for diode and why its 2f when it never says is weird, just do it always i suppose ?
- Also plots
- All in all, pretty solid, showed trouble spots

Exam 2

Very Short Answer

(2) 1. The ________ of an op-amp is typically measured in V/µs.

(2) 2. (________) An op-amp's output can exceed $\pm V_{CC}$.

(2) 3. A non-inverting amplifier with $R_f = 18\,\mathrm{k}$ and $R_i = 2.4\,\mathrm{k}$ has a gain of ________ V/V.

(2) 4. An op-amp with a gain-bandwidth product of 3.6 MHz can provide a gain of 18 V/V over a bandwidth of ________

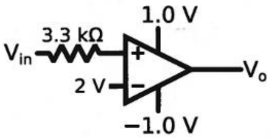
(2) 5. When configured as a voltage follower, an op-amp has a gain of ________.

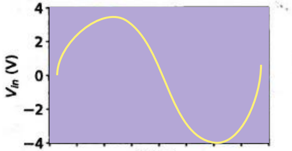
(2) 6. A 6th order filter has a roll-off of ________ dB/decade.

Short Answer

(8) 7. Design a summing amplifier giving $V_{out} = -(3V_A + 4V_B + 6V_C)$. Use a $36\,\mathrm{k\Omega}$ feedback resistor.

(8) 8. For the following circuit, add a plot of $V_o(t)$ to $V_{in}(t)$ below.





Exam 2

Very Short Answer

(2) 1. The Slew Rate of an op-amp is typically measured in V/μs.

(2) 2. (True / False) An op-amp's output can exceed $\pm V_{CC}$.

14 $\frac{RF}{Ri}$ $\frac{18}{2.4}$

(2) 3. A non-inverting amplifier with $R_f = 18\,\text{k}$ and $R_i = 2.4\,\text{k}$ has a gain of 6.5 V/V.

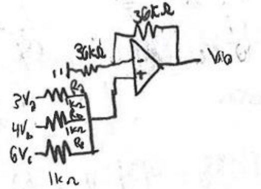
(2) 4. An op-amp with a gain-bandwidth product of 3.6 MHz can provide a gain of 18 V/V over a bandwidth of 100000 Hz.

(2) 5. When configured as a voltage follower, an op-amp has a gain of 1.

(2) 6. A 6th order filter has a roll-off of ~~3~~ dB/decade.

Short Answer

(8) 7. Design a summing amplifier giving $V_{out} = -(3V_A + 4V_B + 6V_C)$. Use a $36\,\text{k}\Omega$ feedback resistor.



$\frac{R_f}{R_i} = -1 = \frac{-36k\Omega}{36k\Omega}$

+ 4 pts Inverting summing amplifier

✔ + 2 pts Partial

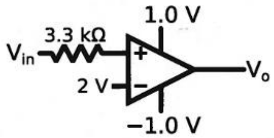
Click here to replace this description.

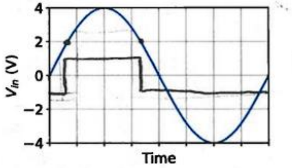
+ 4 pts $R_A = 12\,\text{k}\Omega$
$R_B = 9\,\text{k}\Omega$
$R_C = 6\,\text{k}\Omega$

+ 2 pts (partial)

+ 0 pts Blank / Substantially Incorrect

(8) 8. For the following circuit, add a plot of $V_o(t)$ to $V_{in}(t)$ below.



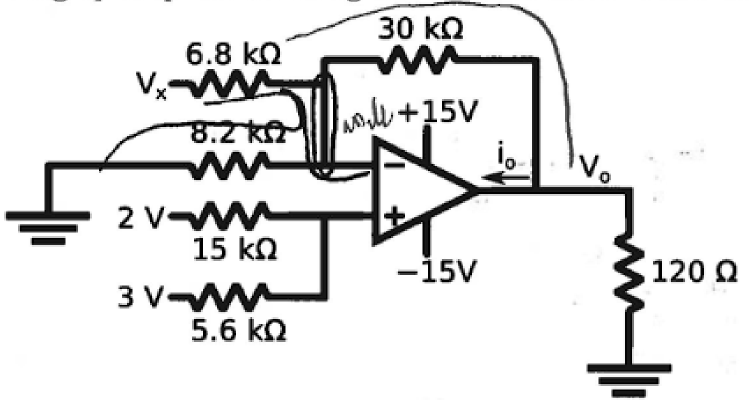


Varies from 1V to -1V based on > or < 2V

Exam 2

Long Answer

9. Analyze the following op-amp circuit using the nodal method. Show all work.



(18) (a) Find an expression for V_o as a function of V_x.

(4) (b) Find an expression for i_o as a function of V_x.

(4) (c) Find the range of V_x over which the op-amp is not saturated.

Exam 2

a)

Vx nodal

$$\frac{(V-)-0}{8.2k\Omega}+\frac{(V-)-V_x}{6.8k\Omega}+\frac{(V-)-V_0}{30k\Omega}=0$$

$$30k\Omega\left(\frac{(V-)}{8.2k\Omega}+\frac{(V-)}{6.8k\Omega}-\frac{V_x}{6.8k\Omega}+\frac{(V-)}{30k\Omega}\right)=V_0$$

$$V-\left(\frac{30k\Omega}{8.2k\Omega}+\frac{30k\Omega}{6.8k\Omega}+1\right)-\frac{30k\Omega\cdot V_x}{6.8k\Omega}=V_0$$

$$2.728V\left(\frac{30k\Omega}{8.2k\Omega}+\frac{30k\Omega}{6.8k\Omega}+1\right)-\frac{30k\Omega\cdot V_x}{6.8k\Omega}=V_0$$

$$\boxed{24.7438-4.412V_x=V_0}$$

V+ nodal

$$\frac{V_+-2V}{15k\Omega}+\frac{V_+-3V}{5.6k\Omega}=0$$

$$V_+\left(\frac{1}{15k\Omega}+\frac{1}{5.6k\Omega}\right)=\frac{2V}{15k\Omega}+\frac{3V}{5.6k\Omega}$$

$$V_+=\frac{\frac{2V}{15k\Omega}+\frac{3V}{5.6k\Omega}}{\frac{1}{15k\Omega}+\frac{1}{5.6k\Omega}}$$

$$V_+=2.728V$$

$$V_+\approx V_-$$

b)

$$i_0+\frac{V_0-V_-}{30k\Omega}+\frac{V_0-0}{120\Omega}=0$$

$$i_0=-\left(\frac{V_0}{30k\Omega}-\frac{V_-}{30k\Omega}+\frac{V_0}{120\Omega}\right)$$

$$\boxed{i_0=-\left(\frac{24.7438-4.412V_x}{30k\Omega}-\frac{2.728V}{30k\Omega}+\frac{24.7438-4.412V_x}{120\Omega}\right)}$$

c)

$$-15V<24.7438-4.412V_x<15V$$

$$-24.7438$$

$$\frac{-39.7438<-4.412V_x<-9.7438}{-4.412}$$

↳ sign flip & < flip

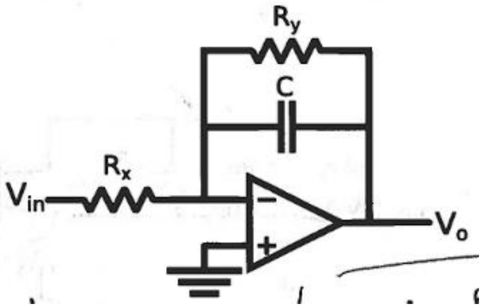
$$9.008>V_x>2.208$$

$$\boxed{2.208<V_x<9.008}$$

Exam 2

10. Transfer Functions

(12) (a) Find the transfer function $H(s)$ of the following circuit as a single fraction and its magnitude $|H(j\omega)|$. Show all work.



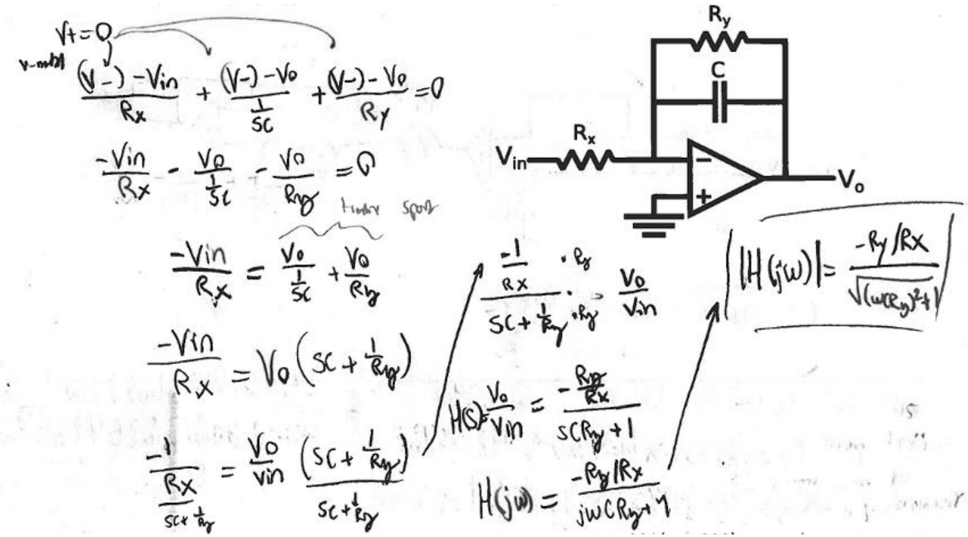
(b) A filter is described by the transfer function $H(s) = \dfrac{20s(s+200)}{(s+20)(s+8000)}$

i. Generate a Bode plot.

ii. Include the approximate shape of the *actual* filter response.

iii. What is its passband gain in dB?

Exam 2

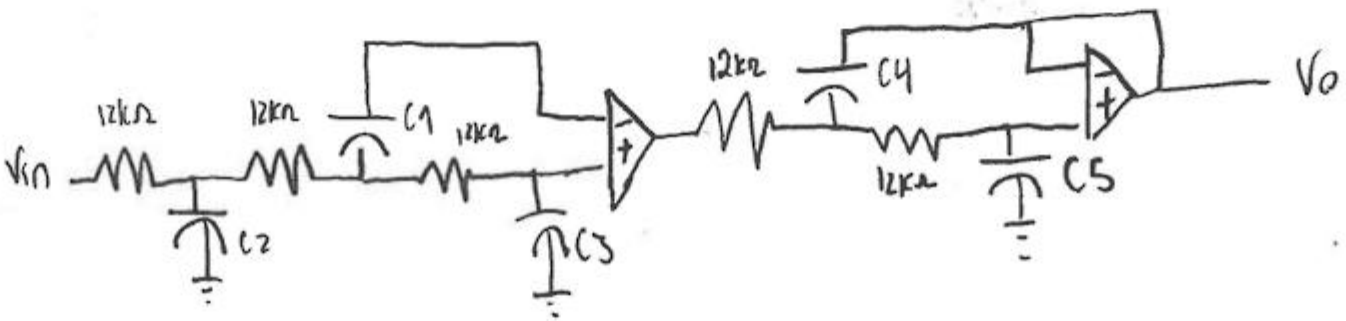


Bode was wrong,
have to try that
again

Exam 2

(20) 11. Design a 5th order Chebyshev low pass filter with 1 dB of passband ripple, a cutoff frequency of 800 Hz, and a passband gain of 7 dB. Use 12 kΩ resistors in the filter stages. Draw the circuit, labeling all component values.

Exam 2



$$C = \frac{1}{2\pi \cdot 800 \cdot 12000} = 1.6579 \times 10^{-8}$$

+ 6 pts Capacitor values (first stage):
$C_1 = 147\,\mathrm{nF}$
$C_2 = 65.2\,\mathrm{nF}$
$C_3 = 4.21\,\mathrm{nF}$
(second stage):
$C_1 = 191.5\,\mathrm{nF}$
$C_2 = 1.55\,\mathrm{nF}$

+ 3 pts Component values (half)

+ 2 pts Gain stage (connected correctly)

+ 2 pts Gain stage resistor ratio 2.2 (inverting) or 1.2 (non-inverting)
Resistors at least hundreds of ohms or kohms

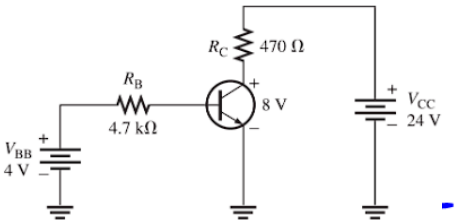
+ 0 pts Blank / substantially incorrect

HW8

8. What is the value of I_C for $I_E = 5.34$ mA and $I_B = 475$ μA?

Delay Cancel Options

~~ection 4–3 BJT Charac~~



~~9. What is the α_{DC} when I_C~~

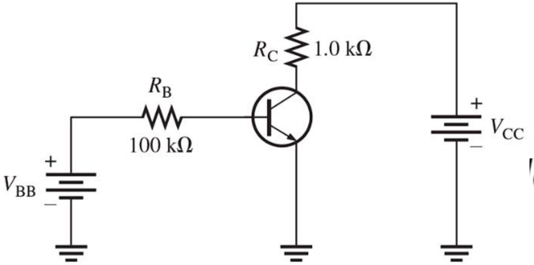
10. A certain transistor has an $I_C = 25$ mA and an $I_B = 200$ μA. Determine the β_{DC}.

11. What is the β_{DC} of a transistor if $I_C = 20.3$ mA and $I_E = 20.5$ mA?

~~12. What is tl~~

13. A certain

~~$I_E = 9.3$~~



I_C when

14. A base current of 50 μA is applied to the transistor in **Figure 4–54**, and a voltage of 5 V is dropped across R_C. Determine the β_{DC} of the transistor.

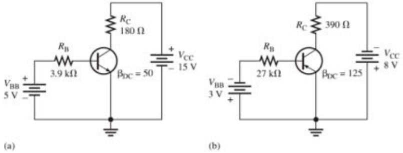
16. Assume that the transistor in the circuit of **Figure 4–54** is replaced with one having a β_{dc} of 200. Determine I_B, I_C, I_E, and V_{CE} given that $V_{CC} = 10$ V and $V_{BB} = 3$ V.

17. If V_{CC} is increased to 15 V in **Figure 4–54**, how much do the currents and V_{CE} change?

18. Determine each current in **Figure 4–55**. What is the β_{DC}?

19. Find V_{CE}, V_{BE}, and V_{CB} in both circuits of **Figure 4–56**.

Figure 4–56



20. Determine whether or not the transistors in **Figure 4–56** are saturated.

HW8

Homework 12 Solution

8. $I_E = I_B + I_C$

$5.34mA = 475\mu A + I_C$

$I_C = 4.865\ mA$

10. $I_C = 25mA \quad I_B = 200\mu A$

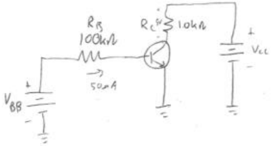
$\beta_{DC} = \frac{I_C}{I_B} = 125$

11. $I_C = 20.3mA \quad I_E = 20.5\ mA$

$I_B = I_E - I_C = 0.2mA$

$\beta_{DC} = \frac{20.3mA}{0.2mA} = 101.5$

14.



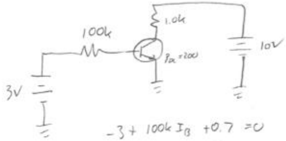
$5V = I_C \cdot R_C$

$= I_C \cdot 1.0k$

$\boxed{I_C = 5mA}$

$\beta_{DC} = \frac{5mA}{50\mu A} = 100$

16.



$-3 + 100k\,I_B + 0.7 = 0$

$I_B = 23\mu A$

$I_C = \beta I_B = 4.6mA$

$I_E = I_C + I_B = 4.623mA$

$-10 + I_C \cdot 1.0k + V_{CE} = 0$

$10 - 4.6V = V_{CE}$

$V_{CE} = 5.4V$

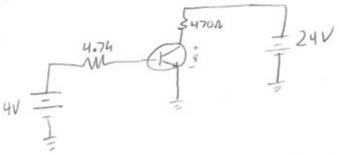
17. Currents stay the same

$-15 + 1.0k \cdot 4.6mA + V_{CE} = 0$

$V_{CE} = 10.4V$

$\Delta V_{CE} = 10.4 - 5.4 = 5V$

18.



$-4 + 4.7k \cdot I_B + 0.7 = 0$

$I_B = 702\mu A$

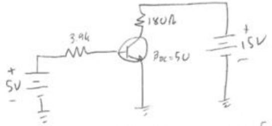
$-24 + I_C(470) + 8 = 0$

$I_C = 34.0mA$

$\beta_{DC} = \frac{I_C}{I_B} = 48.5$

$I_E = I_B + I_C = 34.75\ mA$

19. (a)



$-5 + 3.9k\,I_B + 0.7 = 0$

$I_B = 1.1mA$

$I_C = 55mA$

$-15 + 180I_C \cancel{+ V_{CE} = 0}$

$\boxed{V_{CE} = V_C = 5.08V}$

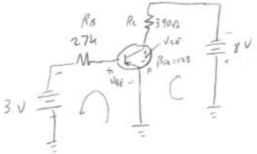
$V_B = 5 - 3.9k \cdot 1.1mA$

$V_B = \boxed{0.7 = V_{BE}}$

$V_{CB} = V_{CE} - V_{BE} = 5.08V - 0.71V$

$\boxed{V_{CB} = 4.37V}$

19(b)



find bias:
$V_{BE} = V_B - V_E = -0.7$

$\boxed{V_{BE} = -0.7}$

$0 - V_{BE} + 27k\,I_B - 3 = 0$

$\frac{2.3V}{27k} = I_B = 85\mu A$

$I_C = \beta I_B = 10.6mA$

$0 - V_{CE} + 390I_C - 8V = 0$

$0 - V_{CE} + 390(10.6mA) - 8V = 0$

$\boxed{V_{CE} = -3.87V}$

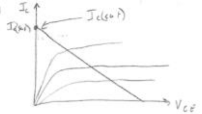
$V_C = -3.87V$

$V_B = -0.7$

$\boxed{V_{CB} = V_C - V_B = -3.166V}$

Assuming $V_{CE(sat)} = 0$

20. (a)



$I_{C(sat)}$ occurs when $V_{CE} = 0$

$-15 + I_{C(sat)}(180) + 0 = 0$

$\boxed{I_{C(sat)} = 83.3mA}$

$-5 + 3.9k\,I_B + 0.7V = 0$

$I_B = 1.1mA$

$I_C = \beta I_B = 55mA$

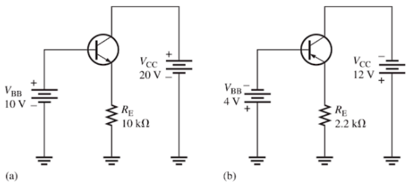
$55mA < I_{C(sat)}$

not saturated

HW8

22. Determine the terminal voltages of each transistor with respect to ground for each circuit in **Figure 4–58**. Also determine V_{CE}, V_{BE}, and V_{CB}.

Figure 4–58



29. Determine the value of the collector resistor in an *npn* transistor amplifier with

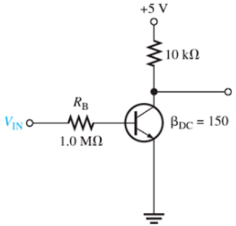
$\beta_{DC} = 250$, $V_{BB} = 2.5$ V, $V_{CC} = 9$ V, $V_{CE} = 4$ V, and $R_B = 100$ kΩ

31. Determine $I_{C(sat)}$ for the transistor in **Figure 4–59**. What is the value of I_B

necessary to produce saturation? What minimum value of V_{IN} is necessar

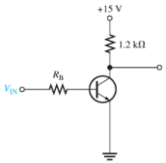
for saturation? Assume $V_{CE(sat)} = 0$ V.

Figure 4–59



32. The transistor in **Figure 4–60** has a β_{DC} of 50. Determine the value of R_B required to ensure saturation when V_{IN} is 5 V. What must V_{IN} be to cut off the transistor? Assume $V_{CE(sat)} = 0$ V.

Figure 4–60



HW8

20(b)

$0 - V_{CE} + 390 I_C - 8V = 0$ $\quad V_{CE(sat)} = 0$

$0 - 0 + 390 I_{C(sat)} = 8V$

$\boxed{I_{C(sat)} = 20.5\text{ mA}}$

$0 - V_{BE} + 27k I_B - 3V = 0$

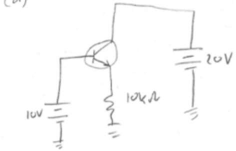
$I_B = 85\mu A$

$I_C = \beta I_B = 10.6\text{ mA}$

$I_{C(sat)} > I_C$. not saturated

22. Determine the terminal voltages of each transistor with respect to ground. Also V_{CE}, V_{BE}, V_{CB}

(a)



$V_B = 10V$

$V_C = 20V$

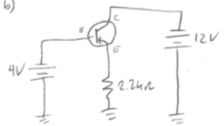
$V_E = 10 - 0.7 = 9.3V$

$V_{CE} = V_C - V_E = 10.7V$

$V_{BE} = V_B - V_E = 0.7V$

$V_{CB} = V_C - V_B = 10V$

(b)



$V_B = -4V$

$V_C = -12V$

$V_E = -3.3V$

$V_{CE} = V_C - V_E = -8.7V$

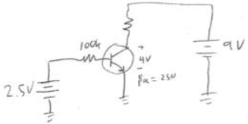
$V_{BE} = V_B - V_E = -0.7V$

$V_{CB} = V_C - V_B = -8V$

28. $A_v = \frac{R_c}{r_e'} = \frac{560}{10} = 56$

$50mV \cdot 56 = 2.8V_{rms}$

29.



$-2.5V + 100k I_B + 0.7 = 0$

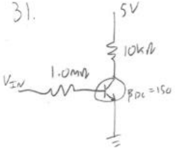
$I_B = 18\mu A$

$I_C = \beta_{DC} I_B = 4.5\text{ mA}$

$-9 + R_C(4.5\text{ mA}) + 4 = 0$

$\boxed{R_C = 1.11\text{ k}\Omega}$

31.



At saturation, $V_{CE} = 0$

$-5V + 10k I_{C(sat)} + 0 = 0$

$I_{C(sat)} = 0.5\text{ mA}$

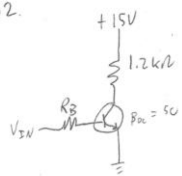
$I_{B_{min}} = \frac{I_{C(sat)}}{150} = 3.33\mu A$

$-V_{IN} + 1.0M\Omega\, I_B + 0.7 = 0$

$-V_{IN} + 1.0M\Omega(3.33\mu A) + 0.7 = 0$

$V_{IN} = 4.03V$

32.



$-15 + 1.2k I_{C(sat)} + 0 = 0$

$\boxed{I_{C(sat)} = 12.5\text{ mA}}$

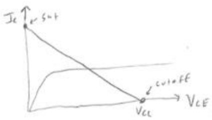
$I_{B_{min}} = 0.25\text{ mA}$

$-V_{IN} + R_B I_{B_{min}} + 0.7 = 0$

$-5V + R_B(0.25\text{ mA}) + 0.7 = 0$

$\boxed{R_B = 17.2\text{ k}\Omega}$

For cutoff



Cutoff occurs when $V_{CE} = V_{CC}$

$-15 + 1.2k \cdot I_C + 15V = 0$

$I_C = 0,$

$I_B = 0,$ $\boxed{V_{IN} = 0}$

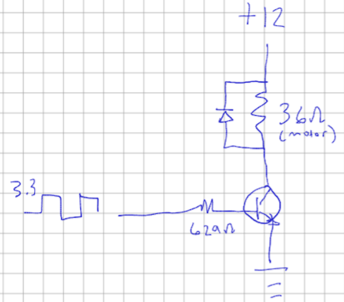
HW8

1. Design a circuit using an NPN transistor ($\beta = 160$, $V_{CE(sat)} = 0.1$ V) as a switch to control a 12 V motor with an on-resistance of $R_M = 36\ \Omega$ using a 3.3 V logic signal. Make sure to include a flyback diode across the motor to protect the transistor from EMF produced by the motor's collapsing magnetic field when it is switched off. Choose R_B such that I_B is at least 2 times the minimum needed for saturation. (3 points)

2. Design a voltage divider bias circuit for an NPN transistor ($\beta = 90$), with $V_{CC} = 15$ V, $R_C = 68\ \Omega$, $R_E = 270\ \Omega$, such that $V_{CE} = 6.0$ V. Find I_C. Use with the Thevenin model, assuming $V_{th} = 9$ V. Find R_{th}, and then use V_{th} and R_{th} to find R_1 and R_2. (3 points)

HW8

1)



$$12 - 36 I_c - 0.1 = 0$$

$$I_c = 330.6\,mA$$

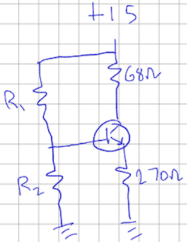
$$I_{B(min)} = \frac{I_c}{\beta} = 2.07\,mA$$

$$2 I_{B(min)} = 4.132\,mA$$

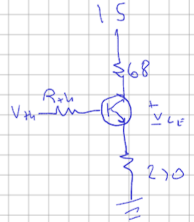
$$3.3 - 4.132\,mA\, R_B - 0.7 = 0$$

$$R_B = 629\,\Omega$$

2)



→



$$15 - 68 I_E - 6.0V - 270 I_E = 0$$

$$I_E = 26.6\,mA \approx I_c$$

$$I_B = \frac{I_c}{\beta} = 296\,\mu A$$

$$9 - I_B \cdot R_{th} - 0.7 - I_E \cdot 270 = 0$$

$$R_{th} = 3754\,\Omega$$

$$V_{th} = 9 = 15 \frac{R_2}{R_1 + R_2}$$

$$R_{th} = \frac{R_1 R_2}{R_1 + R_2} = 3754\,\Omega$$

solve for R_1, R_2

$R_1 = 6257\,\Omega$ $\quad R_2 = 9385\,\Omega$